

Outcome_analysis

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Create the folders

```
dir.create("output")
dir.create("figures")
dir.create("reports")
```

Load necessary libraries

1 Import data and define variables

```
Outcome <- read.csv(
  here("data", "Outcome_data.csv"),
  header = TRUE,
  check.names = FALSE
)

Outcome <- Outcome %>%
  mutate(
    across(c(PID, QCODE), as.character),

    across(c(AGS, YEARS,BLDLOSS, BMI, CATHPERIOD, FUPM, HOSPTIME,OPRTIME,
      POSTUTIFUP, POSTUTI_T), as.numeric),

    across(c(REC011, ANASTOMO,BLDCompleak,BLDCompRetn,BLDCompVoid,BLDSPASM,CATH,CENTER,
      CLAVIENCLASS,EPIDURAL,GENDER, GU30, GU90N, GUER30, GUER90, HEMAT,
      HISTOINJ,ILEUS, INCCOMPBLEED, INCCOMPINF, INCCOMPSCAR, NARC,
      NONARC, PAIN, PREOPEPISODE1, PREOPRNPLN, PREOPRNSCR, PREOPTYPEU1,
      PREOPVURGRADE, PROC365, STENT, SURGIND, SURGTECH2, TRT2, UDR7M,
      URCOMPVUR, URCOMPINJ, URCOMPOBST, URCOMPSTEN, POSTUTI, VURPHASE,
      VURSIDE, RECGRADE_MAX), as.factor)
  )

str(Outcome) # 412 obs. of 55 variables

'data.frame':  412 obs. of  55 variables:
 $ PID      : chr  "P1" "P2" "P3" "P4" ...
 $ QCODE    : chr  "102" "111" "3" "4" ...
 $ REC011   : Factor w/ 2 levels "0","1": 1 1 1 2 1 2 2 1 2 2 ...
```

```

$ AGS : num 5 6 7 10 7 9 10 8 7 11 ...
$ YEARS : num 2011 2014 2017 2014 2011 ...
$ ANASTOMO : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
$ BLDCompleak : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
$ BLDCompRetn : Factor w/ 2 levels "0","1": 1 1 2 1 1 2 1 2 2 2 ...
$ BLDCompVoid : Factor w/ 2 levels "0","1": 2 1 1 1 1 1 1 1 1 1 ...
$ BLDLOSS : num 70 300 150 55 55 190 75 100 55 55 ...
$ BLDSPASM : Factor w/ 2 levels "0","1": 1 2 2 1 1 2 1 2 2 1 ...
$ BMI : num 13.5 13.7 12 13.3 15 15.8 15.3 15.3 14.8 15.3 ...
$ CATH : Factor w/ 2 levels "0","1": 2 2 2 2 2 2 2 2 2 2 ...
$ CATHPERIOD : num 2 7 5 5 4 7 4 5 5 7 ...
$ CENTER : Factor w/ 4 levels "1","2","3","5": 1 1 1 1 1 1 1 1 1 1 ...
$ CLAVIENCLASS : Factor w/ 4 levels "1","2","3","4": 2 2 2 2 1 2 2 1 2 3 ...
$ EPIDURAL : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 2 1 1 1 ...
$ FUPM : num 24 24 44 36 120 36 36 37 12 36 ...
$ GENDER : Factor w/ 2 levels "0","1": 1 1 2 2 2 1 1 2 2 1 ...
$ GU30 : Factor w/ 2 levels "0","1": 1 2 1 1 1 1 2 1 2 1 ...
$ GU90N : Factor w/ 3 levels "0","1","2": 1 2 2 2 1 2 2 1 2 1 ...
$ GUER30 : Factor w/ 2 levels "0","1": 1 2 1 1 1 1 2 1 2 1 ...
$ GUER90 : Factor w/ 2 levels "0","1": 2 2 1 2 1 1 2 1 2 1 ...
$ HEMAT : Factor w/ 2 levels "0","1": 1 2 2 2 1 2 2 1 2 1 ...
$ HISTOINJ : Factor w/ 2 levels "0","1": 2 1 1 1 1 1 1 2 1 1 ...
$ HOSPTIME : num 2 10 5 5 4 5 8 5 5 4 ...
$ ILEUS : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
$ INCCOMPBLEED : Factor w/ 2 levels "0","1": 2 1 2 1 1 1 1 1 1 1 ...
$ INCCOMPINF : Factor w/ 2 levels "0","1": 1 1 1 1 1 2 1 1 2 1 ...
$ INCCOMPSCAR : Factor w/ 2 levels "0","1": 1 2 1 1 1 1 1 1 1 1 ...
$ NARC : Factor w/ 2 levels "0","1": 1 2 2 1 1 2 1 1 2 1 ...
$ NONARC : Factor w/ 2 levels "0","1": 2 2 2 2 2 2 2 2 2 2 ...
$ OPRTIME : num 76 80 80 83 90 75 90 88 85 80 ...
$ PAIN : Factor w/ 4 levels "0","1","2","3": 2 4 4 2 2 3 3 2 4 2 ...
$ PREOPEPISODE1 : Factor w/ 3 levels "1","2","3": 2 3 3 3 3 3 3 2 3 3 ...
$ PREOPRNPLN : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
$ PREOPRNSCR : Factor w/ 2 levels "0","1": 1 2 1 2 1 1 1 1 2 1 ...
$ PREOPTYPEU1 : Factor w/ 2 levels "0","1": 2 2 2 2 2 2 2 2 2 2 ...
$ PREOPVURGRADE : Factor w/ 3 levels "3","4","5": 3 3 3 3 2 3 2 2 3 2 ...
$ PROC365 : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 2 1 1 1 ...
$ STENT : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
$ SURGIND : Factor w/ 11 levels "1","2","3","4",...: 3 3 3 3 4 4 4 3 3 3 ...
$ SURGTECH2 : Factor w/ 3 levels "1","2","3": 3 2 2 2 2 2 2 2 2 2 ...
$ TRT2 : Factor w/ 2 levels "0","1": 2 2 2 2 2 2 2 2 2 2 ...
$ UDR7M : Factor w/ 2 levels "0","1": 2 2 2 2 1 2 2 2 2 2 ...
$ URCOMPVCUR : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
$ URCOMPINJ : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
$ URCOMPOBST : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 2 1 ...
$ URCOMPSTEN : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 2 ...
$ POSTUTI : Factor w/ 2 levels "0","1": 2 1 1 1 1 1 2 1 1 2 ...
$ POSTUTIFUP : num 1 24 44 36 120 36 1 37 12 1 ...
$ POSTUTI_T : num 1 0 0 0 0 0 1 0 0 1 ...
$ VURPHASE : Factor w/ 3 levels "1","2","3": 1 1 3 3 1 1 1 1 1 1 ...
$ VURSIDE : Factor w/ 2 levels "0","1": 1 2 2 2 1 2 2 1 1 2 ...
$ RECGRADE_MAX : Factor w/ 6 levels "0","1","2","3",...: 2 2 1 3 1 3 3 2 3 3 ...

```

2 Data Imputation using KNN (k-nearest neighbors)

```
# Missingness in the dataset
m_outcome <- colSums(is.na(Outcome)) / nrow(Outcome) * 100
print(m_outcome)
```

PID	QCODE	REC011	AGS	YEARS
0.0000000	0.0000000	33.7378641	0.0000000	0.0000000
ANASTOMO	BLDCOMPLEAK	BLDCOMPRETN	BLDCOMPVOID	BLDLOSS
0.0000000	0.0000000	0.0000000	0.0000000	1.9417476
BLDSPASM	BMI	CATH	CATHPERIOD	CENTER
0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
CLAVIENCLASS	EPIDURAL	FUPM	GENDER	GU30
22.0873786	0.0000000	0.0000000	0.0000000	0.0000000
GU90N	GUER30	GUER90	HEMAT	HISTOINJ
0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
HOSPTIME	ILEUS	INCCOMPBLEED	INCCOMPINF	INCCOMPSCAR
0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
NARC	NONARC	OPRTIME	PAIN	PREOPEPISODE1
0.0000000	0.0000000	0.0000000	0.9708738	2.9126214
PREOPRNPLN	PREOPRNSCR	PREOPTYPEU1	PREOPVURGRADE	PROC365
20.6310680	0.0000000	0.2427184	0.0000000	0.0000000
STENT	SURGIND	SURGTECH2	TRT2	UDR7M
0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
URCOMPVCUR	URCOMPINJ	URCOMPOBST	URCOMPSTEN	POSTUTI
8.7378641	0.0000000	0.0000000	0.0000000	0.0000000
POSTUTIFUP	POSTUTI_T	VURPHASE	VURSIDE	RECGRADE_MAX
0.0000000	0.0000000	0.0000000	0.0000000	33.7378641

```
# KNN imputation
Outcome_knn <- Outcome # create a temporary dataset copy for KNN using scaled variables

# Convert categorical variables
cat_vars <- c(
  "GENDER", "PREOPVURGRADE", "VURPHASE", "VURSIDE",
  "UDR7M", "PREOPRNSCR", "HISTOINJ", "SURGIND",
  "CENTER", "TRT2", "SURGTECH2",
  "PREOPEPISODE1", "PREOPRNPLN", "PREOPTYPEU1"
)
Outcome_knn[cat_vars] <- lapply(Outcome_knn[cat_vars], as.factor)

# Scale continuous variables
continuous_vars <- c("AGS", "BMI", "YEARS")
Outcome_knn[continuous_vars] <- scale(Outcome_knn[continuous_vars])

# KNN imputation without subsetting the dataset
Outcome_knn <- knn(
  Outcome_knn,
  variable = c("PREOPEPISODE1", "PREOPRNPLN", "PREOPTYPEU1"), # Variables to impute
  dist_var = c("AGS", "GENDER", "BMI", # Variables used to compute patient similarity
    "PREOPVURGRADE", "VURPHASE", "VURSIDE", "UDR7M",
```

```

      "PREOPRNSCR", "HISTOINJ", "SURGIND", "YEARS",
      "TRT2"
    ),
    k = 5,
    imp_var = FALSE
  )

# CENTER + TRT2 + SURGTECH2 together may overweight institutional structure, which means clustering pat.

Out <- Outcome # New dataset for analysis including imputed variables

Out$PREOPEPISODE1 <- Outcome_knn$PREOPEPISODE1
Out$PREOPRNPLN <- Outcome_knn$PREOPRNPLN
Out$PREOPTYPEU1 <- Outcome_knn$PREOPTYPEU1

```

3 Recode and create (composite) variables

```

# Recode SURGIND into SURGIND2

Out <- Out %>%
  mutate(
    SURGIND2 = case_when(
      SURGIND == 4 ~ 1,
      SURGIND == 3 ~ 2,
      SURGIND == 1 ~ 3,
      SURGIND == 2 ~ 4,
      SURGIND %in% c(5, 6, 7, 8, 9, 10, 11) ~ 5,
      TRUE ~ NA_real_
    ),
    SURGIND2 = factor(SURGIND2)
  )

# Create a new variable LUTD by combining BLDCOMPRETN, BLDCOMPVOID, BLDCOMPLEAK
Out <- Out %>%
  mutate(
    LUTD = case_when(
      BLDCOMPRETN == 1 |
      BLDCOMPVOID == 1 |
      BLDCOMPLEAK == 1 ~ 1,

      BLDCOMPRETN == 0 &
      BLDCOMPVOID == 0 &
      BLDCOMPLEAK == 0 ~ 0,

      TRUE ~ NA_real_
    ),
    LUTD = factor(LUTD)
  )

```

```

# Create a new variable OBST by combining URCOMPOBST, URCOMPSTEN, ANASTOMO
Out <- Out %>%
  mutate(
    OBST = case_when(
      URCOMPOBST == 1 |
      URCOMPSTEN == 1 |
      ANASTOMO == 1 ~ 1,

      URCOMPOBST == 0 &
      URCOMPSTEN == 0 &
      ANASTOMO == 0 ~ 0,

      TRUE ~ NA_real_
    ),
    OBST = factor(OBST)
  )

# Create a new variable SURG by combining INCCOMPBLEED, INCCOMPINF, INCCOMPSCAR
Out <- Out %>%
  mutate(
    SURG = case_when(
      INCCOMPBLEED == 1 |
      INCCOMPINF == 1 |
      INCCOMPSCAR == 1 ~ 1,

      INCCOMPBLEED == 0 &
      INCCOMPINF == 0 &
      INCCOMPSCAR == 0 ~ 0,

      TRUE ~ NA_real_
    ),
    SURG = factor(SURG)
  )

# Create a new variable GU90 by combining no readmission (0) and planned readmission (2)
Out <- Out %>%
  mutate(
    GU90 = recode(as.character(GU90N),
      "0" = "0",
      "1" = "1",
      "2" = "0"),
    GU90 = factor(GU90)
  )

# Create a categorical variable CATHPERIOD_G for the numerical variable CATHPERIOD
Out <- Out %>%
  mutate(CATHPERIOD_G = case_when(
    CATHPERIOD <= 3 ~ "Early removal 3 days",
    CATHPERIOD > 3 & CATHPERIOD <= 7 ~ "Standard removal 4-7 days",
    CATHPERIOD > 7 ~ "Delayed removal >7 days",
  ))

```

```

Out$CATHPERIOD_G <- factor(
  Out$CATHPERIOD_G,
  levels = c("Early removal 3 days", "Standard removal 4-7 days", "Delayed removal >7 days")
)

# For the sensitivity analysis for missing outcome data using two extreme assumptions: Best-case scenario
Out$REC011 <- as.numeric(as.character(Out$REC011))

Out$REC011_best <- ifelse(
  is.na(Out$REC011),
  0,
  Out$REC011
)

Out$REC011_worst <- ifelse(
  is.na(Out$REC011),
  1,
  Out$REC011
)

# Recode complications
comp_vars <- c(
  "LUTD",
  "OBST",
  "URCOMPINJ",
  "SURG",
  "ILEUS"
)

Out[comp_vars] <- lapply(
  Out[comp_vars],
  function(x) as.numeric(as.character(x))
)

Out$ANY <- as.numeric(
  rowSums(
    Out[, comp_vars],
    na.rm = TRUE
  ) > 0
)

minor_vars <- c(
  "LUTD",
  "SURG",
  "ILEUS"
)

Out$MINOR <- as.numeric(
  rowSums(

```

```

    Out[, minor_vars],
    na.rm = TRUE
  ) > 0
)

major_vars <- c(
  "OBST",
  "URCOMPINJ"
)

Out$MAJOR <- as.numeric(
  rowSums(
    Out[, major_vars],
    na.rm = TRUE
  ) > 0
)

# Convert variables to appropriate data types for analysis
Out <- Out %>%
  mutate(
    across(c(PID, QCODE), as.character),

    across(c(AGS, YEARS, BLDLOSS, BMI, CATHPERIOD, FUPM, HOSPTIME, OPRTIME,
              POSTUTIFUP, POSTUTI_T), as.numeric),

    across(c(REC011, REC011_best, REC011_worst, ANASTOMO, BLDCompleak, BLDCompRetn,
              BLDCompVoid, BLDSPASM, CATH, CENTER, CLAVIENCLASS, EPIDURAL, GENDER,
              GU30, GU90N, GUER30, GUER90, HEMAT,
              HISTOINJ, ILEUS, INCCOMPBLEED, INCCOMPINF, INCCOMPSCAR, NARC,
              NONARC, PAIN, PREOPEPISODE1, PREOPRNPLN, PREOPRNSCR, PREOPTYPEU1,
              PREOPVURGRADE, PROC365, STENT, SURGIND, SURGTECH2, TRT2, UDR7M,
              URCOMPVUR, URCOMPINJ, URCOMPOBST, URCOMPSTEN, POSTUTI, VURPHASE,
              VURSIDE, RECGRADE_MAX, SURGIND2, LUTD, OBST, SURG, GU90,
              CATHPERIOD_G, ANY, MINOR, MAJOR), as.factor)
  )

str(Out) # 412 obs. of 66 variables

'data.frame':  412 obs. of  66 variables:
 $ PID      : chr  "P1" "P2" "P3" "P4" ...
 $ QCODE    : chr  "102" "111" "3" "4" ...
 $ REC011   : Factor w/ 2 levels "0","1": 1 1 1 2 1 2 2 1 2 2 ...
 $ AGS      : num   5 6 7 10 7 9 10 8 7 11 ...
 $ YEARS    : num   2011 2014 2017 2014 2011 ...
 $ ANASTOMO : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
 $ BLDCompleak : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
 $ BLDCompRetn : Factor w/ 2 levels "0","1": 1 1 2 1 1 2 1 2 2 2 ...
 $ BLDCompVoid : Factor w/ 2 levels "0","1": 2 1 1 1 1 1 1 1 1 1 ...
 $ BLDLOSS   : num   70 300 150 55 55 190 75 100 55 55 ...
 $ BLDSPASM  : Factor w/ 2 levels "0","1": 1 2 2 1 1 2 1 2 2 1 ...
 $ BMI       : num   13.5 13.7 12 13.3 15 15.8 15.3 15.3 14.8 15.3 ...

```

\$ CATH : Factor w/ 2 levels "0","1": 2 2 2 2 2 2 2 2 2 2 ...
 \$ CATHPERIOD : num 2 7 5 5 4 7 4 5 5 7 ...
 \$ CENTER : Factor w/ 4 levels "1","2","3","5": 1 1 1 1 1 1 1 1 1 1 ...
 \$ CLAVIENCLASS : Factor w/ 4 levels "1","2","3","4": 2 2 2 2 1 2 2 1 2 3 ...
 \$ EPIDURAL : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 2 1 1 1 ...
 \$ FUPM : num 24 24 44 36 120 36 36 37 12 36 ...
 \$ GENDER : Factor w/ 2 levels "0","1": 1 1 2 2 2 1 1 2 2 1 ...
 \$ GU30 : Factor w/ 2 levels "0","1": 1 2 1 1 1 1 2 1 2 1 ...
 \$ GU90N : Factor w/ 3 levels "0","1","2": 1 2 2 2 1 2 2 1 2 1 ...
 \$ GUER30 : Factor w/ 2 levels "0","1": 1 2 1 1 1 1 2 1 2 1 ...
 \$ GUER90 : Factor w/ 2 levels "0","1": 2 2 1 2 1 1 2 1 2 1 ...
 \$ HEMAT : Factor w/ 2 levels "0","1": 1 2 2 2 1 2 2 1 2 1 ...
 \$ HISTOINJ : Factor w/ 2 levels "0","1": 2 1 1 1 1 1 1 2 1 1 ...
 \$ HOSPTIME : num 2 10 5 5 4 5 8 5 5 4 ...
 \$ ILEUS : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
 \$ INCCOMPBLEED : Factor w/ 2 levels "0","1": 2 1 2 1 1 1 1 1 1 1 ...
 \$ INCCOMPINF : Factor w/ 2 levels "0","1": 1 1 1 1 1 2 1 1 2 1 ...
 \$ INCCOMPSCAR : Factor w/ 2 levels "0","1": 1 2 1 1 1 1 1 1 1 1 ...
 \$ NARC : Factor w/ 2 levels "0","1": 1 2 2 1 1 2 1 1 2 1 ...
 \$ NONARC : Factor w/ 2 levels "0","1": 2 2 2 2 2 2 2 2 2 2 ...
 \$ OPRTIME : num 76 80 80 83 90 75 90 88 85 80 ...
 \$ PAIN : Factor w/ 4 levels "0","1","2","3": 2 4 4 2 2 3 3 2 4 2 ...
 \$ PREOPEPISODE1 : Factor w/ 3 levels "1","2","3": 2 3 3 3 3 3 3 2 3 3 ...
 \$ PREOPRNPLN : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
 \$ PREOPRNSCR : Factor w/ 2 levels "0","1": 1 2 1 2 1 1 1 1 2 1 ...
 \$ PREOPTYPEU1 : Factor w/ 2 levels "0","1": 2 2 2 2 2 2 2 2 2 2 ...
 \$ PREOPVURGRADE : Factor w/ 3 levels "3","4","5": 3 3 3 3 2 3 2 2 3 2 ...
 \$ PROC365 : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 2 1 1 1 ...
 \$ STENT : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
 \$ SURGIND : Factor w/ 11 levels "1","2","3","4",...: 3 3 3 3 4 4 4 3 3 3 ...
 \$ SURGTECH2 : Factor w/ 3 levels "1","2","3": 3 2 2 2 2 2 2 2 2 2 ...
 \$ TRT2 : Factor w/ 2 levels "0","1": 2 2 2 2 2 2 2 2 2 2 ...
 \$ UDR7M : Factor w/ 2 levels "0","1": 2 2 2 2 1 2 2 2 2 2 ...
 \$ URCOMPCVUR : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
 \$ URCOMPINJ : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
 \$ URCOMPOBST : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 2 1 ...
 \$ URCOMPSTEN : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 2 ...
 \$ POSTUTI : Factor w/ 2 levels "0","1": 2 1 1 1 1 1 2 1 1 2 ...
 \$ POSTUTIFUP : num 1 24 44 36 120 36 1 37 12 1 ...
 \$ POSTUTI_T : num 1 0 0 0 0 0 1 0 0 1 ...
 \$ VURPHASE : Factor w/ 3 levels "1","2","3": 1 1 3 3 1 1 1 1 1 1 ...
 \$ VURSIDE : Factor w/ 2 levels "0","1": 1 2 2 2 1 2 2 1 1 2 ...
 \$ RECGRADE_MAX : Factor w/ 6 levels "0","1","2","3",...: 2 2 1 3 1 3 3 2 3 3 ...
 \$ SURGIND2 : Factor w/ 5 levels "1","2","3","4",...: 2 2 2 2 1 1 1 2 2 2 ...
 \$ LUTD : Factor w/ 2 levels "0","1": 2 1 2 1 1 2 1 2 2 2 ...
 \$ OBST : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 2 2 ...
 \$ SURG : Factor w/ 2 levels "0","1": 2 2 2 1 1 2 1 1 2 1 ...
 \$ GU90 : Factor w/ 2 levels "0","1": 1 2 2 2 1 2 2 1 2 1 ...
 \$ CATHPERIOD_G : Factor w/ 3 levels "Early removal 3 days",...: 1 2 2 2 2 2 2 2 2 2 ...
 \$ REC011_best : Factor w/ 2 levels "0","1": 1 1 1 2 1 2 2 1 2 2 ...
 \$ REC011_worst : Factor w/ 2 levels "0","1": 1 1 1 2 1 2 2 1 2 2 ...
 \$ ANY : Factor w/ 2 levels "0","1": 2 2 2 1 1 2 1 2 2 2 ...
 \$ MINOR : Factor w/ 2 levels "0","1": 2 2 2 1 1 2 1 2 2 2 ...
 \$ MAJOR : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 2 2 ...

4 Descriptive analysis: BEFORE weighting and matching

```
# Baseline characteristics
table1<- Out %>%
  mutate(YEARS = factor(YEARS)) %>% # Convert only within this pipeline
  dplyr::select(YEARS, AGS, GENDER, BMI, HISTOINJ, SURGIND2, PREOPVURGRADE, VURSIDE, UDR7M,
    VURPHASE, PREOPTYPEU1, PREOPEPISODE1, PREOPRNPLN, PREOPRNSCR,
    SURGTECH2, CENTER, TRT2
  ) %>%
  tbl_summary(
    by = TRT2, # for stratification
    statistic = list(
      all_continuous() ~ "{median} ({p25}-{p75})",
      all_categorical() ~ "{n} ({p}%"
    ),
    digits = all_categorical() ~ c(0, 1), # n = 0 decimals, p = 1 decimal
    missing = "ifany", # shows missing only when present
    missing_text = "Missing"
  ) %>%
  add_overall() %>%
  add_p(
    test = list(
      YEARS ~ "chisq.test"
    ),
    pvalue_fun = ~style_pvalue(.x, digits = 3)) %>% # 3 decimals
  bold_labels()
```

The following warnings were returned during `add\_p()`:

```
! For variable `YEARS` (`TRT2`) and "statistic", "p.value", and "parameter"
  statistics: Chi-squared approximation may be incorrect
```

table1

```
table1 %>%
  as_flex_table() %>%
  flextable::save_as_docx(path = "output/baseline.docx")
```

```
# Baseline characteristics
table1b<- Out %>%
  dplyr::select(YEARS, AGS, GENDER, BMI, HISTOINJ, SURGIND2, PREOPVURGRADE, VURSIDE, UDR7M,
    VURPHASE, PREOPTYPEU1, PREOPEPISODE1, PREOPRNPLN, PREOPRNSCR,
    SURGTECH2, CENTER, TRT2
  ) %>%
  tbl_summary(
    by = TRT2, # for stratification
    statistic = list(
      all_continuous() ~ "{median} ({p25}-{p75})",
      all_categorical() ~ "{n} ({p}%"
    ),
    digits = all_categorical() ~ c(0, 1), # n = 0 decimals, p = 1 decimal
```

Characteristic	Overall N = 412 ¹	0 N = 197 ¹	1 N = 215 ¹	p-value ²
YEARS				<0.001
2010	5 (1.2%)	0 (0.0%)	5 (2.3%)	
2011	10 (2.4%)	0 (0.0%)	10 (4.7%)	
2012	30 (7.3%)	12 (6.1%)	18 (8.4%)	
2013	45 (10.9%)	22 (11.2%)	23 (10.7%)	
2014	40 (9.7%)	20 (10.2%)	20 (9.3%)	
2015	32 (7.8%)	14 (7.1%)	18 (8.4%)	
2016	45 (10.9%)	15 (7.6%)	30 (14.0%)	
2017	34 (8.3%)	12 (6.1%)	22 (10.2%)	
2018	59 (14.3%)	25 (12.7%)	34 (15.8%)	
2019	36 (8.7%)	22 (11.2%)	14 (6.5%)	
2020	44 (10.7%)	29 (14.7%)	15 (7.0%)	
2021	18 (4.4%)	15 (7.6%)	3 (1.4%)	
2022	14 (3.4%)	11 (5.6%)	3 (1.4%)	
AGS	8.00 (6.00-9.00)	8.00 (4.00-9.00)	8.00 (6.00-9.00)	0.122
GENDER				0.334
0	236 (57.3%)	108 (54.8%)	128 (59.5%)	
1	176 (42.7%)	89 (45.2%)	87 (40.5%)	
BMI	14.1 (13.6-18.1)	15.1 (13.5-20.8)	13.9 (13.6-15.0)	<0.001
HISTOINJ				0.515
0	368 (89.3%)	178 (90.4%)	190 (88.4%)	
1	44 (10.7%)	19 (9.6%)	25 (11.6%)	
SURGIND2				<0.001
1	215 (52.2%)	94 (47.7%)	121 (56.3%)	
2	132 (32.0%)	57 (28.9%)	75 (34.9%)	
3	19 (4.6%)	10 (5.1%)	9 (4.2%)	
4	13 (3.2%)	11 (5.6%)	2 (0.9%)	
5	33 (8.0%)	25 (12.7%)	8 (3.7%)	
PREOPVURGRADE				0.194
3	20 (4.9%)	11 (5.6%)	9 (4.2%)	
4	126 (30.6%)	52 (26.4%)	74 (34.4%)	
5	266 (64.6%)	134 (68.0%)	132 (61.4%)	
VURSIDE				0.608
0	162 (39.3%)	80 (40.6%)	82 (38.1%)	
1	250 (60.7%)	117 (59.4%)	133 (61.9%)	
UDR7M				<0.001
0	97 (23.5%)	32 (16.2%)	65 (30.2%)	
1	315 (76.5%)	165 (83.8%)	150 (69.8%)	
VURPHASE				<0.001
1	226 (54.9%)	112 (56.9%)	114 (53.0%)	
2	69 (16.7%)	1 (0.5%)	68 (31.6%)	
3	117 (28.4%)	84 (42.6%)	33 (15.3%)	
PREOPTYPEU1				0.031
0	8 (1.9%)	7 (3.6%)	1 (0.5%)	
1	404 (98.1%)	190 (96.4%)	214 (99.5%)	
PREOPEPISODE1				<0.001
1	24 (5.8%)	16 (8.1%)	8 (3.7%)	
2	98 (23.8%)	65 (33.0%)	33 (15.3%)	
3	290 (70.4%)	116 (58.9%)	174 (80.9%)	
PREOPRNPLN				<0.001

```

    missing = "ifany", # shows missing only when present
    missing_text = "Missing"
  ) %>%
  add_overall() %>%
  add_p(pvalue_fun = ~style_pvalue(.x, digits = 3)) %>% # 3 decimals
  bold_labels()

table1b

table1b %>%
  as_flex_table() %>%
  flextable::save_as_docx(path = "output/baseline2.docx")

# Procedural factors
table2<- Out %>%
  dplyr::select(TRT2, OPRTIME, STENT, CATH, CATHPERIOD, CATHPERIOD_G, HOSPTIME) %>%
  tbl_summary(
    by = TRT2, # for stratification
    statistic = list(
      all_continuous() ~ "{median} ({p25}-{p75})", # or ~ "{mean} ± {sd}",
      all_categorical() ~ "{n} ({p}%)"
    ),
    digits = all_categorical() ~ c(0, 1), # n = 0 decimals, p = 1 decimal
    missing = "ifany",
    missing_text = "Missing"
  ) %>%
  add_overall() %>%
  add_p(pvalue_fun = ~style_pvalue(.x, digits = 3)) %>% # 3 decimals
  bold_labels()

table2

table2 %>%
  as_flex_table() %>%
  flextable::save_as_docx(path = "output/procedural.docx")

# Plot catheterization distribution
p <- ggplot(
  Out,
  aes(
    x = factor(
      TRT2,
      levels = c(0, 1),
      labels = c("Cohen Group", "Lich-Gregoir Group")
    ),
    y = CATHPERIOD
  )
) +
  geom_boxplot() +
  labs(

```

Characteristic	Overall N = 412 ¹	0 N = 197 ¹	1 N = 215
YEARS	2,016.00 (2,014.00-2,019.00)	2,018.00 (2,014.00-2,020.00)	2,016.00 (2,013.00-2,019.00)
AGS	8.00 (6.00-9.00)	8.00 (4.00-9.00)	8.00 (6.00-9.00)
GENDER			
0	236 (57.3%)	108 (54.8%)	128 (59.5%)
1	176 (42.7%)	89 (45.2%)	87 (40.5%)
BMI	14.1 (13.6-18.1)	15.1 (13.5-20.8)	13.9 (13.6-18.1)
HISTOINJ			
0	368 (89.3%)	178 (90.4%)	190 (88.4%)
1	44 (10.7%)	19 (9.6%)	25 (11.6%)
SURGIND2			
1	215 (52.2%)	94 (47.7%)	121 (56.3%)
2	132 (32.0%)	57 (28.9%)	75 (34.9%)
3	19 (4.6%)	10 (5.1%)	9 (4.2%)
4	13 (3.2%)	11 (5.6%)	2 (0.9%)
5	33 (8.0%)	25 (12.7%)	8 (3.7%)
PREOPVURGRADE			
3	20 (4.9%)	11 (5.6%)	9 (4.2%)
4	126 (30.6%)	52 (26.4%)	74 (34.4%)
5	266 (64.6%)	134 (68.0%)	132 (61.4%)
VURSIDE			
0	162 (39.3%)	80 (40.6%)	82 (38.1%)
1	250 (60.7%)	117 (59.4%)	133 (61.9%)
UDR7M			
0	97 (23.5%)	32 (16.2%)	65 (30.2%)
1	315 (76.5%)	165 (83.8%)	150 (69.8%)
VURPHASE			
1	226 (54.9%)	112 (56.9%)	114 (53.0%)
2	69 (16.7%)	1 (0.5%)	68 (31.6%)
3	117 (28.4%)	84 (42.6%)	33 (15.3%)
PREOPTYPEU1			
0	8 (1.9%)	7 (3.6%)	1 (0.5%)
1	404 (98.1%)	190 (96.4%)	214 (99.5%)
PREOPEPISODE1			
1	24 (5.8%)	16 (8.1%)	8 (3.7%)
2	98 (23.8%)	65 (33.0%)	33 (15.3%)
3	290 (70.4%)	116 (58.9%)	174 (80.9%)
PREOPRNPLN			
0	332 (80.6%)	131 (66.5%)	201 (93.5%)
1	80 (19.4%)	66 (33.5%)	14 (6.5%)
PREOPRNSCR			
0	275 (66.7%)	104 (52.8%)	171 (79.5%)
1	137 (33.3%)	93 (47.2%)	44 (20.5%)
SURGTECH2			
1	85 (20.6%)	15 (7.6%)	70 (32.6%)
2	273 (66.3%)	137 (69.5%)	136 (63.3%)
3	54 (13.1%)	45 (22.8%)	9 (4.2%)
CENTER			
1	290 (70.4%) ¹²	103 (52.3%)	187 (87.0%)
2	4 (1.0%)	4 (2.0%)	0 (0.0%)
3	37 (9.0%)	31 (15.7%)	6 (2.8%)

Characteristic	Overall N = 412 ¹	0 N = 197 ¹	1 N = 215 ¹	p-value ²
OPRTIME	80 (70-113)	115 (85-150)	70 (67-80)	<0.001
STENT				<0.001
0	270 (65.5%)	92 (46.7%)	178 (82.8%)	
1	142 (34.5%)	105 (53.3%)	37 (17.2%)	
CATH				0.179
0	9 (2.2%)	2 (1.0%)	7 (3.3%)	
1	403 (97.8%)	195 (99.0%)	208 (96.7%)	
CATHPERIOD	4 (1-5)	4 (4-38)	1 (1-4)	<0.001
CATHPERIOD_G				<0.001
Early removal 3 days	164 (39.8%)	12 (6.1%)	152 (70.7%)	
Standard removal 4-7 days	165 (40.0%)	126 (64.0%)	39 (18.1%)	
Delayed removal >7 days	83 (20.1%)	59 (29.9%)	24 (11.2%)	
HOSPTIME	4.00 (2.00-5.50)	5.00 (4.00-6.00)	2.00 (2.00-5.00)	<0.001

¹Median (Q1-Q3); n (%)

²Wilcoxon rank sum test; Pearson's Chi-squared test; Fisher's exact test

```

    x = "Treatment Group",
    y = "Catheterization Duration (days)"
) +
theme_bw() #optional

ggsave(filename = "figures/catheterization.png",
        plot = p,
        width = 8, # width in inches
        height = 6, # height in inches
        dpi = 300) # resolution

# Plot catheterization per enrollment year

okabe_ito <- c("#E69F00", "#56B4E9", "#009E73") # Define the color palette vector

p2 <-ggplot(
  Out,
  aes(x = CATHPERIOD_G, y = YEARS, fill = CATHPERIOD_G)) +
  geom_boxplot(alpha = 0.7, outlier.shape = NA, staplewidth = 0.2) +
  geom_jitter(width = 0.2, alpha = 0.3) + # Adds raw data points so you can see sample size
  # scale_fill_manual(values = okabe_ito) + # modern palette colorblind-friendly
  # scale_fill_brewer(palette = "Set2") + Classic palette color
  scale_fill_viridis(option = "mako", begin = 0.3, end = 0.8) + # palette like the high-end scientific
  # scale_fill_nejm() +
  # scale_fill_jama() +
  scale_y_continuous(
    limits = c(2010, 2023), # Set the min and max limits of the axis
    breaks = seq(2010, 2023, by = 2), # Create tick marks every 2 years
    expand = expansion(mult = c(0.05, 0.05)) # Adds a clean 5% padding at top and bottom
  ) +

```

```

labs(
  #title = "Year of Surgery Distribution across Catheter Removal Categories",
  x = "Catheter Removal Category",
  y = "Year of Surgery"
) +
theme_bw() + # Apply the base theme first...
theme(
  legend.position = "none",      # Removes the legend
  panel.grid = element_blank()  # Removes all major and minor gridlines
)

ggsave(filename = "figures/catheterization_enrollment.png",
  plot = p2,
  width = 8,    # width in inches
  height = 6,  # height in inches
  dpi = 300)   # resolution

# If you want to use the exact color schemes used by major medical journals, you can install the ggsci package
# scale_fill_nejm(): The New England Journal of Medicine palette (clean, professional, muted jewel tones)
# scale_fill_jama(): The Journal of the American Medical Association palette (highly modern and bold).

# Plot catheterization per enrollment year and treatment: Faceted Plots (recommended)
Out$Treatment <- factor(
  Out$TRT2,
  levels = c(0,1),
  labels = c("Cohen", "Lich-Gregoir")
)

p3 <- ggplot(
  Out,
  aes(x = CATHPERIOD_G, y = YEARS, fill = CATHPERIOD_G)
) +
  geom_boxplot(alpha = 0.7, outlier.shape = NA, staplewidth = 0.2) +
  geom_jitter(width = 0.1, alpha = 0.4) +
  scale_fill_brewer(palette = "Set2") +
  facet_wrap(~ Treatment) + # Splits the plot into two columns based on treatment
  labs(
    title = "Year of Surgery Distribution across Catheter Removal Categories",
    subtitle = "Split by Treatment Technique",
    x = "Catheter Removal Category",
    y = "Year of Surgery"
  ) +
  theme_bw() +
  theme(
    legend.position = "none",      # No legend needed as x-axis labels are clear
    panel.grid = element_blank(),
    strip.background = element_rect(fill = "gray95"), # Colors the "Cohen/Lich" header boxes
    strip.text = element_text(face = "bold", size = 11)
  )

ggsave(filename = "figures/catheterization_enrollment_trt.png",
  plot = p3, width = 10, height = 6, dpi = 300) # Slightly wider to accommodate two panels

```

```

# Plot catheterization per enrollment year and treatment: Side-by-Side Box Plots (Grouped)
Out$Treatment <- factor(
  Out$TRT2,
  levels = c(0,1),
  labels = c("Cohen", "Lich-Gregoir")
)

p4 <- ggplot(
  Out,
  aes(x = CATHPERIOD_G, y = YEARS, fill = Treatment) # Grouping by treatment technique
) +
  geom_boxplot(alpha = 0.7, outlier.shape = NA, position = position_dodge(0.8), staplewidth = 0.2) +
  geom_jitter(
    position = position_jitterdodge(jitter.width = 0.1, dodge.width = 0.8),
    alpha = 0.4
  ) +
  scale_fill_brewer(palette = "Set1") + # Different color palette to distinguish treatment groups
  labs(
    title = "Year of Surgery Distribution by Catheter Removal & Treatment Group",
    x = "Catheter Removal Category",
    y = "Year of Surgery",
    fill = "Treatment Technique" # Title for the legend
  ) +
  theme_bw() +
  theme(
    legend.position = "bottom", # Keeps the legend but places it neatly at the bottom
    panel.grid = element_blank()
  )

ggsave(filename = "figures/catheterization_enrollment_trt_2.png",
  plot = p4, width = 9, height = 6, dpi = 300)

# Outcomes
table3 <- Out %>%
  dplyr::select(TRT2, FUPM, LUTD, OBST, URCOMPINJ, SURG, ILEUS, CLAVIENCLASS, URCOMPVUR,
    GU30, GU90, GU90N, GUER30, GUER90, PROC365,
    REC011, REC011_best, REC011_worst, RECGRADE_MAX,
    BLDLOSS, BLDSPASM, HEMAT,
    POSTUTI) %>%
  tbl_summary(
    by = TRT2, # for stratification
    statistic = list(
      all_continuous() ~ "{median} ({p25}--{p75})", # or ~ "{mean} ± {sd}",
      all_categorical() ~ "{n} ({p}%)",
    ),
    digits = all_categorical() ~ c(0, 1), # n = 0 decimals, p = 1 decimal
    missing = "ifany",
    missing_text = "Missing"
  ) %>%
  add_overall() %>%
  add_p(pvalue_fun = ~style_pvalue(.x, digits = 3)) %>% # 3 decimals

```

```

bold_labels()

table3

table3 %>%
  as_flex_table() %>%
  flextable::save_as_docx(path = "output/outcomes.docx")

## Treatment distribution change over time using a chi-square test

tab_year <- table(Out$YEARS, Out$Treatment)

chisq.test(tab_year)

      Pearson's Chi-squared test

data:  tab_year
X-squared = 44.138, df = 12, p-value = 1.447e-05

p_value_year <- chisq.test(tab_year)$p.value

p_label <- ifelse(
  p_value_year < 0.001,
  "Chi-square p < 0.001",
  paste0(
    "Chi-square p = ",
    format.pval(p_value_year, digits = 3)
  )
)

p_label

[1] "Chi-square p < 0.001"

# Proportion plot for treatment across

plot_data1 <- Out %>%
  count(YEARS, Treatment) %>%
  group_by(Treatment) %>%
  mutate(
    Percent = 100 * n / sum(n)
  ) %>%
  ungroup()

pt <- ggplot(
  plot_data1,
  aes(
    x = Percent,
    y = factor(YEARS),

```

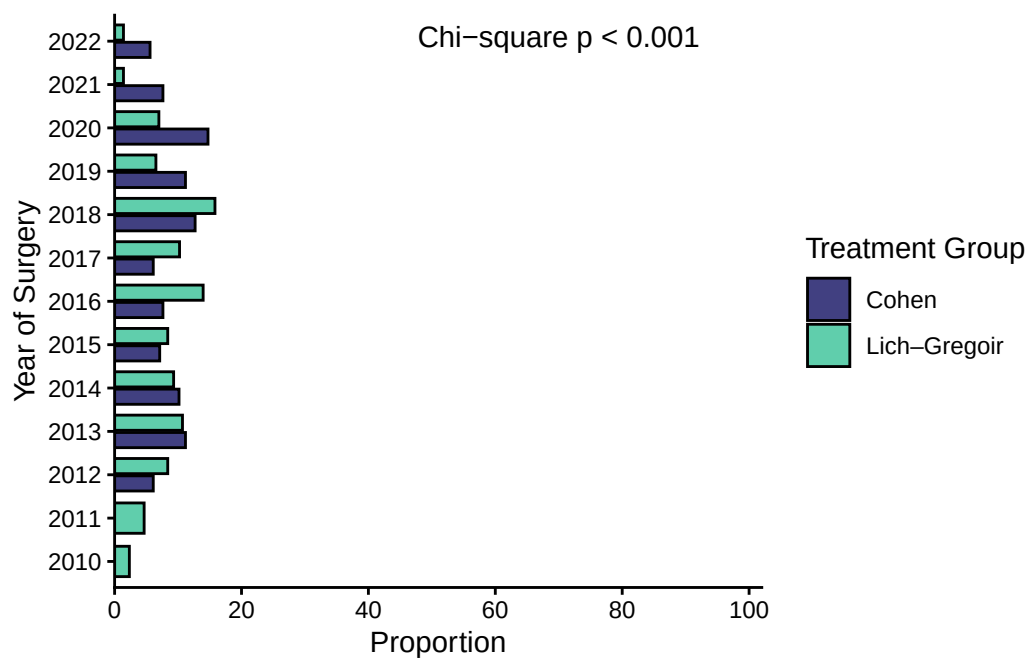

Characteristic	Overall N = 412 ¹	0 N = 197 ¹	1 N = 215 ¹	p-value ²
FUPM	28 (24-40)	30 (24-48)	26 (24-39)	0.140
LUTD				0.048
0	335 (81.3%)	168 (85.3%)	167 (77.7%)	
1	77 (18.7%)	29 (14.7%)	48 (22.3%)	
OBST				0.002
0	371 (90.0%)	187 (94.9%)	184 (85.6%)	
1	41 (10.0%)	10 (5.1%)	31 (14.4%)	
URCOMPINJ				0.714
0	405 (98.3%)	193 (98.0%)	212 (98.6%)	
1	7 (1.7%)	4 (2.0%)	3 (1.4%)	
SURG				0.175
0	314 (76.2%)	156 (79.2%)	158 (73.5%)	
1	98 (23.8%)	41 (20.8%)	57 (26.5%)	
ILEUS				0.218
0	385 (93.4%)	181 (91.9%)	204 (94.9%)	
1	27 (6.6%)	16 (8.1%)	11 (5.1%)	
CLAVIENCLASS				<0.001
1	201 (62.6%)	102 (76.1%)	99 (52.9%)	
2	83 (25.9%)	19 (14.2%)	64 (34.2%)	
3	36 (11.2%)	13 (9.7%)	23 (12.3%)	
4	1 (0.3%)	0 (0.0%)	1 (0.5%)	
Missing	91	63	28	
URCOMPVCUR				0.012
0	367 (97.6%)	159 (95.2%)	208 (99.5%)	
1	9 (2.4%)	8 (4.8%)	1 (0.5%)	
Missing	36	30	6	
GU30				<0.001
0	349 (84.7%)	144 (73.1%)	205 (95.3%)	
1	63 (15.3%)	53 (26.9%)	10 (4.7%)	
GU90				<0.001
0	342 (83.0%)	146 (74.1%)	196 (91.2%)	
1	70 (17.0%)	51 (25.9%)	19 (8.8%)	
GU90N				<0.001
0	261 (63.3%)	87 (44.2%)	174 (80.9%)	
1	70 (17.0%)	51 (25.9%)	19 (8.8%)	
2	81 (19.7%)	59 (29.9%)	22 (10.2%)	
GUER30				<0.001
0	316 (76.7%)	116 (58.9%)	200 (93.0%)	
1	96 (23.3%)	81 (41.1%)	15 (7.0%)	
GUER90				<0.001
0	305 (74.0%)	112 (56.9%)	193 (89.8%)	
1	107 (26.0%)	85 (43.1%)	22 (10.2%)	
PROC365				0.658
0	399 (96.8%)	190 (96.4%)	209 (97.2%)	
1	13 (3.2%)	7 (3.6%)	6 (2.8%)	
REC011				0.090
0	249 (91.2%)	91 (87.5%)	158 (93.5%)	
1	24 (8.8%)	13 (12.5%)	11 (6.5%)	
Missing	139	93	46	
REC011_best				0.521

```

    fill = Treatment
  )
) +
geom_col(
  position = position_dodge(width = 0.8),
  width = 0.7,
  colour = "black"
) +
scale_x_continuous(
  limits = c(0, 100),
  breaks = seq(0, 100, 20),
  expand = expansion(mult = c(0, 0.02))
) +
scale_fill_viridis_d(option = "mako", begin = 0.3, end = 0.8) +
labs(
  x = "Proportion",
  y = "Year of Surgery",
  fill = "Treatment Group"
) +
annotate(
  "text",
  x = 70,
  y = Inf,
  label = p_label,
  vjust = 1.5,
  size = 4
) +
theme_classic()

```

pt



```

ggsave(filename = "figures/treatment.png",
        plot = pt,
        width = 8, # width in inches
        height = 10, # height in inches
        dpi = 300) # resolution

##### Postoperative UTI outcome as time-to-event outcome

# Create labeled TRT2 variable only for survival analysis
Out$Treatment <- factor(
  Out$TRT2,
  levels = c(0,1),
  labels = c("Cohen", "Lich-Gregoir")
)

surv_object <- Surv(time = Out$POSTUTIFUP, event = Out$POSTUTI== 1)
fit <- survfit(surv_object ~ Treatment, data = Out)

p_u <- ggsurvplot(
  fit,
  data = Out,
  fun="event", # Show cumulative incidence (1 - survival)
  pval = FALSE,
  pval.size = 4,
  pval.coord = c(0, 0.9), # c(x,y) according to my x and y scale
  conf.int = TRUE,
  risk.table = TRUE,
  risk.table.y.text = FALSE, # to remove text from y-axis in the risk table
  # risk.table.title = "At risk",
  risk.table.fontsize = 4, # controls numbers in table
  xlim = c(0, 135),
  break.time.by = 12,
  xlab = "Follow-up time",
  ylab = "Cumulative incidence of postopetative symptomatic UTI",
  palette = c("#ADD8E6", "#00008B"),
  legend.title = "Treatment",
  legend.labs = c("Cohen Group", "Lich-Gregoir Group"),
  ylim = c(0, 1) # Force y-axis from 0 to 1
)

# calculate P-value from ggsurvplot (pval = TRUE), then turn it off

p_u$plot <- p_u$plot +
  annotate(
    "text",
    x = 20,
    y = 0.90,
    label = "Log rank test p = 0.058",
    size = 4
  )

#-----

```

```

# Customize the plot
#-----
p_u$plot <- p_u$plot +
  scale_x_continuous(
    limits = c(0, 135),
    breaks = seq(0, 135, 12)
  ) +
  scale_y_continuous(
    labels = function(x) paste0(round(x * 100, 1), "%"),
    limits = c(0, 1)
  ) +
  theme_classic() + # Many clinical epidemiology journals prefer theme_classic() instead of theme_bw()
  theme(
    axis.title.x = element_text(size = 12),
    axis.title.y = element_text(size = 12),
    axis.text.x = element_text(size = 10),
    axis.text.y = element_text(size = 10),
    legend.title = element_text(size = 11),
    legend.text = element_text(size = 10)
  )

```

Scale for x is already present.

Adding another scale for x, which will replace the existing scale.

Scale for y is already present.

Adding another scale for y, which will replace the existing scale.

```

#-----
# Customize the risk table
#-----
p_u$table <- p_u$table +
  scale_x_continuous(
    limits = c(0, 135),
    breaks = seq(0, 135, 12)
  ) +
  theme(
    axis.title.x = element_blank(),
    axis.text.x = element_blank(),
    axis.ticks.x = element_blank(),
    axis.line.x = element_blank(),
    axis.line.y = element_blank(),
    plot.title = element_text(size = 12),
    text = element_text(size = 10),
    axis.title.y = element_text(size = 12)
  )

```

Scale for x is already present.

Adding another scale for x, which will replace the existing scale.

```

#-----
# Combine plot and risk table using patchwork
#-----
final_plot <- p_u$plot / p_u$table +

```

```

plot_layout(heights = c(4, 1))

#-----
# Save the figure
#-----
ggsave(
  filename = "figures/UTI.png",
  plot = final_plot,
  width = 8,
  height = 6,
  dpi = 300
)

```

```

Ignoring unknown labels:
* colour : "Treatment"

```

6 Do weighting: using ATO (Recommended)

```

# In R formula syntax, use *, not just : if you want both main effects and interaction.
# Ex: PREOPVURGRADE + VURPHASE + PREOPVURGRADE:VURPHASE = PREOPVURGRADE * VURPHASE

```

```

set.seed(123)

w.out <- weightit(
  TRT2 ~ AGS + YEARS + GENDER + BMI + PREOPRNPLN + HISTOINJ +
    SURGIND2 + PREOPVURGRADE + VURSIDE + UDR7M + VURPHASE +
    PREOPEPISODE1 + PREOPTYPEU1 + PREOPRNSCR + SURGTECH2,
  data = Out,
  method = "ps",
  estimand = "ATO", # ATE = Average Treatment Effect = this is standard inverse probability of treatment
  stabilize = FALSE # To weight based on stabilized IPTW to reduce variability in weights (reduce ext.
)
Out$weights <- w.out$weights # Add weights to the dataset
write.csv(Out, "data/weighted_data.csv", row.names = FALSE)

summary(w.out$weights)

```

```

      Min.   1st Qu.   Median     Mean   3rd Qu.    Max.
0.0007522 0.0688498 0.2352262 0.2909478 0.4649644 0.9527184

```

```

# Table for balance
# This gives: SMD before and after weighting, and flagged imbalance (>0.1)
bal <- cobalt::bal.tab(w.out, un = TRUE, ESS = TRUE, m.threshold = 0.1) # ESS to calculate ESS separately
# un=TRUE shows imbalance before AND after weighting, un=FALSE shows only after weighting
bal

```

Balance Measures

```

      Type Diff.Un Diff.Adj   M.Threshold
prop.score Distance  1.8173  -0.0102 Balanced, <0.1

```

AGS	Contin.	0.2801	-0.0000	Balanced, <0.1
YEARS	Contin.	-0.4813	0.0000	Balanced, <0.1
GENDER	Binary	-0.0471	-0.0000	Balanced, <0.1
BMI	Contin.	-0.6526	-0.0000	Balanced, <0.1
PREOPRNPLN	Binary	-0.2699	0.0000	Balanced, <0.1
HISTOINJ	Binary	0.0198	-0.0000	Balanced, <0.1
SURGIND2_1	Binary	0.0856	0.0000	Balanced, <0.1
SURGIND2_2	Binary	0.0595	-0.0000	Balanced, <0.1
SURGIND2_3	Binary	-0.0089	-0.0000	Balanced, <0.1
SURGIND2_4	Binary	-0.0465	-0.0000	Balanced, <0.1
SURGIND2_5	Binary	-0.0897	-0.0000	Balanced, <0.1
PREOPVURGRADE_3	Binary	-0.0140	-0.0000	Balanced, <0.1
PREOPVURGRADE_4	Binary	0.0802	-0.0000	Balanced, <0.1
PREOPVURGRADE_5	Binary	-0.0662	0.0000	Balanced, <0.1
VURSIDE	Binary	0.0247	-0.0000	Balanced, <0.1
UDR7M	Binary	-0.1399	0.0000	Balanced, <0.1
VURPHASE_1	Binary	-0.0383	-0.0000	Balanced, <0.1
VURPHASE_2	Binary	0.3112	0.0000	Balanced, <0.1
VURPHASE_3	Binary	-0.2729	-0.0000	Balanced, <0.1
PREOPEPISODE1_1	Binary	-0.0440	-0.0000	Balanced, <0.1
PREOPEPISODE1_2	Binary	-0.1765	-0.0000	Balanced, <0.1
PREOPEPISODE1_3	Binary	0.2205	0.0000	Balanced, <0.1
PREOPTYPEU1	Binary	0.0309	0.0000	Balanced, <0.1
PREOPRNSCR	Binary	-0.2674	-0.0000	Balanced, <0.1
SURGTECH2_1	Binary	0.2494	0.0000	Balanced, <0.1
SURGTECH2_2	Binary	-0.0629	-0.0000	Balanced, <0.1
SURGTECH2_3	Binary	-0.1866	-0.0000	Balanced, <0.1

Balance tally for mean differences

	count
Balanced, <0.1	28
Not Balanced, >0.1	0

Variable with the greatest mean difference

Variable	Diff.Adj	M.Threshold
BMI	-0	Balanced, <0.1

Effective sample sizes

	Control	Treated
Unadjusted	197.	215.
Adjusted	126.08	113.73

```
bal_df <- as.data.frame(bal$Balance)
```

```
smd_table <- bal_df %>%
```

```
  mutate(
```

```
    Variable = rownames(bal_df),
```

```
    SMD_Before = round(abs(Diff.Un), 3),
```

```
    SMD_After = round(abs(Diff.Adj), 3)
```

```
) %>% # you are taking the absolute value of the standardized mean differences before and after weight
```

```
dplyr::select(Variable, SMD_Before, SMD_After) # SMDs for each covariate individually, before and after
```

```
smd_table %>%
```

```
  flextable() %>%
```

```
  save_as_docx(path = "output/SMD_w.docx")
```

```

# Figure for balance - Love plot
# Review guidelines to decide on plotting Standardized or Absolute Mean Differences
pl <- love.plot(
  w.out,
  threshold = 0.1,
  abs = TRUE, # As SMD sign is meaningless for balance, show only absolute SMDs using abs = TRUE (the c
  stats = "mean.diffs", # forces SMD instead of raw differences
  var.order = "unadjusted", # Orders variables based on imbalance before matching
  drop.distance = TRUE, # To exclude the distance from the plot
  sample.names = c("Before Overlap Weighting", "After Overlap Weighting"),
  var.names = c(
    AGS = "Age at surgery",
    YEARS = "Year of surgery",
    BMI = "Body mass index",
    GENDER = "Female gender",
    PREOPRNPLN = "Preop pyelonephritis",
    HISTOINJ = "History of injection",
    VURSIDE = "Right-sided VUR",
    UDR7M = "Ureteral dilation 7mm",
    PREOPTYPEU1 = "Preop febrile UTIs",
    PREOPRNSCR = "Preop renal scarring",
    VURPHASE_1 = "Reflux: voiding phase",
    VURPHASE_2 = "Reflux: filling phase",
    VURPHASE_3 = "Reflux: filling + voiding",
    SURGTECH2_1 = "Surgeon experience 100 operations",
    SURGTECH2_2 = "Surgeon experience 50-100 operations",
    SURGTECH2_3 = "Surgeon experience <50 operations",
    PREOPEPISODE1_1 = "Preop UTI episodes: 1",
    PREOPEPISODE1_2 = "Preop UTI episodes: 2",
    PREOPEPISODE1_3 = "Preop UTI episodes: 3",
    SURGIND2_1 = "Indication: breakthrough UTI",
    SURGIND2_2 = "Indication: high-grade reflux (III-V)",
    SURGIND2_3 = "Indication: poor compliance to medical treatment",
    SURGIND2_4 = "Indication: ineffective/poorly tolerated CAP",
    SURGIND2_5 = "Indication: other",
    PREOPVURGRADE_3 = "Preop VUR grade: III",
    PREOPVURGRADE_4 = "Preop VUR grade: IV",
    PREOPVURGRADE_5 = "Preop VUR grade: V"
  )
)

```

Warning: Standardized mean differences and raw mean differences are present in the same plot. Use the `stars` argument to distinguish between them and appropriately label the x-axis. See `love.plot()` for details.

```

pl <- pl +
  labs(
    x = "Absolute Standardized Differences",
    subtitle = "Overlap population of all children"
  ) +
  theme(
    plot.margin = margin(10, 10, 10, 10),

```

```

axis.text = element_text(size = 16),
axis.title = element_text(size = 16),
plot.title = element_text(size = 16),
legend.text = element_text(size = 16),
legend.title = element_text(size = 16),
axis.ticks = element_line(size = 0.5, color = "black"),
axis.line = element_line(size = 0.5, color = "black"),
panel.background = element_blank(),
panel.grid.major = element_blank(),
panel.grid.minor = element_blank(),
panel.border = element_blank(),
axis.ticks.length = unit(0.3, "cm"),
panel.spacing = unit(2, "lines")
)

```

Warning: The `size` argument of `element\_line()` is deprecated as of ggplot2 3.4.0.
 i Please use the `linewidth` argument instead.

```
ggsave ("figures/love_plot_w.png", plot = pl, width = 12, height = 10, dpi = 300)
```

```

# Compute mean SMD
smd_df <- bal$Balance
mean_smd_before <- mean(abs(smd_df$Diff.Un), na.rm = TRUE) # calculate mean SMD before matching
mean_smd_after <- mean(abs(smd_df$Diff.Adj), na.rm = TRUE) # calculate mean SMD before matching
mean_smd <- data.frame(
  Mean_SMD_Before = mean_smd_before,
  Mean_SMD_After = mean_smd_after
)

```

```
mean_smd
```

```

  Mean_SMD_Before Mean_SMD_After
1      0.2158582    0.0003644521

```

```

# how much effective sample size remains (overall ESS: calculated manually)?
summary(w.out$weights)

```

```

      Min.   1st Qu.   Median     Mean   3rd Qu.     Max.
0.0007522 0.0688498 0.2352262 0.2909478 0.4649644 0.9527184

```

```
bal.tab(w.out)$Effective.Sample.Size
```

```
NULL
```

```
sum(w.out$weights)^2 / sum(w.out$weights^2)
```

```
[1] 239.181
```



```

# propensity score distribution
summary(w.out$ps)

      Min.   1st Qu.   Median     Mean   3rd Qu.     Max.
0.006799 0.218309 0.504244 0.521845 0.803450 0.999248

pl2 <-plot(summary(w.out, type = "hist"))
ggsave ("figures/propensity_w.png", plot = pl2, width = 12, height = 10, dpi = 300)

# Distribution of overlap weights: histogram and density

## Histogram
summary(w.out$weights)

      Min.   1st Qu.   Median     Mean   3rd Qu.     Max.
0.0007522 0.0688498 0.2352262 0.2909478 0.4649644 0.9527184

png("figures/weights_histo_w.png",
    width = 6,
    height = 4,
    units = "in",
    res = 300)

hist(
  w.out$weights,
  breaks = 30,
  col = "lightblue",
  border = "black",
  main = "Distribution of Overlap Weights",
  xlab = "Overlap Weight"
)
dev.off()

pdf
2

## Density
# owd<- ggplot(Out,
#   aes(weights,
#     colour = Treatment,
#     fill = Treatment)) +
#   geom_density(alpha = 0.3, linewidth = 1) +
#   labs(
#     x = "Overlap weight",
#     y = "Density"
#   ) +
#   theme_classic()
# ggsave ("figures/weights_dens_w.png", plot = owd, width = 12, height = 8, dpi = 300)

```

```

# Here, you cannot write:
# "We estimated the average treatment effect in all children."
# You should write: "We estimated the average treatment effect among children with overlapping treatment"
# "Among children with comparable probability of receiving either surgical technique, overlap weighting"

# How many patients are essentially excluded?
sum(w.out$weights < 0.05)

[1] 90

sum(w.out$weights < 0.01)

[1] 39

# If only a few patients have tiny weights → excellent.
# If 50-100 patients have tiny weights → your overlap population is narrower.

# Compare treatment group sizes after weighting
aggregate(weights ~ TRT2, data=Out, sum)

  TRT2  weights
1     0 59.93525
2     1 59.93525

# Check whether treatment effect changes
# The final question:
# Does the treatment effect estimate remain similar?
# For example:
# ATE IPTW: OR = 0.65 (95% CI 0.40-1.05)
# ATO: OR = 0.70 (95% CI 0.45-1.10)
# → robust.
# But:
# ATE: OR=0.30
# ATO: OR=0.90
# → treatment effect differs by population.

# Your ATO results are actually better behaved than your ATE IPTW results:
# better balance
# fewer extreme weights
# higher ESS
# I would not interpret ATO as "weighting a small population." I would interpret it as focusing the analysis on the population with comparable probability of receiving either surgical technique.

# Distributional balance for propensity score (purpose: to evaluate overlap/positivity):

# After weighting
# p_ps <- bal.plot(w.out, var.name = "prop.score", type = "density")

# p_ps <- p_ps +
#   scale_color_viridis_d(

```

```

#   option = "mako",
#   begin = 0.3,
#   end = 0.8,
#   labels = c("Cohen group", "Lich-Gregoir group")
# ) +
# scale_fill_viridis_d(
#   option = "mako",
#   begin = 0.3,
#   end = 0.8,
#   labels = c("Cohen group", "Lich-Gregoir group")
# )

```

```

# ggsave(
#   filename = "figures/PS_overlap_weighting.png",
#   plot = p_ps,
#   width = 12,
#   height = 6,
#   dpi = 300
# )

```

```

# Before-After weighting
p_ps <- bal.plot(
  w.out,
  var.name = "prop.score",
  type = "density",
  which = "both"
)

```

```

p_ps <- p_ps +
  scale_color_viridis_d(
    option = "mako",
    begin = 0.3,
    end = 0.8,
    labels = c("Cohen group", "Lich-Gregoir group")
  ) +
  scale_fill_viridis_d(
    option = "mako",
    begin = 0.3,
    end = 0.8,
    labels = c("Cohen group", "Lich-Gregoir group")
  )

```

Scale for fill is already present.

Adding another scale for fill, which will replace the existing scale.

```

ggsave(
  "figures/PS_density_before_after_ATO.png",
  p_ps,
  width = 10,
  height = 7,
  dpi = 300
)

```

```
# Identify extreme patients using common standard thresholds: PS<0.10 or PS>0.90 (Broad definition), PS
# Out[w.out$ps < 0.1 | w.out$ps > 0.9, ]

# The only thing I would still investigate is which clinical characteristics define the patients with PS
# "Who was excluded from the overlap population?"
# You can answer that by comparing those extreme-propensity patients with the overlap patients.
```

- CENTER will not included in the propensity score because there is one center that did not contribute cases in one treatment technique + two other centers did not show a great overlap: a good overlap is only exist in one center.
- What does the ESS of 239 mean clinically? It does not mean that only 239 children were analyzed. You still include all 412 children, but their contributions differ: A child with a high overlap weight contributes more information. A child with a near-zero overlap weight contributes very little. The analysis is effectively equivalent to having a randomized comparison in approximately 239 patients.
- Why did you lose about 42% of effective information? Because your original cohort likely contains patients where treatment choice was almost predetermined. For example: very severe VUR → almost always treated with one technique certain surgeon experience levels → almost always associated with one technique certain patient characteristics → strongly favor one technique Those patients cannot provide much information about comparative effectiveness, because there is little realistic alternative treatment. ATO appropriately reduces their influence.
- One thing I would definitely report Because ATO changes the target population, describe it explicitly: Instead of: “We evaluated all 412 children...” write something like: “The original cohort included 412 children. Treatment effects were estimated using overlap weighting, which emphasizes patients with comparable probabilities of receiving either surgical technique. The resulting effective sample size was 239 patients.”
- Your results suggest good overlap The combination of: Perfect balance No extreme weights Equal weighted treatment groups ESS 239 suggests that your data contain a meaningful overlap population. This is actually a very favorable situation for ATO.
- Because conventional inverse probability of treatment weighting resulted in residual covariate imbalance and extreme weights, overlap weighting was used as the primary approach. Overlap weighting estimates the average treatment effect among patients with clinical equipoise, where either treatment technique could plausibly have been selected.
- Comparing matching and weighting: Because ATO avoids discarding patients, it often provides: narrower confidence intervals better statistical power provided the propensity model is appropriate.
- ATO did not reduce your analysis to a tiny selected subgroup. It focused the causal comparison on the substantial proportion of children for whom both surgical techniques were clinically plausible.
- The core issue of estimand choice: does ATO answer the clinical question you actually want to answer, or are you using it only because it gives better balance? In my case: I can apply ATO, and my clinical question can actually provide a strong justification for it — but I need to frame the research question appropriately. I should not present ATO as merely a technical solution to improve SMDs.

What is your clinical question? You are not asking: “What is the effect of Technique A versus Technique B in all children with unilateral VUR?” You are asking something closer to: “Among children with unilateral VUR who are candidates for either surgical technique, is one technique associated with better outcomes?”

Therefore, the clinically relevant question becomes: “When surgeons have a genuine choice between techniques, which one performs better?” That is exactly what ATO estimates.

The key is that you must define this before analysis. Do not write: “We used overlap weighting because IPTW failed.” That makes it look like a data-driven rescue method. Instead write something like: “Because the objective was to compare surgical techniques among children for whom either approach could reasonably be selected, the target estimand was the average treatment effect in the overlap population (ATO). Overlap weighting was selected a priori to estimate this effect.” This is a much stronger justification.

How I would frame your objective Instead of: “To compare outcomes between two surgical techniques for unilateral VUR.” I would write: “To compare postoperative outcomes between two surgical techniques among children with unilateral VUR who were eligible for either approach.” This aligns naturally with ATO.

7 Descriptive analysis: AFTER weighting

```
# Out is the dataset with added weights
# Simply having a weights column in your dataset is not sufficient for tbl_svsummary(). The survey des.

# Create survey design object using overlap weights
design_ato <- svydesign(
  ids = ~1, # ids = CENTER means that patients are clustered within centers, use ids = ~1 if there is
# however, the survey package's cluster-robust variance estimation is not very reliable with very few c.
  data = Out,
  weights = ~weights
)

# Weighted baseline characteristics
table1w <- design_ato %>%
  tbl_svsummary(
    by = TRT2, # for stratification
    include = c(
      YEARS, AGS, GENDER, BMI, HISTOINJ, SURGIND2, PREOPVURGRADE, VURSIDE, UDR7M,
      VURPHASE, PREOPTYPEU1, PREOPEPISODE1, PREOPRNPLN, PREOPRNSCR,
      SURGTECH2, CENTER # generally we should not include TRT2 in include list
    ),
    statistic = list(
      all_continuous() ~ "{median} ({p25}-{p75})",
      all_categorical() ~ "{n} ({p}%"
    ),
    digits = all_categorical() ~ c(0, 1),
    missing = "ifany", # shows missing only when present
    missing_text = "Missing"
  ) %>%
  add_overall() %>%
  # add_p(pvalue_fun = ~style_pvalue(.x, digits = 3)) %>% # add_p() is not informative after weighting
  bold_labels()

table1w

table1w %>%
  as_flex_table() %>%
  flextable::save_as_docx(
    path = "output/baseline_w.docx"
  )
```

Characteristic	Overall N = 120 ^I	0 N = 60 ^I	1 N = 60
YEARS	2,017.00 (2,014.00-2,019.00)	2,017.00 (2,014.00-2,019.00)	2,017.00 (2,014.00-2,019.00)
AGS	8.00 (6.00-9.00)	8.00 (7.00-9.00)	7.00 (6.00-9.00)
GENDER			
0	65 (54.1%)	32 (54.1%)	32 (54.1%)
1	55 (45.9%)	28 (45.9%)	28 (45.9%)
BMI	14.1 (13.5-17.4)	14.0 (13.2-17.7)	14.1 (13.6-16.9)
HISTOINJ			
0	104 (87.1%)	52 (87.1%)	52 (87.1%)
1	16 (12.9%)	8 (12.9%)	8 (12.9%)
SURGIND2			
1	55 (46.2%)	28 (46.2%)	28 (46.2%)
2	48 (40.1%)	24 (40.1%)	24 (40.1%)
3	6 (5.2%)	3 (5.2%)	3 (5.2%)
4	2 (1.9%)	1 (1.9%)	1 (1.9%)
5	8 (6.6%)	4 (6.6%)	4 (6.6%)
PREOPVURGRADE			
3	7 (5.9%)	4 (5.9%)	4 (5.9%)
4	35 (29.0%)	17 (29.0%)	17 (29.0%)
5	78 (65.1%)	39 (65.1%)	39 (65.1%)
VURSIDE			
0	45 (37.4%)	22 (37.4%)	22 (37.4%)
1	75 (62.6%)	38 (62.6%)	38 (62.6%)
UDR7M			
0	28 (23.5%)	14 (23.5%)	14 (23.5%)
1	92 (76.5%)	46 (76.5%)	46 (76.5%)
VURPHASE			
1	87 (72.9%)	44 (72.9%)	44 (72.9%)
2	2 (1.5%)	1 (1.5%)	1 (1.5%)
3	31 (25.5%)	15 (25.5%)	15 (25.5%)
PREOPTYPEU1			
0	1 (1.2%)	1 (1.2%)	1 (1.2%)
1	118 (98.8%)	59 (98.8%)	59 (98.8%)
PREOPEPISODE1			
1	7 (5.7%)	3 (5.7%)	3 (5.7%)
2	35 (29.5%)	18 (29.5%)	18 (29.5%)
3	78 (64.9%)	39 (64.9%)	39 (64.9%)
PREOPRNPLN			
0	101 (84.2%)	50 (84.2%)	50 (84.2%)
1	19 (15.8%)	9 (15.8%)	9 (15.8%)
PREOPRNSCR			
0	84 (69.8%)	42 (69.8%)	42 (69.8%)
1	36 (30.2%)	18 (30.2%)	18 (30.2%)
SURGTECH2			
1	7 (6.2%)	4 (6.2%)	4 (6.2%)
2	99 (82.3%)	49 (82.3%)	49 (82.3%)
3	14 (11.5%)	7 (11.5%)	7 (11.5%)
CENTER			
1	87 (72.9%) ³⁰	42 (69.4%)	46 (76.4%)
2	1 (0.9%)	1 (1.8%)	0 (0.0%)
3	9 (7.1%)	5 (8.5%)	3 (5.8%)

Characteristic	Overall N = 120 ¹	0 N = 60 ¹	1 N = 60 ¹	p-value ²
OPRTIME	85 (70-130)	130 (95-160)	70 (68-80)	<0.001
STENT				0.075
0	78 (64.7%)	35 (59.1%)	42 (70.4%)	
1	42 (35.3%)	25 (40.9%)	18 (29.6%)	
CATH				0.555
0	2 (2.1%)	1 (1.5%)	2 (2.6%)	
1	117 (97.9%)	59 (98.5%)	58 (97.4%)	
CATHPERIOD	4 (3-5)	4 (4-5)	3 (1-5)	<0.001
CATHPERIOD_G				<0.001
Early removal 3 days	38 (31.8%)	5 (8.1%)	33 (55.6%)	
Standard removal 4-7 days	58 (48.5%)	43 (71.6%)	15 (25.4%)	
Delayed removal >7 days	24 (19.7%)	12 (20.3%)	11 (19.1%)	
HOSPTIME	5.00 (3.00-6.00)	5.00 (5.00-6.00)	3.00 (2.00-5.00)	<0.001

¹Median (Q1-Q3); n (%)

²Design-based KruskalWallis test; Pearson's X²: Rao & Scott adjustment

```
# Weighted procedural factors
table2w<- design_ato %>%
  tbl_svsummary(
    by = TRT2, # for stratification
    include = c (
      OPRTIME, STENT, CATH, CATHPERIOD, CATHPERIOD_G, HOSPTIME
    ),
    statistic = list(
      all_continuous() ~ "{median} ({p25}-{p75})", # or ~ "{mean} ± {sd}",
      all_categorical() ~ "{n} ({p}%)"
    ),
    digits = all_categorical() ~ c(0, 1), # n = 0 decimals, p = 1 decimal
    missing = "ifany",
    missing_text = "Missing"
  ) %>%
  add_overall() %>%
  add_p(pvalue_fun = ~style_pvalue(.x, digits = 3)) %>% # 3 decimals
  bold_labels()
```

The following warnings were returned during `add\_p()`:

```
! For variable `CATH` (`TRT2`) and "statistic" and "p.value" statistics:
  Chi-squared approximation may be incorrect
```

table2w

```
table2w %>%
  as_flex_table() %>%
  flextable::save_as_docx(path = "output/procedural_w.docx")
```

```

# Plot catheterization distribution: better visualization for weighted causal analysis: weighted density
p_w <- ggplot(
  Out,
  aes(
    x = CATHPERIOD,
    weight = weights,
    fill = factor(
      TRT2,
      levels = c(0, 1),
      labels = c("Cohen Group", "Lich-Gregoir Group")
    )
  )
) +
scale_fill_viridis_d(option = "mako", begin = 0.3, end = 0.8) +
geom_density(alpha = 0.4) +
labs(
  x = "Catheterization Duration (days)",
  y = "Weighted density",
  fill = "Treatment Group"
) +
theme_bw() + #optional
theme(
  panel.grid = element_blank() # Removes all major and minor gridlines
)

ggsave(filename = "figures/catheterization_w.png",
  plot = p_w,
  width = 8, # width in inches
  height = 6, # height in inches
  dpi = 300) # resolution

# Outcomes
table3w<- design_ato %>%
tbl_svsummary(
  by = TRT2, # for stratification
  include = c (
    FUPM, LUTD, OBST, URCOMPINJ, SURG, ILEUS, CLAVIENCLASS, URCOMPCVUR,
    GU30, GU90, GU90N, GUER30, GUER90, PROC365,
    REC011, REC011_best, REC011_worst, RECGRADE_MAX, BLDLOSS, BLDSPASM, HEMAT, POSTUTI
  ),
  statistic = list(
    all_continuous() ~ "{median} ({p25}-{p75})", # or ~ "{mean} ± {sd}",
    all_categorical() ~ "{n} ({p}%"
  ),
  digits = all_categorical() ~ c(0, 1), # n = 0 decimals, p = 1 decimal
  missing = "ifany",
  missing_text = "Missing"
) %>%
add_overall() %>%
add_p(pvalue_fun = ~style_pvalue(.x, digits = 3)) %>% # 3 decimals

```



```

bold_labels()

The following warnings were returned during `add_p()``
! For variable `CLAVIENCLASS` (`TRT2`) and "statistic" and "p.value"
  statistics: Chi-squared approximation may be incorrect
! For variable `RECGRADE_MAX` (`TRT2`) and "statistic" and "p.value"
  statistics: Chi-squared approximation may be incorrect
! For variable `URCOMPINJ` (`TRT2`) and "statistic" and "p.value" statistics:
  Chi-squared approximation may be incorrect

table3w

table3w %>%
  as_flex_table() %>%
  flextable::save_as_docx(path = "output/outcomes_w.docx")

##### Postoperative UTI outcome as time-to-event outcome: Weighted survival curve

fit_w <- survfit(
  Surv(POSTUTIFUP, POSTUTI == 1) ~ Treatment,
  data = Out,
  weights = w.out$weights
)

p_u_w <- ggsurvplot(
  fit_w,
  data = Out,
  fun = "event",
  pval = FALSE,
  pval.size = 4,
  pval.coord = c(0, 0.9), # c(x,y) according to my x and y scale
  conf.int = TRUE,
  risk.table = TRUE,
  risk.table.y.text = FALSE, # to remove text from y-axis in the risk table
  risk.table.fontsize = 4, # controls numbers in table
  xlim = c(0,135),
  break.time.by = 12,
  xlab = "Follow-up time",
  ylab = "Cumulative incidence of postoperative symptomatic UTI",
  palette = c("#ADD8E6", "#00008B"),
  legend.title = "Treatment",
  legend.labs = c("Cohen Group", "Lich-Gregoir Group"),
  ylim = c(0,1)
)

# ggsurvplot(pval = TRUE) calculates the p-value using a log-rank test based on the data supplied, and
# So the curve may be weighted, but the p-value may still come from an unweighted comparison.
# So, it is better to use a weighted Cox model then add the p-value from your weighted Cox model manually.
cox_w <- svycoxph(
  Surv(POSTUTIFUP, POSTUTI == 1) ~ Treatment,

```

Characteristic	Overall N = 120 ¹	0 N = 60 ¹	1 N = 60 ¹	p-value ²
FUPM	26 (24-38)	30 (24-39)	24 (13-36)	0.007
LUTD				0.696
0	99 (82.7%)	50 (83.6%)	49 (81.8%)	
1	21 (17.3%)	10 (16.4%)	11 (18.2%)	
OBST				0.028
0	108 (90.2%)	57 (94.3%)	52 (86.0%)	
1	12 (9.8%)	3 (5.7%)	8 (14.0%)	
URCOMPINJ				0.165
0	117 (97.8%)	58 (96.5%)	59 (99.1%)	
1	3 (2.2%)	2 (3.5%)	1 (0.9%)	
SURG				0.469
0	86 (71.5%)	42 (69.3%)	44 (73.7%)	
1	34 (28.5%)	18 (30.7%)	16 (26.3%)	
ILEUS				0.004
0	111 (92.7%)	53 (88.1%)	58 (97.3%)	
1	9 (7.3%)	7 (11.9%)	2 (2.7%)	
CLAVIENCLASS				0.002
1	57 (60.7%)	35 (72.6%)	22 (48.2%)	
2	25 (26.7%)	7 (15.5%)	18 (38.5%)	
3	11 (12.1%)	6 (11.9%)	6 (12.2%)	
4	1 (0.5%)	0 (0.0%)	1 (1.1%)	
Missing	26	12	14	
URCOMPVCUR				<0.001
0	108 (97.3%)	52 (94.7%)	56 (99.8%)	
1	3 (2.7%)	3 (5.3%)	0 (0.2%)	
Missing	9	5	3	
GU30				<0.001
0	97 (81.1%)	40 (66.6%)	57 (95.6%)	
1	23 (18.9%)	20 (33.4%)	3 (4.4%)	
GU90				<0.001
0	92 (77.0%)	40 (66.3%)	53 (87.7%)	
1	28 (23.0%)	20 (33.7%)	7 (12.3%)	
GU90N				<0.001
0	69 (58.0%)	28 (46.1%)	42 (69.9%)	
1	28 (23.0%)	20 (33.7%)	7 (12.3%)	
2	23 (19.1%)	12 (20.3%)	11 (17.8%)	
GUER30				<0.001
0	84 (70.3%)	29 (48.0%)	56 (92.6%)	
1	36 (29.7%)	31 (52.0%)	4 (7.4%)	
GUER90				<0.001
0	79 (66.3%)	27 (45.6%)	52 (86.9%)	
1	40 (33.7%)	33 (54.4%)	8 (13.1%)	
PROC365				0.630
0	114 (95.3%)	57 (94.6%)	58 (96.0%)	
1	6 (4.7%)	3 (5.4%)	2 (4.0%)	
REC011				0.303
0	73 (89.6%)	36 (87.2%)	37 (92.2%)	
1	8 (10.4%)	5 (12.8%)	3 (7.8%)	
Missing	39	19	20	
REC011_best				0.271

```
design = design_ato
)
```

```
summary(cox_w)
```

```
Independent Sampling design (with replacement)
svydesign(ids = ~1, data = Out, weights = ~weights)
```

Call:

```
svycoxph(formula = Surv(POSTUTIFUP, POSTUTI == 1) ~ Treatment,
  design = design_ato)
```

```
n= 412, number of events= 112
```

	coef	exp(coef)	se(coef)	robust se	z	Pr(> z)
TreatmentLich-Gregoir	-0.2859	0.7513	0.1903	0.2483	-1.151	0.25

	exp(coef)	exp(-coef)	lower .95	upper .95
TreatmentLich-Gregoir	0.7513	1.331	0.4618	1.222

```
Concordance= 0.54 (se = 0.035 )
```

```
Likelihood ratio test= NA on 1 df, p=NA
```

```
Wald test = 1.33 on 1 df, p=0.2
```

```
Score (logrank) test = NA on 1 df, p=NA
```

(Note: the likelihood ratio and score tests assume independence of observations within a cluster, the Wald and robust score tests do not).

```
p_u_w$plot <- p_u_w$plot +
  annotate(
    "text",
    x = 20,
    y = 0.90,
    label = "Weighted Cox p = 0.250",
    size = 4
  )
```

```
#-----
# Customize the plot
#-----
```

```
p_u_w$plot <- p_u_w$plot +
  scale_x_continuous(
    limits = c(0, 135),
    breaks = seq(0, 135, 12)
  ) +
  scale_y_continuous(
    labels = function(x) paste0(round(x * 100, 1), "%"),
    limits = c(0, 1)
  ) +
  theme_classic() + # Many clinical epidemiology journals prefer theme_classic() instead of theme_bw()
  theme(
    axis.title.x = element_text(size = 12),
    axis.title.y = element_text(size = 12),
```

```

axis.text.x = element_text(size = 10),
axis.text.y = element_text(size = 10),
legend.title = element_text(size = 11),
legend.text = element_text(size = 10)
)

```

Scale for x is already present.

Adding another scale for x, which will replace the existing scale.

Scale for y is already present.

Adding another scale for y, which will replace the existing scale.

```

#-----
# Customize the risk table
#-----
p_u_w$table <- p_u_w$table +
  scale_x_continuous(
    limits = c(0, 135),
    breaks = seq(0, 135, 12)
  ) +
  theme(
    axis.title.x = element_blank(),
    axis.text.x = element_blank(),
    axis.ticks.x = element_blank(),
    axis.line.x = element_blank(),
    axis.line.y = element_blank(),
    plot.title = element_text(size = 12),
    text = element_text(size = 10),
    axis.title.y = element_text(size = 12)
  )

```

Scale for x is already present.

Adding another scale for x, which will replace the existing scale.

```

#-----
# Combine plot and risk table
#-----
final_plot_w <- p_u_w$plot / p_u_w$table +
  plot_layout(heights = c(4, 1))

#-----
# Save the figure
#-----
ggsave(
  filename = "figures/UTI_w.png",
  plot = final_plot_w,
  width = 8,
  height = 6,
  dpi = 300
)

```

Ignoring unknown labels:

```
* colour : "Treatment"
```

```
# Figure: Weighted Kaplan-Meier cumulative incidence curves after overlap weighting.
# Table: Weighted Cox proportional hazards model estimating the association between surgical technique and
```

Pearson's χ^2 test with Rao-Scott adjustment is a version of the usual chi-square test that is modified for weighted survey data (or other analyses involving complex weighting). In your case, it appears because you are using `tbl_svysummary()` with an object created using `svydesign()` and your ATO weights.

Although follow-up duration differed between groups, it was not included as a covariate because it is a post-treatment variable and may introduce bias if adjusted for. Outcomes defined within fixed follow-up windows were analyzed according to their prespecified time periods, while time-to-event outcomes incorporated varying follow-up through survival analysis.

8 Regression models

Regression blocks:

- Block 1: technique effect on postoperative outcomes (complications, readmissions, reoperation, VUR and UTI recurrence)
- Block 2: exposure-effect analysis (stent, operative time, catheterization) on postoperative outcomes
- Block 3: Prognostic factors for readmission and reoperation: Complications and VUR/UTI recurrence effects on readmissions and reoperation

Discarded: CLAVIENCLASS URCOMPCVUR GU90N GUER30 GUER90 RECGRADE_MAX BLDLOSS BLDSPASM HEMAT

5 Complications: LUTD OBST URCOMPINJ SURG ILEUS

3 Readmissions and reoperations: GU30 GU90 PROC365

2 Success (VUR and UTI recurrence): REC011 (Binary outcome)
POSTUTI (Time-to-event outcome)

After ATO weighting, your model is not really a conventional “univariable analysis.” It is better described as: “Weighted marginal treatment effect estimation using GEE.” Because the confounding control occurred during weighting, not regression adjustment.

Multiple comparison correction is not dependent on the type of regression model. It is applied to the set of p-values that you consider part of the same family of hypotheses.

- Why AUC is not commonly reported for causal models AUC is mainly used for:
 - clinical prediction models
- risk scores
- diagnostic models

Examples: “Can we predict which child will develop postoperative obstruction?” For those models, discrimination is the goal.

Your question is: “Does Cohen vs Lich-Gregoir change the risk of postoperative obstruction/pain?” Here, prediction performance is secondary.

- For a randomized trial or comparative effectiveness observational study, reviewers usually expect:
 - balance diagnostics,
 - confounder selection rationale,
 - sensitivity analyses, not AUC.

Reporting AUC may confuse the interpretation: A reviewer may ask: “Are you developing a prediction model or estimating a causal effect?”

Block 1

```
# GEE model after overlap weighting - Accounting for center-clustering

# Block 1: technique effect on postoperative outcomes = Univariable analyses

# Question: What is the difference in outcome risk between techniques among patients who could have rec

outcomes <- c(
  "LUTD",
  "OBST",
  "URCOMPINJ",
  "SURG",
  "ILEUS",
  "GU30",
  "GU90",
  "PROC365",
  "REC011",
  "REC011_best",
  "REC011_worst"
)

# convert outcomes to numeric
Out[outcomes] <- lapply(
  Out[outcomes],
  function(x) as.numeric(as.character(x))
)

# Apply GEE model after overlap weighting for all binary outcomes
gee_results <- lapply(outcomes, function(y) {
  formula <- as.formula(
    paste(y, "~ TRT2")
  )
  model <- geeglm(
    formula,
    data = Out,
    id = CENTER,
    weights = weights,
    family = binomial(link = "logit"),
    corstr = "exchangeable"
  )
  tidy(model, exponentiate = TRUE, conf.int = TRUE) # estimate = OR because exponentiate = TRUE
})

Warning in eval(family$initialize): non-integer #successes in a binomial glm!
Warning in eval(family$initialize): non-integer #successes in a binomial glm!
Warning in eval(family$initialize): non-integer #successes in a binomial glm!
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Warning in eval(family$initialize): non-integer #successes in a binomial glm!
Warning in eval(family$initialize): non-integer #successes in a binomial glm!
Warning in eval(family$initialize): non-integer #successes in a binomial glm!
```

```
Warning in eval(family$initialize): non-integer #successes in a binomial glm!
Warning in eval(family$initialize): non-integer #successes in a binomial glm!
```

```
names(gee_results) <- outcomes
```

```
# Combine results into one table
gee_table <- do.call(
  rbind,
  lapply(names(gee_results), function(x){
    tmp <- gee_results[[x]]
    tmp <- tmp %>%
      filter(grepl("^TRT2", term)) %>%
      mutate(outcome = x)
    tmp
  })
)
```

```
# Create final manuscript table
gee_table_final <- gee_table %>%
  mutate(
    Effect_measure = "OR",
    Estimate = sprintf("%.2f", estimate),
    CI = paste0(
      sprintf("%.2f", conf.low),
      "-",
      sprintf("%.2f", conf.high)
    ),
    P_value = p.value
  ) %>%
  dplyr::select(
    Outcome = outcome,
    Effect_measure,
    Estimate,
    CI,
    P_value
  )
```

```
# the warnings are a consequence of valid ATO weighting, not an error. Your models ran successfully.
# The model internally detects that the weighted outcome contribution is fractional and gives the warning.
# You are not using frequency weights to represent replicated patients. You are using sampling/analysis weights.
# With propensity score weighting, fractional contributions are expected.
```

```
# weighted Cox model for POSTUTI
```

```
# The main issue with the clustered Cox model (using survey design object with ids = CENTER) was the unbalanced design
```

```
cox_uti <- svycoxph(
  Surv(POSTUTIFUP, POSTUTI == 1) ~ TRT2 + strata(CENTER),
  design = design_ato
)
```

```
# your earlier binary GEE models are estimating a population-average effect, while this stratified Cox model is estimating an individual-level effect
```

```

# If you wanted HRs for each center, you would need: + factor(CENTER)

# `strata(CENTER)` allows each center to have its own baseline hazard function. It estimates the treatment effect for each center.

cox_summary <- summary(cox_uti)

Independent Sampling design (with replacement)
svydesign(ids = ~1, data = Out, weights = ~weights)

cox_row <- grep(
  "^TRT2",
  rownames(cox_summary$coefficients)
)

POSTUTI_table <- data.frame(
  Outcome = "POSTUTI",
  Effect_measure = "HR",
  Estimate = sprintf(
    "%.2f",
    cox_summary$coefficients[
      cox_row,
      "exp(coef)"
    ]
  ),
  CI = paste0(
    sprintf("%.2f",
      cox_summary$conf.int[cox_row, "lower .95"]),
    "-",
    sprintf("%.2f",
      cox_summary$conf.int[cox_row, "upper .95"])
  ),
  P_value = cox_summary$coefficients[
    cox_row,
    "Pr(>|z|)"
  ]
)

# Combine final results
final_results <- rbind(
  gee_table_final,
  POSTUTI_table
)

# Multiple correction

## Bonferroni correction
final_results$p_bonferroni <- p.adjust(
  final_results$P_value,
  method = "bonferroni"
)

```



```

## Holm-Bonferroni
final_results$p_holm <- p.adjust(
  final_results$P_value,
  method = "holm"
)

## Benjamini-Hochberg FDR
final_results$p_FDR <- p.adjust(
  final_results$P_value,
  method = "BH"
)

final_results <- final_results %>%
  mutate(
    across(
      c(P_value, p_bonferroni, p_holm, p_FDR),
      ~case_when(
        .x < 0.001 ~ "<0.001",
        TRUE ~ sprintf("%.3f", .x)
      )
    )
  )

final_results

# A tibble: 12 x 8
  Outcome      Effect_measure Estimate CI      P_value p_bonferroni p_holm p_FDR
  <chr>        <chr>          <chr> <chr> <chr>    <chr>      <chr> <chr>
1 LUTD        OR            1.16  0.80~ 0.446    1.000      0.446  0.446
2 OBST        OR            3.67  2.21~ <0.001  <0.001     <0.001 <0.0~
3 URCOMPINJ   OR            0.43  0.27~ <0.001  0.005      0.004  0.001
4 SURG        OR            0.72  0.58~ 0.004    0.051      0.025  0.007
5 ILEUS       OR            0.10  0.04~ <0.001  <0.001     <0.001 <0.0~
6 GU30        OR            0.02  0.00~ 0.090    1.000      0.362  0.121
7 GU90        OR            0.20  0.16~ <0.001  <0.001     <0.001 <0.0~
8 PROC365     OR            0.79  0.60~ 0.109    1.000      0.362  0.125
9 REC011      OR            0.57  0.39~ 0.002    0.026      0.015  0.004
10 REC011_best OR            0.50  0.33~ <0.001  0.006      0.004  0.001
11 REC011_worst OR           1.30  0.99~ 0.056    0.670      0.279  0.084
12 POSTUTI    HR            0.67  0.41~ 0.114    1.000      0.362  0.125

final_results %>%
  flextable() %>%
  theme_booktabs() %>%
  autofit() %>%
  align(
    j = c("Estimate", "CI", "P_value"),
    align = "center",
    part = "all"
  ) %>%
  save_as_docx(
    path = "output/treatment_unadj_w.docx"
  )

```

```

)

# Evaluate whether CENTER distribution is balanced after weighting, despite that CENTER was not included

# bal.tab(
#   w.out,
#   data = Out,
#   add1 = ~ CENTER
# )

# Results showed that all CENTER SMD < 0.05 → excellent balance
# Therefore, after ATO weighting, CENTER is well balanced between treatment groups.
# This is an important finding because your original data showed strong center-treatment imbalance.
# Why did CENTER become balanced although it was not in the propensity score model?
# This is possible because CENTER was correlated with the covariates you included in the propensity model.

```

Block 2

```

# Block 2: exposure-effect analysis (stent, operative time, catheterization) on postoperative outcomes

# Question: Is there still an association between technique and outcome after accounting for differences?

# Catheterization duration was considered a post-treatment variable and was not included in the outcome

# Operative time was included in secondary analyses as a procedural characteristic reflecting both technique and outcome

# Final decision: I will remove OPRTIME from multivariable models.

# Your justification for keeping STENT is relatively strong:
## The literature recognizes stent use as a variable component of ureteral reimplantation technique.
## Stent placement may directly influence postoperative outcomes (e.g., obstruction, urinary symptoms, etc.).
## It may represent a modifiable aspect of perioperative management.

# When you adjust for OPRTIME, you are asking: "Among patients with the same operative duration, is there a difference in outcomes?"

outcomes <- c(
  "LUTD",
  "OBST",
  "URCOMPINJ",
  "SURG",
  "ILEUS",
  "GU30",
  "GU90",
  "PROC365",
  "REC011",
  "REC011_best",
  "REC011_worst"
)

```

```

)

# convert outcomes to numeric
Out[outcomes] <- lapply(
  Out[outcomes],
  function(x) as.numeric(as.character(x))
)

# Predictors to include in the outcome models
predictors <- c("TRT2", "STENT")

# Apply GEE model after overlap weighting for all binary outcomes
gee_results <- lapply(outcomes, function(y) {
  formula <- as.formula(
    paste(y, "~",
          paste(predictors, collapse = " + ")
    )
  )
  model <- geeglm(
    formula,
    data = Out,
    id = CENTER,
    weights = weights,
    family = binomial(link = "logit"),
    corstr = "exchangeable"
  )
  tidy(model, exponentiate = TRUE, conf.int = TRUE) # estimate = OR because exponentiate = TRUE
})

Warning in eval(family$initialize): non-integer #successes in a binomial glm!
Warning in eval(family$initialize): non-integer #successes in a binomial glm!
Warning in eval(family$initialize): non-integer #successes in a binomial glm!
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Warning in eval(family$initialize): non-integer #successes in a binomial glm!
Warning in eval(family$initialize): non-integer #successes in a binomial glm!
Warning in eval(family$initialize): non-integer #successes in a binomial glm!

names(gee_results) <- outcomes

# Combine results into one table
gee_table <- do.call(
  rbind,
  lapply(names(gee_results), function(x){
    tmp <- gee_results[[x]]
    tmp %>%
      mutate(outcome = x)
  })
)

```

```

# Create final manuscript table
gee_table_final <- gee_table %>%
  mutate(
    Effect_measure = "OR",
    Estimate = sprintf("%.2f", estimate),
    CI = paste0(
      sprintf("%.2f", conf.low),
      "-",
      sprintf("%.2f", conf.high)
    ),
    P_value = p.value
  ) %>%
  dplyr::select(
    Outcome = outcome,
    Predictor = term,
    Effect_measure,
    Estimate,
    CI,
    P_value
  )

# the warnings are a consequence of valid ATO weighting, not an error. Your models ran successfully.
# The model internally detects that the weighted outcome contribution is fractional and gives the warning
# You are not using frequency weights to represent replicated patients. You are using sampling/analysis weights
# With propensity score weighting, fractional contributions are expected.

# weighted Cox model for POSTUTI

# The main issue with the clustered Cox model (using survey design object with ids = CENTER) was the uninformative
# baseline hazard function.

cox_uti <- svycoxph(
  Surv(POSTUTIFUP, POSTUTI == 1) ~ TRT2 + STENT + strata(CENTER),
  design = design_ato
)

# your earlier binary GEE models are estimating a population-average effect, while this stratified Cox model
# estimates a subject-specific effect.

# If you wanted HRs for each center, you would need: + factor(CENTER)

# `strata(CENTER)` allows each center to have its own baseline hazard function. It estimates the treatment effect
# relative to the baseline hazard function for each center.

cox_summary <- summary(cox_uti)

Independent Sampling design (with replacement)
svydesign(ids = ~1, data = Out, weights = ~weights)

cox_row <- rownames(cox_summary$coefficients)

POSTUTI_table <- data.frame(
  Outcome = "POSTUTI",
  Predictor = rownames(cox_summary$coefficients),

```

```

Effect_measure = "HR",
Estimate = sprintf(
  "%.2f",
  cox_summary$coefficients[
    cox_row,
    "exp(coef)"
  ]
),
CI = paste0(
  sprintf("%.2f",
    cox_summary$conf.int[cox_row, "lower .95"]),
  "-",
  sprintf("%.2f",
    cox_summary$conf.int[cox_row, "upper .95"])
),
P_value = cox_summary$coefficients[
  cox_row,
  "Pr(>|z|)"
]
)

# Combine final results
final_results <- rbind(
  gee_table_final,
  POSTUTI_table
)

final_results

# A tibble: 35 x 6
  Outcome Predictor Effect_measure Estimate CI P_value
* <chr> <chr> <chr> <chr> <chr> <dbl>
1 LUTD (Intercept) OR 0.32 0.24-0.42 2.78e-15
2 LUTD TRT21 OR 1.01 0.69-1.50 9.43e- 1
3 LUTD STENT1 OR 0.35 0.14-0.84 1.85e- 2
4 OBST (Intercept) OR 0.02 0.01-0.06 2.05e-13
5 OBST TRT21 OR 3.71 2.50-5.52 8.75e-11
6 OBST STENT1 OR 1.09 0.50-2.40 8.24e- 1
7 URCOMPINJ (Intercept) OR 0.03 0.02-0.05 0
8 URCOMPINJ TRT21 OR 0.36 0.27-0.49 6.58e-11
9 URCOMPINJ STENT1 OR 2.03 0.52-7.91 3.06e- 1
10 SURG (Intercept) OR 1.04 0.75-1.45 8.01e- 1
# i 25 more rows

final_results <- final_results %>%
  mutate(
    across(
      c(P_value),
      ~case_when(
        .x < 0.001 ~ "<0.001",
        TRUE ~ sprintf("%.3f", .x)
      )
    )
  )

```

```

    )
  )
)

final_results %>%
  flextable() %>%
  theme_booktabs() %>%
  autofit() %>%
  align(
    j = c("Estimate", "CI", "P_value"),
    align = "center",
    part = "all"
  ) %>%
  save_as_docx(
    path = "output/treatment_adj_all_w.docx"
  )

# Multiple correction

# Extract TRT2 rows only

gee_trt2 <- gee_table_final %>%
  filter(grepl("^TRT2", Predictor))

POSTUTI_trt2 <- POSTUTI_table %>%
  filter(grepl("^TRT2", Predictor))

# Combine only treatment effects
treatment_results <- rbind(
  gee_trt2,
  POSTUTI_trt2
)

# Apply Correction

treatment_results$p_bonferroni <- p.adjust(
  as.numeric(treatment_results$P_value),
  method = "bonferroni"
) # Bonferroni correction

treatment_results$p_holm <- p.adjust(
  as.numeric(treatment_results$P_value),
  method = "holm"
) # Holm-Bonferroni

treatment_results$p_FDR <- p.adjust(
  as.numeric(treatment_results$P_value),
  method = "BH"
) # Benjamini-Hochberg FDR

```

```

treatment_results <- treatment_results %>%
  mutate(
    across(
      c(P_value, p_bonferroni, p_holm, p_FDR),
      ~case_when(
        .x < 0.001 ~ "<0.001",
        TRUE ~ sprintf("%.3f", .x)
      )
    )
  )

treatment_results

# A tibble: 12 x 9
  Outcome Predictor Effect_measure Estimate CI P_value p_bonferroni p_holm
  <chr> <chr> <chr> <chr> <chr> <chr> <chr> <chr>
1 LUTD TRT21 OR 1.01 0.69~ 0.943 1.000 0.943
2 OBST TRT21 OR 3.71 2.50~ <0.001 <0.001 <0.001
3 URCOMPINJ TRT21 OR 0.36 0.27~ <0.001 <0.001 <0.001
4 SURG TRT21 OR 0.63 0.50~ <0.001 0.003 0.002
5 ILEUS TRT21 OR 0.20 0.17~ <0.001 <0.001 <0.001
6 GU30 TRT21 OR 0.03 0.01~ <0.001 <0.001 <0.001
7 GU90 TRT21 OR 0.18 0.15~ <0.001 <0.001 <0.001
8 PROC365 TRT21 OR 0.71 0.56~ 0.003 0.042 0.010
9 REC011 TRT21 OR 0.61 0.46~ <0.001 0.009 0.004
10 REC011_b~ TRT21 OR 0.55 0.45~ <0.001 <0.001 <0.001
11 REC011_w~ TRT21 OR 1.47 1.15~ 0.002 0.027 0.009
12 POSTUTI TRT21 HR 0.67 0.41~ 0.114 1.000 0.228
# i 1 more variable: p_FDR <chr>

treatment_results %>%
  flextable() %>%
  theme_booktabs() %>%
  autofit() %>%
  align(
    j = c("Estimate", "CI", "P_value"),
    align = "center",
    part = "all"
  ) %>%
  save_as_docx(
    path = "output/treatment_adj_w.docx"
  )

# ADDITIONAL MULTIPLE COMPARISONS

# Extract STENT rows only

gee_st <- gee_table_final %>%
  filter(grepl("^STENT", Predictor))

POSTUTI_st <- POSTUTI_table %>%
  filter(grepl("^STENT", Predictor))

```

```

# Combine only stent effects
st_results <- rbind(
  gee_st,
  POSTUTI_st
)

# Apply Correction

st_results$p_bonferroni <- p.adjust(
  as.numeric(st_results$P_value),
  method = "bonferroni"
) # Bonferroni correction

st_results$p_holm <- p.adjust(
  as.numeric(st_results$P_value),
  method = "holm"
) # Holm-Bonferroni

st_results$p_FDR <- p.adjust(
  as.numeric(st_results$P_value),
  method = "BH"
) # Benjamini-Hochberg FDR

st_results <- st_results %>%
  mutate(
    across(
      c(P_value, p_bonferroni, p_holm, p_FDR),
      ~case_when(
        .x < 0.001 ~ "<0.001",
        TRUE ~ sprintf("%.3f", .x)
      )
    )
  )

st_results

# A tibble: 12 x 9
  Outcome Predictor Effect_measure Estimate CI P_value p_bonferroni p_holm
  <chr> <chr> <chr> <chr> <chr> <chr> <chr>
1 LUTD STENT1 OR 0.35 0.14~ 0.019 0.223 0.111
2 OBST STENT1 OR 1.09 0.50~ 0.824 1.000 1.000
3 URCOMPINJ STENT1 OR 2.03 0.52~ 0.306 1.000 1.000
4 SURG STENT1 OR 0.11 0.03~ <0.001 0.010 0.009
5 ILEUS STENT1 OR 0.06 0.01~ <0.001 0.003 0.003
6 GU30 STENT1 OR 0.25 0.09~ 0.007 0.087 0.069
7 GU90 STENT1 OR 0.22 0.07~ 0.013 0.161 0.094
8 PROC365 STENT1 OR 0.50 0.11~ 0.387 1.000 1.000
9 REC011 STENT1 OR 2.09 1.22~ 0.007 0.083 0.069
10 REC011_b~ STENT1 OR 0.35 0.08~ 0.181 1.000 0.903
11 REC011_w~ STENT1 OR 3.83 1.42~ 0.008 0.096 0.069
12 POSTUTI STENT1 HR 0.93 0.42~ 0.854 1.000 1.000
# i 1 more variable: p_FDR <chr>

```



```

st_results %>%
  flextable() %>%
  theme_booktabs() %>%
  autofit() %>%
  align(
    j = c("Estimate", "CI", "P_value"),
    align = "center",
    part = "all"
  ) %>%
  save_as_docx(
    path = "output/stent_adj_w.docx"
  )

```

Block 3

```

# Block 3: Prognostic factors for readmission and reoperation = Multivariable analyses including (Treatment)

# IMPORTANT: The model is valid statistically, but interpretation depends on whether the complication p

# Your analysis seems to be asking: "Are postoperative complications associated with subsequent outcomes

# Multiplicity adjustment was performed separately for each prespecified outcome because outcomes represent

# Why temporality is an issue: For a predictor to be a causal risk factor, it must occur before the outcome

# A limitation for this analysis: "Because exact timing of postoperative complications was not available

# As the change of reversed temporality is higher with GU30, I would restrict the analysis to GU90 and GU10

# Outcomes to be explained
outcomes <- c(
  "GU90",
  "PROC365"
)

# Complications evaluated separately
complications <- c(
  "LUTD",
  "OBST",
  "URCOMPINJ",
  "SURG",
  "ILEUS",
  "POSTUTI",          # here, no cox regression because the outcome is binary and POSTUTI is an exposure
  "REC011",
  "REC011_best",
  "REC011_worst"
)

# Adjustment variables

```

```

adjusters <- c(
  "TRT2",
  "STENT"
)

# Convert binary variables to numeric
Out[c(outcomes, complications)] <- lapply(
  Out[c(outcomes, complications)],
  function(x) as.numeric(as.character(x))
)

gee_results <- list()

for (outcome in outcomes) {

  for (complication in complications) {

    print(paste("Running:", outcome, complication))

    vars <- c(
      outcome,
      complication,
      adjusters,
      "CENTER",
      "weights"
    )

    data_model <- Out %>%
      dplyr::select(all_of(vars)) %>%
      na.omit()

    formula <- as.formula(
      paste(
        outcome,
        "~",
        paste(
          c(complication, adjusters),
          collapse = " + "
        )
      )
    )

    model <- geeglm(
      formula,
      data = data_model,
      id = CENTER,
      weights = weights,
      family = binomial(link = "logit"),
      corstr = "exchangeable"
    )
  }
}

```

```

result <- tidy(
  model,
  exponentiate = TRUE,
  conf.int = TRUE
) %>%
  filter(term == complication) %>%
  mutate(
    Outcome = outcome,
    Predictor = complication,
    N = nrow(data_model)
  )

gee_results[[paste(outcome, complication, sep="_")] <- result
}
}

[1] "Running: GU90 LUTD"

Warning in eval(family$initialize): non-integer #successes in a binomial glm!

[1] "Running: GU90 OBST"

Warning in eval(family$initialize): non-integer #successes in a binomial glm!

[1] "Running: GU90 URCOMPINJ"

Warning in eval(family$initialize): non-integer #successes in a binomial glm!

[1] "Running: GU90 SURG"

Warning in eval(family$initialize): non-integer #successes in a binomial glm!

[1] "Running: GU90 ILEUS"

Warning in eval(family$initialize): non-integer #successes in a binomial glm!

[1] "Running: GU90 POSTUTI"

Warning in eval(family$initialize): non-integer #successes in a binomial glm!

[1] "Running: GU90 REC011"

Warning in eval(family$initialize): non-integer #successes in a binomial glm!

[1] "Running: GU90 REC011_best"

Warning in eval(family$initialize): non-integer #successes in a binomial glm!

[1] "Running: GU90 REC011_worst"

```

```

Warning in eval(family$initialize): non-integer #successes in a binomial glm!

[1] "Running: PROC365 LUTD"

Warning in eval(family$initialize): non-integer #successes in a binomial glm!

[1] "Running: PROC365 OBST"

Warning in eval(family$initialize): non-integer #successes in a binomial glm!

[1] "Running: PROC365 URCOMPINJ"

Warning in eval(family$initialize): non-integer #successes in a binomial glm!

[1] "Running: PROC365 SURG"

Warning in eval(family$initialize): non-integer #successes in a binomial glm!

[1] "Running: PROC365 ILEUS"

Warning in eval(family$initialize): non-integer #successes in a binomial glm!

[1] "Running: PROC365 POSTUTI"

Warning in eval(family$initialize): non-integer #successes in a binomial glm!

[1] "Running: PROC365 REC011"

Warning in eval(family$initialize): non-integer #successes in a binomial glm!

[1] "Running: PROC365 REC011_best"

Warning in eval(family$initialize): non-integer #successes in a binomial glm!

[1] "Running: PROC365 REC011_worst"

Warning in eval(family$initialize): non-integer #successes in a binomial glm!

gee_table <- bind_rows(gee_results)

gee_table_final <- gee_table %>%
  mutate(
    Effect_measure = "OR",
    Estimate = sprintf("%.2f", estimate),
    CI = paste0(
      sprintf("%.2f", conf.low),
      "-",
      sprintf("%.2f", conf.high)
    )
  )

```

```

    ),
    P_value = p.value
  ) %>%
  dplyr::select(
    Outcome,
    Predictor,
    N,
    Effect_measure,
    Estimate,
    CI,
    P_value
  )

gee_table_final <- gee_table_final %>%
  group_by(Outcome) %>%
  mutate(
    p_bonferroni = p.adjust(
      P_value,
      method = "bonferroni"
    ),
    p_holm = p.adjust(
      P_value,
      method = "holm"
    ),
    p_FDR = p.adjust(
      P_value,
      method = "BH"
    )
  ) %>%
  ungroup()

gee_table_final <- gee_table_final %>%
  mutate(
    across(
      c(P_value, p_bonferroni, p_holm, p_FDR),
      ~case_when(
        .x < 0.001 ~ "<0.001",
        TRUE ~ sprintf("%.3f", .x)
      )
    )
  )

# GU90 table
gee_table_final %>%
  filter(Outcome == "GU90") %>%
  dplyr::select(-Outcome) %>%
  flextable() %>%
  theme_booktabs() %>%
  autofit() %>%
  align(

```

```

      j = c("N", "Estimate", "CI", "P_value"),
      align = "center",
      part = "all"
    ) %>%
    save_as_docx(
      path = "output/GU90_adj_weighted.docx"
    )

# PROC365 table
gee_table_final %>%
  filter(Outcome == "PROC365") %>%
  dplyr::select(-Outcome) %>%
  flextable() %>%
  theme_booktabs() %>%
  autofit() %>%
  align(
    j = c("N", "Estimate", "CI", "P_value"),
    align = "center",
    part = "all"
  ) %>%
  save_as_docx(
    path = "output/PROC365_adj_weighted.docx"
  )

```

Block 4 - exploratory

```

# Block 3: Prognostic factors for readmission and reoperation = Multivariable analyses including (Treatment, Complications)

# IMPORTANT: The model is valid statistically, but interpretation depends on whether the complication p-values are significant

# Your analysis seems to be asking: "Are postoperative complications associated with subsequent outcomes?"

# Multiplicity adjustment was performed separately for each prespecified outcome because outcomes represent different clinical questions

# Outcomes to be explained
outcomes <- c(
  "GU90",
  "PROC365"
)

# Complications evaluated separately
complications <- c(
  "ANY",
  "MAJOR",
  "MINOR",
  "POSTUTI",
  "REC011",
  "REC011_best",
  "REC011_worst"
)

```

```

# Adjustment variables
adjusters <- c(
  "TRT2",
  "STENT"
)

# Convert binary variables to numeric
Out[c(outcomes, complications)] <- lapply(
  Out[c(outcomes, complications)],
  function(x) as.numeric(as.character(x))
)

gee_results <- list()

for (outcome in outcomes) {
  for (complication in complications) {
    print(paste("Running:", outcome, complication))

    vars <- c(
      outcome,
      complication,
      adjusters,
      "CENTER",
      "weights"
    )

    data_model <- Out %>%
      dplyr::select(all_of(vars)) %>%
      na.omit()

    formula <- as.formula(
      paste(
        outcome,
        "~",
        paste(
          c(complication, adjusters),
          collapse = " + "
        )
      )
    )

    model <- geeglm(
      formula,
      data = data_model,
      id = CENTER,
      weights = weights,
      family = binomial(link = "logit"),
      corstr = "exchangeable"
    )
  }
}

```

```

result <- tidy(
  model,
  exponentiate = TRUE,
  conf.int = TRUE
) %>%
  filter(term == complication) %>%
  mutate(
    Outcome = outcome,
    Predictor = complication,
    N = nrow(data_model)
  )

gee_results[[paste(outcome, complication, sep="_")] <- result
}
}

gee_table <- bind_rows(gee_results)

gee_table_final <- gee_table %>%
  mutate(
    Effect_measure = "OR",
    Estimate = sprintf("%.2f", estimate),
    CI = paste0(
      sprintf("%.2f", conf.low),
      "-",
      sprintf("%.2f", conf.high)
    ),
    P_value = p.value
  ) %>%
  dplyr::select(
    Outcome,
    Predictor,
    N,
    Effect_measure,
    Estimate,
    CI,
    P_value
  )

gee_table_final <- gee_table_final %>%
  group_by(Outcome) %>%
  mutate(
    p_bonferroni = p.adjust(
      P_value,
      method = "bonferroni"
    ),
    p_holm = p.adjust(
      P_value,
      method = "holm"
    ),
    p_FDR = p.adjust(

```



```

      P_value,
      method = "BH"
    )
  ) %>%
  ungroup()

gee_table_final <- gee_table_final %>%
  mutate(
    across(
      c(P_value, p_bonferroni, p_holm, p_FDR),
      ~case_when(
        .x < 0.001 ~ "<0.001",
        TRUE ~ sprintf("%.3f", .x)
      )
    )
  )

# GU90 table
gee_table_final %>%
  filter(Outcome == "GU90") %>%
  dplyr::select(-Outcome) %>%
  flextable() %>%
  theme_booktabs() %>%
  autofit() %>%
  align(
    j = c("N", "Estimate", "CI", "P_value"),
    align = "center",
    part = "all"
  ) %>%
  save_as_docx(
    path = "output/GU90_adj2_weighted.docx"
  )

# PROC365 table
gee_table_final %>%
  filter(Outcome == "PROC365") %>%
  dplyr::select(-Outcome) %>%
  flextable() %>%
  theme_booktabs() %>%
  autofit() %>%
  align(
    j = c("N", "Estimate", "CI", "P_value"),
    align = "center",
    part = "all"
  ) %>%
  save_as_docx(
    path = "output/PROC365_adj2_weighted.docx"
  )

```

PAIN - Non-epidural Cohort

Here, we estimated: Effect of surgical technique on pain in the overlap population of children who were not given epidural.

Above: we estimated: Effect of surgical technique on other outcomes in the overlap population of all children.

Whenever the target population changes because of an inclusion/exclusion criterion, the propensity score model should generally be re-estimated.

Overlap weighting only guarantees balance in the population on which the propensity score was estimated. Once you remove observations after weighting, that guarantee disappears -> The correct approach is to restrict the cohort first and then estimate a new propensity score model and new ATO (overlap) weights within that restricted cohort.

In your case, the scientific question is: Among children who did not receive epidural injection, does surgical technique affect postoperative pain?

This is not the same estimand as: Among all children, what is the effect of surgical technique?

1 Filter & weight the new population

```
# Filter dataset to only those who did not receive EPIDURAL
Outp <- Out %>%
  filter(EPIDURAL == "0") %>%
  mutate(
    PAIN = factor(
      PAIN,
      levels = c("0", "1", "2", "3"),
      labels = c(
        "No pain",
        "Mild pain (score 1-3)",
        "Moderate pain (score 4-6)",
        "Severe pain (score 7-10)"
      ),
      ordered = TRUE
    )
  )

# Do weighting: using ATO (Recommended)

set.seed(123)

w.outp <- weightit(
  TRT2 ~ AGS + YEARS + GENDER + BMI + PREOPRNPLN + HISTOINJ +
    SURGIND2 + PREOPVURGRADE + VURSIDE + UDR7M + VURPHASE +
    PREOPEPISODE1 + PREOPTYPEU1 + PREOPRNSCR + SURGTECH2,
  data = Outp,
  method = "ps",
  estimand = "ATO",
  stabilize = FALSE # used in ATE
)

Outp$weights <- w.outp$weights # Add weights to the dataset
write.csv(Outp, "data/weighted_nonepidural_data.csv", row.names = FALSE)
```

```
summary(w.outp$weights)
```

```
      Min.    1st Qu.    Median      Mean   3rd Qu.      Max.
0.0007272 0.0555583 0.2395282 0.2875446 0.4552000 0.9418707
```

```
# Table for balance
```

```
# This gives: SMD before and after weighting, and flagged imbalance (>0.1)
```

```
balp <- cobalt::bal.tab(w.outp, un = TRUE, ESS = TRUE, m.threshold = 0.1) # ESS to calculate ESS separately
```

```
# un=TRUE shows imbalance before AND after weighting, un=FALSE shows only after weighting
```

```
balp
```

Balance Measures

	Type	Diff.Un	Diff.Adj	M.Threshold
prop.score	Distance	1.8310	-0.0259	Balanced, <0.1
AGS	Contin.	0.3389	0.0000	Balanced, <0.1
YEARS	Contin.	-0.5189	-0.0000	Balanced, <0.1
GENDER	Binary	-0.0397	-0.0000	Balanced, <0.1
BMI	Contin.	-0.6798	-0.0000	Balanced, <0.1
PREOPRNPLN	Binary	-0.2680	-0.0000	Balanced, <0.1
HISTOINJ	Binary	0.0111	-0.0000	Balanced, <0.1
SURGIND2_1	Binary	0.0673	0.0000	Balanced, <0.1
SURGIND2_2	Binary	0.0864	-0.0000	Balanced, <0.1
SURGIND2_3	Binary	-0.0123	-0.0000	Balanced, <0.1
SURGIND2_4	Binary	-0.0520	-0.0000	Balanced, <0.1
SURGIND2_5	Binary	-0.0894	-0.0000	Balanced, <0.1
PREOPVURGRADE_3	Binary	-0.0179	-0.0000	Balanced, <0.1
PREOPVURGRADE_4	Binary	0.0999	-0.0000	Balanced, <0.1
PREOPVURGRADE_5	Binary	-0.0820	0.0000	Balanced, <0.1
VURSIDE	Binary	0.0233	-0.0000	Balanced, <0.1
UDR7M	Binary	-0.1485	0.0000	Balanced, <0.1
VURPHASE_1	Binary	-0.0702	-0.0000	Balanced, <0.1
VURPHASE_2	Binary	0.3261	0.0000	Balanced, <0.1
VURPHASE_3	Binary	-0.2559	-0.0000	Balanced, <0.1
PREOPEPISODE1_1	Binary	-0.0509	-0.0000	Balanced, <0.1
PREOPEPISODE1_2	Binary	-0.1592	-0.0000	Balanced, <0.1
PREOPEPISODE1_3	Binary	0.2101	0.0000	Balanced, <0.1
PREOPTYPEU1	Binary	0.0120	0.0000	Balanced, <0.1
PREOPRNSCR	Binary	-0.2746	-0.0000	Balanced, <0.1
SURGTECH2_1	Binary	0.2538	0.0000	Balanced, <0.1
SURGTECH2_2	Binary	-0.0625	-0.0000	Balanced, <0.1
SURGTECH2_3	Binary	-0.1913	-0.0000	Balanced, <0.1

```
Balance tally for mean differences
```

	count
Balanced, <0.1	28
Not Balanced, >0.1	0

```
Variable with the greatest mean difference
```

Variable	Diff.Adj	M.Threshold
YEARS	-0	Balanced, <0.1

```
Effective sample sizes
```

	Control	Treated
Unadjusted	178.	205.
Adjusted	113.59	105.65

```

balp_df <- as.data.frame(balp$Balance)
smd_table <- balp_df %>%
  mutate(
    Variable = rownames(balp_df),
    SMD_Before = round(abs(Diff.Un), 3),
    SMD_After = round(abs(Diff.Adj), 3)
  ) %>%
  dplyr::select(Variable, SMD_Before, SMD_After) # SMDs for each covariate individually, before and after

smd_table %>%
  flextable() %>%
  save_as_docx(path = "output/SMD_pw.docx")

# Figure for balance - Love plot
# Review guidelines to decide on plotting Standardized or Absolute Mean Differences
plp <- love.plot(
  w.outp,
  threshold = 0.1,
  abs = TRUE, # As SMD sign is meaningless for balance, show only absolute SMDs using abs = TRUE (the c
  stats = "mean.diffs", # forces SMD instead of raw differences
  var.order = "unadjusted", # Orders variables based on imbalance before matching
  drop.distance = TRUE, # To exclude the distance from the plot
  sample.names = c("Before Overlap Weighting", "After Overlap Weighting"),
  var.names = c(
    AGS = "Age at surgery",
    YEARS = "Year of surgery",
    BMI = "Body mass index",
    GENDER = "Female gender",
    PREOPRNPLN = "Preop pyelonephritis",
    HISTOINJ = "History of injection",
    VURSIDE = "Right-sided VUR",
    UDR7M = "Ureteral dilation 7mm",
    PREOPTYPEU1 = "Preop febrile UTIs",
    PREOPRNSCR = "Preop renal scarring",
    VURPHASE_1 = "Reflux: voiding phase",
    VURPHASE_2 = "Reflux: filling phase",
    VURPHASE_3 = "Reflux: filling + voiding",
    SURGTECH2_1 = "Surgeon experience 100 operations",
    SURGTECH2_2 = "Surgeon experience 50-100 operations",
    SURGTECH2_3 = "Surgeon experience <50 operations",
    PREOPEPISODE1_1 = "Preop UTI episodes: 1",
    PREOPEPISODE1_2 = "Preop UTI episodes: 2",
    PREOPEPISODE1_3 = "Preop UTI episodes: 3",
    SURGIND2_1 = "Indication: breakthrough UTI",
    SURGIND2_2 = "Indication: high-grade reflux (III-V)",
    SURGIND2_3 = "Indication: poor compliance to medical treatment",
    SURGIND2_4 = "Indication: ineffective/poorly tolerated CAP",
    SURGIND2_5 = "Indication: other",
    PREOPVURGRADE_3 = "Preop VUR grade: III",

```

```

PREOPVURGRADE_4 = "Preop VUR grade: IV",
PREOPVURGRADE_5 = "Preop VUR grade: V"
)
)

```

Warning: Standardized mean differences and raw mean differences are present in the same plot. Use the `stars` argument to distinguish between them and appropriately label the x-axis. See `love.plot()` for details.

```

plp <- plp +
  labs(
    x = "Absolute Standardized Differences",
    subtitle = "Overlap population of children who did not receive epidural analgesia"
  )
  theme(
    plot.margin = margin(10, 10, 10, 10),
    axis.text = element_text(size = 16),
    axis.title = element_text(size = 16),
    plot.title = element_text(size = 16),
    legend.text = element_text(size = 16),
    legend.title = element_text(size = 16),
    axis.ticks = element_line(size = 0.5, color = "black"),
    axis.line = element_line(size = 0.5, color = "black"),
    panel.background = element_blank(),
    panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(),
    panel.border = element_blank(),
    axis.ticks.length = unit(0.3, "cm"),
    panel.spacing = unit(2, "lines")
  )

```

<theme> List of 14

```

$ axis.title      : <ggplot2::element_text>
..@ family       : NULL
..@ face         : NULL
..@ italic       : chr NA
..@ fontweight   : num NA
..@ fontwidth    : num NA
..@ colour       : NULL
..@ size         : num 16
..@ hjust        : NULL
..@ vjust        : NULL
..@ angle        : NULL
..@ lineheight   : NULL
..@ margin       : NULL
..@ debug        : NULL
..@ inherit.blank: logi FALSE
$ axis.text       : <ggplot2::element_text>
..@ family       : NULL
..@ face         : NULL
..@ italic       : chr NA
..@ fontweight   : num NA
..@ fontwidth    : num NA

```

```

..@ colour      : NULL
..@ size        : num 16
..@ hjust       : NULL
..@ vjust       : NULL
..@ angle       : NULL
..@ lineheight  : NULL
..@ margin      : NULL
..@ debug       : NULL
..@ inherit.blank: logi FALSE
$ axis.ticks    : <ggplot2::element_line>
..@ colour      : chr "black"
..@ linewidth   : num 0.5
..@ linetype    : NULL
..@ lineend     : NULL
..@ linejoin    : NULL
..@ arrow       : logi FALSE
..@ arrow.fill  : chr "black"
..@ inherit.blank: logi FALSE
$ axis.ticks.length: 'simpleUnit' num 0.3cm
..- attr(*, "unit")= int 1
$ axis.line     : <ggplot2::element_line>
..@ colour      : chr "black"
..@ linewidth   : num 0.5
..@ linetype    : NULL
..@ lineend     : NULL
..@ linejoin    : NULL
..@ arrow       : logi FALSE
..@ arrow.fill  : chr "black"
..@ inherit.blank: logi FALSE
$ legend.text   : <ggplot2::element_text>
..@ family      : NULL
..@ face        : NULL
..@ italic      : chr NA
..@ fontweight  : num NA
..@ fontwidth   : num NA
..@ colour      : NULL
..@ size        : num 16
..@ hjust       : NULL
..@ vjust       : NULL
..@ angle       : NULL
..@ lineheight  : NULL
..@ margin      : NULL
..@ debug       : NULL
..@ inherit.blank: logi FALSE
$ legend.title  : <ggplot2::element_text>
..@ family      : NULL
..@ face        : NULL
..@ italic      : chr NA
..@ fontweight  : num NA
..@ fontwidth   : num NA
..@ colour      : NULL
..@ size        : num 16
..@ hjust       : NULL
..@ vjust       : NULL

```

```

..@ angle      : NULL
..@ lineheight : NULL
..@ margin     : NULL
..@ debug      : NULL
..@ inherit.blank: logi FALSE
$ panel.background : <ggplot2::element_blank>
$ panel.border     : <ggplot2::element_blank>
$ panel.spacing    : 'simpleUnit' num 2lines
..- attr(*, "unit")= int 3
$ panel.grid.major : <ggplot2::element_blank>
$ panel.grid.minor : <ggplot2::element_blank>
$ plot.title       : <ggplot2::element_text>
..@ family        : NULL
..@ face          : NULL
..@ italic        : chr NA
..@ fontweight    : num NA
..@ fontwidth     : num NA
..@ colour        : NULL
..@ size          : num 16
..@ hjust         : NULL
..@ vjust         : NULL
..@ angle         : NULL
..@ lineheight    : NULL
..@ margin        : NULL
..@ debug         : NULL
..@ inherit.blank: logi FALSE
$ plot.margin      : <ggplot2::margin> num [1:4] 10 10 10 10
@ complete: logi FALSE
@ validate: logi TRUE

ggsave ("figures/love_plot_pw.png", plot = plp, width = 12, height = 10, dpi = 300)

# Compute mean SMD
smdp_df <- balp$Balance
mean_smdp_before <- mean(abs(smdp_df$Diff.Un), na.rm = TRUE) # calculate mean SMD before matching
mean_smdp_after <- mean(abs(smdp_df$Diff.Adj), na.rm = TRUE) # calculate mean SMD before matching
mean_smdp <- data.frame(
  Mean_SMDp_Before = mean_smdp_before,
  Mean_SMDp_After = mean_smdp_after
)

mean_smdp

  Mean_SMDp_Before Mean_SMDp_After
1      0.2226086      0.0009262931

# how much effective sample size remains (overall ESS: calculated manually)?
summary(w.outp$weights)

  Min.   1st Qu.   Median     Mean   3rd Qu.    Max.
0.0007272 0.0555583 0.2395282 0.2875446 0.4552000 0.9418707

```

```

bal.tab(w.outp)$Effective.Sample.Size

NULL

sum(w.outp$weights)^2 / sum(w.outp$weights^2)

[1] 218.9503

# propensity score distribution
summary(w.outp$ps)

      Min.   1st Qu.   Median     Mean   3rd Qu.     Max.
0.006964 0.240110 0.543399 0.535248 0.819747 0.999273

pl2p <-plot(summary(w.outp, type = "hist"))
ggsave("figures/propensity_pw.png", plot = pl2p, width = 12, height = 10, dpi = 300)

# Distribution of overlap weights: histogram and density

## Histogram
summary(w.outp$weights)

      Min.   1st Qu.   Median     Mean   3rd Qu.     Max.
0.0007272 0.0555583 0.2395282 0.2875446 0.4552000 0.9418707

png("figures/weights_histo_pw.png",
    width = 6,
    height = 4,
    units = "in",
    res = 300)

hist(
  w.outp$weights,
  breaks = 30,
  col = "lightblue",
  border = "black",
  main = "Distribution of Overlap Weights",
  xlab = "Overlap Weight"
)
dev.off()

pdf
2

## Density
# owdp<- ggplot(Outp,
#   aes(weights,
#     colour = Treatment,
#     fill = Treatment)) +

```



```

# geom_density(alpha = 0.3, linewidth = 1) +
# labs(
#   x = "Overlap weight",
#   y = "Density"
# ) +
# theme_classic()
# ggsave ("figures/weights_dens_pw.png", plot = owdp, width = 12, height = 8, dpi = 300)

# How many patients are essentially excluded?
sum(w.outp$weights < 0.05)

[1] 90

sum(w.outp$weights < 0.01)

[1] 35

# If only a few patients have tiny weights → excellent.
# If 50-100 patients have tiny weights → your overlap population is narrower.

# Compare treatment group sizes after weighting
aggregate(weights ~ TRT2, data=Outp, sum)

  TRT2 weights
1    0 55.0648
2    1 55.0648

# Distributional balance for propensity score (purpose: to evaluate overlap/positivity):

# After weighting
# p_ps_pain <- bal.plot(w.outp, var.name = "prop.score", type = "density")

# p_ps_pain <- p_ps_pain +
#   scale_color_viridis_d(
#     option = "mako",
#     begin = 0.3,
#     end = 0.8,
#     labels = c("Cohen group", "Lich-Gregoir group")
#   ) +
#   scale_fill_viridis_d(
#     option = "mako",
#     begin = 0.3,
#     end = 0.8,
#     labels = c("Cohen group", "Lich-Gregoir group")
#   )

# ggsave(
#   filename = "figures/propensity_density_pain_overlap_weighting.png",
#   plot = p_ps_pain,
#   width = 12,

```

```

# height = 6,
# dpi = 300
# )

# Before-After weighting
p_ps_pain <- bal.plot(
  w.outp,
  var.name = "prop.score",
  type = "density",
  which = "both"
)

p_ps_pain <- p_ps_pain +
  scale_color_viridis_d(
    option = "mako",
    begin = 0.3,
    end = 0.8,
    labels = c("Cohen group", "Lich-Gregoir group")
  ) +
  scale_fill_viridis_d(
    option = "mako",
    begin = 0.3,
    end = 0.8,
    labels = c("Cohen group", "Lich-Gregoir group")
  )

```

Scale for fill is already present.

Adding another scale for fill, which will replace the existing scale.

```

ggsave(
  "figures/PS_density_before_after_pATO.png",
  p_ps_pain,
  width = 10,
  height = 7,
  dpi = 300
)

```

2 Descriptive analysis

```

# Baseline characteristics in the population for pain
table1p <- Outp %>%
  dplyr::select(YEARS, AGS, GENDER, BMI, HISTOINJ, SURGIND2, PREOPVURGRADE, VURSIDE, UDR7M,
    VURPHASE, PREOPTYPEU1, PREOPEPISODE1, PREOPRNPLN, PREOPRNSCR,
    SURGTECH2, CENTER, TRT2
  ) %>%
  tbl_summary(
    by = TRT2, # for stratification
    statistic = list(
      all_continuous() ~ "{median} ({p25}-{p75})",
      all_categorical() ~ "{n} ({p}%"
    ),
  )

```

```

    digits = all_categorical() ~ c(0, 1), # n = 0 decimals, p = 1 decimal
    missing = "ifany", # shows missing only when present
    missing_text = "Missing"
  ) %>%
  add_overall() %>%
  add_p(pvalue_fun = ~style_pvalue(.x, digits = 3)) %>% # 3 decimals
  bold_labels()

table1p

table1p %>%
  as_flex_table() %>%
  flextable::save_as_docx(path = "output/baseline_p.docx")

# Create survey design object using overlap weights
design_pato <- svydesign(
  ids = ~1,
  data = Outp,
  weights = ~weights
)

# Weighted baseline characteristics
table1pw <- design_pato %>%
  tbl_svysummary(
    by = TRT2, # for stratification
    include = c(
      YEARS, AGS, GENDER, BMI, HISTOINJ, SURGIND2, PREOPVURGRADE, VURSIDE, UDR7M,
      VURPHASE, PREOPTYPEU1, PREOPEPISODE1, PREOPRNPLN, PREOPRNSCR,
      SURGTECH2, CENTER # generally we should not include TRT2 in include list
    ),
    statistic = list(
      all_continuous() ~ "{median} ({p25}-{p75})",
      all_categorical() ~ "{n} ({p}%"
    ),
    digits = all_categorical() ~ c(0, 1),
    missing = "ifany", # shows missing only when present
    missing_text = "Missing"
  ) %>%
  add_overall() %>%
  # add_p(pvalue_fun = ~style_pvalue(.x, digits = 3)) %>% # add_p() is not informative after weighting
  bold_labels()

table1pw

table1pw %>%
  as_flex_table() %>%
  flextable::save_as_docx(
    path = "output/baseline_pw.docx"
  )

```

Characteristic	Overall N = 383 ^I	0 N = 178 ^I	1 N = 205 ^I
YEARS	2,017.00 (2,014.00-2,019.00)	2,018.00 (2,014.00-2,020.00)	2,016.00 (2,013.00-2,019.00)
AGS	7.00 (6.00-9.00)	8.00 (4.00-9.00)	7.00 (6.00-9.00)
GENDER			
0	219 (57.2%)	98 (55.1%)	121 (59.0%)
1	164 (42.8%)	80 (44.9%)	84 (41.0%)
BMI	14.1 (13.5-17.7)	14.9 (13.4-21.0)	13.9 (13.6-14.4)
HISTOINJ			
0	342 (89.3%)	160 (89.9%)	182 (88.8%)
1	41 (10.7%)	18 (10.1%)	23 (11.2%)
SURGIND2			
1	201 (52.5%)	87 (48.9%)	114 (55.6%)
2	121 (31.6%)	48 (27.0%)	73 (35.6%)
3	19 (5.0%)	10 (5.6%)	9 (4.4%)
4	13 (3.4%)	11 (6.2%)	2 (1.0%)
5	29 (7.6%)	22 (12.4%)	7 (3.4%)
PREOPVURGRADE			
3	20 (5.2%)	11 (6.2%)	9 (4.4%)
4	113 (29.5%)	43 (24.2%)	70 (34.1%)
5	250 (65.3%)	124 (69.7%)	126 (61.5%)
VURSIDE			
0	148 (38.6%)	71 (39.9%)	77 (37.6%)
1	235 (61.4%)	107 (60.1%)	128 (62.4%)
UDR7M			
0	95 (24.8%)	30 (16.9%)	65 (31.7%)
1	288 (75.2%)	148 (83.1%)	140 (68.3%)
VURPHASE			
1	218 (56.9%)	108 (60.7%)	110 (53.7%)
2	69 (18.0%)	1 (0.6%)	68 (33.2%)
3	96 (25.1%)	69 (38.8%)	27 (13.2%)
PREOPTYPEU1			
0	4 (1.0%)	3 (1.7%)	1 (0.5%)
1	379 (99.0%)	175 (98.3%)	204 (99.5%)
PREOPEPISODE1			
1	24 (6.3%)	16 (9.0%)	8 (3.9%)
2	90 (23.5%)	57 (32.0%)	33 (16.1%)
3	269 (70.2%)	105 (59.0%)	164 (80.0%)
PREOPRNPLN			
0	311 (81.2%)	119 (66.9%)	192 (93.7%)
1	72 (18.8%)	59 (33.1%)	13 (6.3%)
PREOPRNSCR			
0	265 (69.2%)	97 (54.5%)	168 (82.0%)
1	118 (30.8%)	81 (45.5%)	37 (18.0%)
SURGTECH2			
1	80 (20.9%)	13 (7.3%)	67 (32.7%)
2	254 (66.3%)	124 (69.7%)	130 (63.4%)
3	49 (12.8%)	41 (23.0%)	8 (3.9%)
CENTER			
1	282 (73.6%) ⁶⁸	99 (55.6%)	183 (89.3%)
2	0 (0.0%)	0 (0.0%)	0 (0.0%)
3	37 (9.7%)	31 (17.4%)	6 (2.9%)

Characteristic	Overall N = 110 ^I	0 N = 55 ^I	1 N = 55
YEARS	2,017.00 (2,014.00-2,019.00)	2,017.00 (2,014.00-2,019.00)	2,017.00 (2,015.00-2,019.00)
AGS	8.00 (6.00-9.00)	8.00 (6.00-9.00)	7.00 (6.00-9.00)
GENDER			
0	60 (54.2%)	30 (54.2%)	30 (54.2%)
1	50 (45.8%)	25 (45.8%)	25 (45.8%)
BMI	14.0 (13.4-16.3)	13.9 (13.2-17.5)	14.1 (13.6-14.6)
HISTOINJ			
0	96 (86.8%)	48 (86.8%)	48 (86.8%)
1	15 (13.2%)	7 (13.2%)	7 (13.2%)
SURGIND2			
1	51 (46.5%)	26 (46.5%)	26 (46.5%)
2	44 (39.8%)	22 (39.8%)	22 (39.8%)
3	6 (5.6%)	3 (5.6%)	3 (5.6%)
4	2 (1.9%)	1 (1.9%)	1 (1.9%)
5	7 (6.2%)	3 (6.2%)	3 (6.2%)
PREOPVURGRADE			
3	7 (6.5%)	4 (6.5%)	4 (6.5%)
4	30 (27.6%)	15 (27.6%)	15 (27.6%)
5	72 (65.8%)	36 (65.8%)	36 (65.8%)
VURSIDE			
0	39 (35.7%)	20 (35.7%)	20 (35.7%)
1	71 (64.3%)	35 (64.3%)	35 (64.3%)
UDR7M			
0	27 (24.8%)	14 (24.8%)	14 (24.8%)
1	83 (75.2%)	41 (75.2%)	41 (75.2%)
VURPHASE			
1	84 (76.4%)	42 (76.4%)	42 (76.4%)
2	2 (1.6%)	1 (1.6%)	1 (1.6%)
3	24 (22.0%)	12 (22.0%)	12 (22.0%)
PREOPTYPEU1			
0	1 (0.8%)	0 (0.8%)	0 (0.8%)
1	109 (99.2%)	55 (99.2%)	55 (99.2%)
PREOPEPISODE1			
1	7 (6.1%)	3 (6.1%)	3 (6.1%)
2	33 (30.3%)	17 (30.3%)	17 (30.3%)
3	70 (63.6%)	35 (63.6%)	35 (63.6%)
PREOPRNPLN			
0	93 (84.6%)	47 (84.6%)	47 (84.6%)
1	17 (15.4%)	8 (15.4%)	8 (15.4%)
PREOPRNSCR			
0	80 (73.0%)	40 (73.0%)	40 (73.0%)
1	30 (27.0%)	15 (27.0%)	15 (27.0%)
SURGTECH2			
1	6 (5.4%)	3 (5.4%)	3 (5.4%)
2	92 (83.3%)	46 (83.3%)	46 (83.3%)
3	12 (11.3%)	6 (11.3%)	6 (11.3%)
CENTER			
1	84 (76.3%) ⁶⁹	40 (72.9%)	44 (79.6%)
2	0 (0.0%)	0 (0.0%)	0 (0.0%)
3	9 (8.0%)	5 (9.5%)	4 (6.4%)

Characteristic	Overall N = 110 ¹	0 N = 55 ¹	1 N = 55 ¹	p-value ²
OPRTIME	85 (70-135)	135 (90-160)	70 (67-80)	<0.001
STENT				0.062
0	74 (67.1%)	34 (61.0%)	40 (73.1%)	
1	36 (32.9%)	21 (39.0%)	15 (26.9%)	
CATH				0.516
0	2 (2.3%)	1 (1.6%)	2 (2.9%)	
1	108 (97.7%)	54 (98.4%)	53 (97.1%)	
CATHPERIOD	4 (3-5)	4 (4-5)	3 (1-5)	<0.001
CATHPERIOD_G				<0.001
Early removal 3 days	36 (33.1%)	4 (6.6%)	33 (59.5%)	
Standard removal 4-7 days	56 (50.5%)	42 (75.8%)	14 (25.1%)	
Delayed removal >7 days	18 (16.5%)	10 (17.6%)	8 (15.4%)	
HOSPTIME	5.00 (3.00-6.00)	5.00 (5.00-6.00)	3.00 (2.00-5.00)	<0.001

¹Median (Q1-Q3); n (%)

²Design-based KruskalWallis test; Pearson's X²: Rao & Scott adjustment

```
# Weighted procedural factors
table2p<- design_pato %>%
  tbl_svsummary(
    by = TRT2, # for stratification
    include = c (
      OPRTIME, STENT, CATH, CATHPERIOD, CATHPERIOD_G, HOSPTIME
    ),
    statistic = list(
      all_continuous() ~ "{median} ({p25}-{p75})", # or ~ "{mean} ± {sd}",
      all_categorical() ~ "{n} ({p}%)"
    ),
    digits = all_categorical() ~ c(0, 1), # n = 0 decimals, p = 1 decimal
    missing = "ifany",
    missing_text = "Missing"
  ) %>%
  add_overall() %>%
  add_p(pvalue_fun = ~style_pvalue(.x, digits = 3)) %>% # 3 decimals
  bold_labels()
```

The following warnings were returned during `add\_p()`:

! For variable `CATH` (`TRT2`) and "statistic" and "p.value" statistics:

Chi-squared approximation may be incorrect

table2p

```
table2p %>%
  as_flex_table() %>%
  flextable::save_as_docx(path = "output/procedural_pw.docx")
```

Weighted pain

```

# Before weighting

table4 <- Outp %>%
  dplyr::select(PAIN, NARC, NONARC, Treatment) %>%
  tbl_summary(
    by = Treatment,
    statistic = list(
      all_categorical() ~ "{n} ({p}%"
    ),
    label = list(
      PAIN ~ "Pain score",
      NARC ~ "Narcotic use",
      NONARC ~ "Non-narcotic use"
    ),
    digits = all_categorical() ~ c(0, 1),
    missing = "ifany",
    missing_text = "Missing"
  ) %>%
  add_overall() %>%
  add_p(pvalue_fun = ~style_pvalue(.x, digits = 3)) %>%
  bold_labels()

# table4

# table4 %>%
#   as_flex_table() %>%
#   flextable::save_as_docx(path = "output/pain_before.docx")

# After ATO weighting

design_atop <- svydesign(
  ids = ~1,
  data = Outp,
  weights = ~weights
)

# Weighted pain characteristics
table4w <- design_atop %>%
  tbl_svsummary(
    by = Treatment,      # for stratification
    include = c(
      PAIN, NARC, NONARC
    ),
    statistic = list(
      all_categorical() ~ "{n} ({p}%"
    ),
    label = list(
      PAIN ~ "Pain score",
      NARC ~ "Narcotic use",
      NONARC ~ "Non-narcotic use"
    ),
    digits = all_categorical() ~ c(0, 1),

```

```

    missing = "ifany", # shows missing only when present
    missing_text = "Missing"
  ) %>%
  add_overall() %>%
  add_p(pvalue_fun = ~style_pvalue(.x, digits = 3)) %>%
  bold_labels()

The following warnings were returned during `add_p()`:
! For variable `NONARC` (`Treatment`) and "statistic" and "p.value" statistics:
  Chi-squared approximation may be incorrect
! For variable `PAIN` (`Treatment`) and "statistic" and "p.value" statistics:
  Chi-squared approximation may be incorrect

# table4w

# table4w %>%
#   as_flex_table() %>%
#   flextable::save_as_docx(
#     path = "output/pain_w.docx"
#   )

# MERGE TABLES
table_combined <- tbl_merge(
  tbls = list(table4, table4w),
  tab_spanner = c("**Before overlap weighting**", "**After overlap weighting**")
) %>%
  modify_caption(
    "Table 4. Comparison of pain intensity and analgesic use between treatment groups before and after p
  ) %>%
  modify_source_note(
    "Analysis restricted to patients not receiving epidural analgesia."
  )

table_combined

table_combined %>%
  as_flex_table() %>%
  flextable::save_as_docx(path = "output/pain_w.docx")

```

3 Plot: BEFORE-AFTER weighting

```

# label datasets
before <- Outp %>%
  mutate(
    plot_weight = 1,
    Status = "Before overlap weighting (N=383), p<0.001"
  )

after <- Outp %>%

```


Characteristic	Before overlap weighting			p-value
	Overall N = 383 ¹	Cohen N = 178 ¹	Lich-Gregoir N = 205 ¹	
Pain score				<0.001
No pain	3 (0.8%)	0 (0.0%)	3 (1.5%)	
Mild pain (score 1-3)	127 (33.2%)	45 (25.3%)	82 (40.0%)	
Moderate pain (score 4-6)	123 (32.1%)	53 (29.8%)	70 (34.1%)	
Severe pain (score 7-10)	130 (33.9%)	80 (44.9%)	50 (24.4%)	
Narcotic use				0.2
0	215 (56.1%)	105 (59.0%)	110 (53.7%)	
1	168 (43.9%)	73 (41.0%)	95 (46.3%)	
Non-narcotic use				0.2
0	2 (0.5%)	2 (1.1%)	0 (0.0%)	
1	381 (99.5%)	176 (98.9%)	205 (100.0%)	

¹n (%)

²Fisher's exact test; Pearson's Chi-squared test

³Pearson's X²: Rao & Scott adjustment

Analysis restricted to patients not receiving epidural analgesia.

```
mutate(
  plot_weight = weights,
  Status = "After overlap weighting (ESS=219), p<0.001")

# Combine
plot_data <- bind_rows(before, after)

# pain labels
plot_data <- plot_data %>%
  mutate(
    PAIN_plot = factor(
      PAIN,
      levels = levels(PAIN),
      labels = c("No pain", "Mild pain", "Moderate pain", "Severe pain")
    )
  )

# To define order of data
plot_data <- plot_data %>%
  mutate(
    Status = factor(
      Status,
      levels = c(
        "Before overlap weighting (N=383), p<0.001",
        "After overlap weighting (ESS=219), p<0.001"
      )
    )
  )

# Pain distribution stacked by Treatment (within facet)
```

```

# summary
plot_summary <- plot_data %>%
  group_by(Status, Treatment, PAIN_plot) %>%
  summarise(
    weighted_n = sum(plot_weight, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  group_by(Status, Treatment) %>%
  mutate(
    Percent = 100 * weighted_n / sum(weighted_n)
  )

s0 <- ggplot(plot_summary,
  aes(x = PAIN_plot, y = Percent, fill = Treatment)) +
  geom_bar(stat = "identity", position = "stack") +
  facet_wrap(~Status) +
  labs(x = "
    1-Day Postoperative Pain Score",
    y = "Percentage of Patients not receiving epidural analgesia (%)") +
  scale_fill_manual(values = c(
    "Cohen" = "#ADD8E6",
    "Lich-Gregoir" = "#00008B"),
    name = NULL) +
  scale_y_continuous(limits = c(0, 100)) +
  theme_minimal() +
  theme(
    panel.grid = element_blank(), # Remove gridlines
    axis.title = element_text(size = 14), # Set axis labels size
    axis.text = element_text(size = 8), # Set axis text size
    panel.border = element_rect(color = "black", fill = NA), # Add frame to the plot
    plot.title = element_text(hjust = 0.5, size = 16), # Center the title
    axis.ticks = element_line(size = 0.5, color = "black"), # Set tick line properties
    axis.ticks.length = unit(-0.2, "cm"), # Move ticks inside the plot
    axis.text.y = element_text(size = 12), # Adjust the y-axis text size
    panel.spacing = unit(0, "lines") # Reduce space between panels
  )

ggsave(filename = "figures/pain_before_after_w.png",
  plot = s0,
  width = 8, # width in inches
  height = 6, # height in inches
  dpi = 300) # resolution

# 100% stacked grouped bar plot

# summary
plot_summary1 <- plot_data %>%
  group_by(Status, Treatment, PAIN) %>%
  summarise(
    weighted_n = sum(plot_weight, na.rm = TRUE),
    .groups = "drop"
  )

```

```

) %>%
group_by(Status, Treatment) %>%
mutate(
  Percent = 100 * weighted_n / sum(weighted_n)
)

s1 <- ggplot(plot_summary1,
  aes(x = Treatment, y = Percent, fill = PAIN)) +
geom_bar(stat = "identity", position = "fill", width = 0.7) +
facet_wrap(~Status) +
labs(
  x = "
  Treatment groups",
  y = "Percentage of patients not receiving epidural analgesia (%)",
  fill = "1-Day postoperative pain score"
) +
scale_fill_manual(values = c(
  "No pain" = "#f5f5f5", # Intead of pure white "#ffffff"
  "Mild pain (score 1-3)" = "#fee8c8",
  "Moderate pain (score 4-6)" = "#fdbb84",
  "Severe pain (score 7-10)" = "#e34a33"
)) +
scale_y_continuous(labels = scales::percent_format(scale = 100)) +
theme_minimal() +
theme(
  panel.grid = element_blank(),
  axis.title = element_text(size = 12),
  axis.text = element_text(size = 11),
  panel.border = element_rect(color = "black", fill = NA),
  strip.text = element_text(face = "bold"),
  panel.spacing = unit(0, "lines")
)

ggsave(
  filename = "figures/pain_stacked_w.png", # Title: Distribution of pain intensity (score) in treatment
  plot = s1,
  width = 9,
  height = 6,
  dpi = 300
)

```

4 Regression

```

# As PAIN is an ordinal variable -> Ordinal logistic GEE model is appropriate but ordinal logistic GEE

# PAIN will not be added to the first set of hypothesis because:
## Set 1: Main postoperative outcomes (full cohort)
## Set 2: Pain outcomes (non-epidural population)

# Because postoperative pain was evaluated among children who did not receive epidural analgesia, a sep

```

```
# After overlap weighting, Lich-Gregoir was associated with 80% lower odds of severe postoperative pain
```

```
# -----  
### Moderate-severe pain vs. Other levels  
# -----
```

```
Outp <- Outp %>%  
  mutate(  
    PAIN_ms = factor(  
      ifelse(  
        PAIN %in% c(  
          "Moderate pain (score 4-6)",  
          "Severe pain (score 7-10)"  
        ),  
        1,  
        0  
      ),  
      levels = c(0, 1),  
      labels = c(  
        "No/Mild pain (score 0-3)",  
        "Moderate-Severe pain (score 4-10)"  
      )  
    )  
  )  
)
```

```
Outp$PAIN_ms <- ifelse(  
  Outp$PAIN_ms == "Moderate-Severe pain (score 4-10)",  
  1,  
  0  
)
```

```
# -----  
### Exploratory analysis: Severe pain vs. Other levels  
# -----
```

```
Outp <- Outp %>%  
  mutate(  
    PAIN_s = factor(  
      ifelse(  
        PAIN == "Severe pain (score 7-10)",  
        1,  
        0  
      ),  
      levels = c(0, 1),  
      labels = c(  
        "No severe pain (score 0-6)",  
        "Severe pain (score 7-10)"  
      )  
    )  
  )  
)
```

```

# a numeric binary outcome:
Outp$PAIN_s <- ifelse(
  Outp$PAIN_s == "Severe pain (score 7-10)",
  1,
  0
)

#-----
# Block 1
#-----

outcomes <- c(
  "PAIN_ms",
  "PAIN_s"
)

# convert outcomes to numeric
Outp[outcomes] <- lapply(
  Outp[outcomes],
  function(x) as.numeric(as.character(x))
)

# Apply GEE model after overlap weighting for all binary outcomes
gee_results <- lapply(outcomes, function(y) {
  formula <- as.formula(
    paste(y, "~ TRT2")
  )
  model <- geeglm(
    formula,
    data = Outp,
    id = CENTER,
    weights = weights,
    family = binomial(link = "logit"),
    corstr = "exchangeable"
  )
  tidy(model, exponentiate = TRUE, conf.int = TRUE) # estimate = OR because exponentiate = TRUE
})

```

```

Warning in eval(family$initialize): non-integer #successes in a binomial glm!
Warning in eval(family$initialize): non-integer #successes in a binomial glm!

```

```

names(gee_results) <- outcomes

```

```

# Combine results into one table
gee_table <- do.call(
  rbind,
  lapply(names(gee_results), function(x){
    tmp <- gee_results[[x]]
    tmp <- tmp %>%
      filter(grepl("^TRT2", term)) %>%
      mutate(outcome = x)
    tmp
  })
)

```

```

    })
  )

# Create final manuscript table
gee_table_final <- gee_table %>%
  mutate(
    Effect_measure = "OR",
    Estimate = sprintf("%.2f", estimate),
    CI = paste0(
      sprintf("%.2f", conf.low),
      "-",
      sprintf("%.2f", conf.high)
    ),
    P_value = ifelse(p.value < 0.001, "<0.001", sprintf("%.3f", p.value))
  ) %>%
  dplyr::select(
    Outcome = outcome,
    Effect_measure,
    Estimate,
    CI,
    P_value
  )

gee_table_final

# A tibble: 2 x 5
  Outcome Effect_measure Estimate CI      P_value
<chr>    <chr>          <chr> <chr>    <chr>
1 PAIN_ms OR            0.32  0.14-0.73 0.007
2 PAIN_s  OR            0.20  0.15-0.26 <0.001

gee_table_final %>%
  flextable() %>%
  theme_booktabs() %>%
  autofit() %>%
  align(
    j = c("Estimate", "CI", "P_value"),
    align = "center",
    part = "all"
  ) %>%
  save_as_docx(
    path = "output/pain_unadj_w.docx"
  )

#-----
# Block 2
#-----

outcomes <- c(
  "PAIN_ms",
  "PAIN_s"
)

```

```

# convert outcomes to numeric
Outp[outcomes] <- lapply(
  Outp[outcomes],
  function(x) as.numeric(as.character(x))
)

# Predictors to include in the outcome models
predictors <- c("TRT2", "STENT")

# Apply GEE model after overlap weighting for all binary outcomes
gee_results <- lapply(outcomes, function(y) {
  formula <- as.formula(
    paste(y, "~",
          paste(predictors, collapse = " + ")
    )
  )
  model <- geeglm(
    formula,
    data = Outp,
    id = CENTER,
    weights = weights,
    family = binomial(link = "logit"),
    corstr = "exchangeable"
  )
  tidy(model, exponentiate = TRUE, conf.int = TRUE) # estimate = OR because exponentiate = TRUE
})

Warning in eval(family$initialize): non-integer #successes in a binomial glm!
Warning in eval(family$initialize): non-integer #successes in a binomial glm!

names(gee_results) <- outcomes

# Combine results into one table
gee_table <- do.call(
  rbind,
  lapply(names(gee_results), function(x){
    tmp <- gee_results[[x]]
    tmp %>%
      mutate(outcome = x)
  })
)

# Create final manuscript table
gee_table_final <- gee_table %>%
  mutate(
    Effect_measure = "OR",
    Estimate = sprintf("%.2f", estimate),
    CI = paste0(
      sprintf("%.2f", conf.low),
      "-",
      sprintf("%.2f", conf.high)
    )
  )

```

```

    ),
    P_value = ifelse(p.value < 0.001, "<0.001", sprintf("%.3f", p.value))
  ) %>%
  dplyr::select(
    Outcome = outcome,
    Predictor = term,
    Effect_measure,
    Estimate,
    CI,
    P_value
  )

```

```
gee_table_final
```

```

# A tibble: 6 x 6
  Outcome Predictor Effect_measure Estimate CI P_value
<chr> <chr> <chr> <chr> <chr> <chr>
1 PAIN_ms (Intercept) OR 3.78 1.37-10.37 0.010
2 PAIN_ms TRT21 OR 0.32 0.15-0.69 0.004
3 PAIN_ms STENT1 OR 0.87 0.45-1.68 0.680
4 PAIN_s (Intercept) OR 3.74 2.46-5.70 <0.001
5 PAIN_s TRT21 OR 0.14 0.10-0.19 <0.001
6 PAIN_s STENT1 OR 0.08 0.02-0.27 <0.001

```

```

gee_table_final %>%
  flextable() %>%
  theme_booktabs() %>%
  autofit() %>%
  align(
    j = c("Estimate", "CI", "P_value"),
    align = "center",
    part = "all"
  ) %>%
  save_as_docx(
    path = "output/pain_adj_w.docx"
  )

```

SENSITIVITY ANALYSIS 1

1 E-value for outcomes

```

# E-value explains away a specific treatment-outcome association.
# Code to compute E-value and its confidence interval
# "Default E-value (vs null)" is simply the standard output when true = 1 (implicitly)
# For "Non-null E-value (shifted)", use true = ....
# The E-value for the confidence interval was computed using the limit closest to the null value.

# General explanation
# evalues.OR(est = OR, lo = CI_lower, hi = CI_upper, rare = TRUE) # rare=TRUE means outcome prevalence

```



```

# Example: 0.48 (0.36 - 0.63)
# evalues.OR(est = 0.48,
#             lo = 0.36,
#             hi = 0.63,
#             rare = TRUE)

# If using Cox regression
# evalues.HR(est, lo, hi)

# Link for the calculator: https://www.evalue-calculator.com/evalue/

# E-value for the main treatment effect estimates

## Before covariates adjustment - significant outcomes (when adjustment include hospital time), prevalence
# 9.8% OBST          OR  3.67    2.21-6.10
# evalues.OR(est = 3.67,
#             lo = 2.21,
#             hi = 6.10,
#             rare = TRUE)

# 3.2% URCOMPINJ      OR   0.43    0.27-0.69
# 28.5% SURG          OR   0.72    0.58-0.90
# 7.3% ILEUS          OR   0.10    0.04-0.25
# 23% GU90            OR  0.20    0.16-0.26
# 10.4% REC011        OR  0.57    0.39-0.81
# 7% REC011_best      OR   0.50    0.33-0.74
# 72.6% PAIN_ms       OR  0.32    0.14-0.73

## After covariates adjustment - significant outcomes:
# 9.8% OBSTadj        OR   4.85    2.37-9.92
# 18.9% GU30adj       OR   0.22    0.09-0.54
# 72.6% PAIN_msadj    OR   0.32    0.15-0.69

# evaluate_data <- data.frame(
#   Outcome = c(
#     "OBST",
#     "URCOMPINJ",
#     "SURG",
#     "ILEUS",
#     "GU90",
#     "REC011",
#     "REC011_best",
#     "OBSTadj",
#     "GU30adj",
#     "PAIN_ms",
#     "PAIN_msadj"
#   ),
#   OR = c(
#     3.67,
#     0.43,
#     0.72,

```

```

# 0.10,
# 0.20,
# 0.57,
# 0.50,
# 4.85,
# 0.22,
# 0.32,
# 0.32
# ),
# CI_low = c(
# 2.21,
# 0.27,
# 0.58,
# 0.04,
# 0.16,
# 0.39,
# 0.33,
# 2.37,
# 0.09,
# 0.14,
# 0.15
# ),
# CI_high = c(
# 6.10,
# 0.69,
# 0.90,
# 0.25,
# 0.26,
# 0.81,
# 0.74,
# 9.92,
# 0.54,
# 0.73,
# 0.69
# )
# )

# Specify rare outcome assumption
# evaluate_data <- evaluate_data %>%
#   mutate(
#     rare = ifelse(
#       Outcome %in% c("SURG", "GU90", "GU30adj", "Pain_ms", "Pain_msadj"),
#       FALSE,
#       TRUE
#     )
#   )

# Calculate E-values
# evaluate_results <- evaluate_data %>%
#   rowwise() %>%
#   mutate(

```

```

#   OR_input = ifelse(OR < 1, 1/OR, OR),
#   CI_low_input = ifelse(OR < 1, 1/CI_high, CI_low),
#   CI_high_input = ifelse(OR < 1, 1/CI_low, CI_high),

#   E_value = list(
#     values.OR(
#       est = OR_input,
#       lo = CI_low_input,
#       hi = CI_high_input,
#       rare = rare
#     )
#   )
# ) %>%
#   ungroup()

# evaluate_table <- evaluate_results %>%
#   mutate(
#     E_value_point = sapply(
#       E_value,
#       function(x) x["E-values", "point"]
#     ),
#     E_value_CI = sapply(
#       E_value,
#       function(x) x["E-values", "lower"]
#     )
#   ) %>%
#   dplyr::select(
#     Outcome,
#     OR,
#     CI_low,
#     CI_high,
#     E_value_point,
#     E_value_CI
#   )

# evaluate_table

# evaluate_table_formatted <- evaluate_table %>%
#   mutate(
#     OR = sprintf("%.2f", OR),
#     CI = paste0(
#       sprintf("%.2f", CI_low),
#       "_",
#       sprintf("%.2f", CI_high)
#     ),
#     E_value_point = sprintf("%.2f", E_value_point),
#     E_value_CI = sprintf("%.2f", E_value_CI)
#   ) %>%
#   dplyr::select(
#     Outcome,
#     OR,
#     CI,

```

```

#     E_value_point,
#     E_value_CI
# )

# evaluate_table_formatted %>%
#   flextable() %>%
#   theme_booktabs() %>%
#   autofit() %>%
#   align(
#     j = c("OR", "CI", "E_value_point", "E_value_CI"),
#     align = "center",
#     part = "all"
#   ) %>%
#   save_as_docx(
#     path = "output/E_value_hosp.docx"
#   )

# -----

## Before covariates adjustment - significant outcomes (when adjustment include only stenting), prevalence
# 9.8% OBST                OR 3.67    2.21-6.10
# 2.2% URCOMPINJ           OR 0.43    0.27-0.69
# 28.5% SURG               OR 0.72    0.58-0.90
# 7.3% ILEUS               OR 0.10    0.04-0.25
# 23% GU90                 OR 0.20    0.16-0.26
# 10.4% RECO11             OR 0.57    0.39-0.81
# 7% RECO11_best           OR 0.50    0.33-0.74
# 39.3% RECO11_worst       OR 1.30    0.99-1.69 (NS)
# 18.9% GU30               OR 0.02    0.00-1.87 (NS)
# 4.7% PROC365             OR 0.79    0.60-1.05 (NS)

# 72.6% PAIN_ms            OR 0.32    0.14-0.73
# 37.9% PAIN_s             OR 0.20    0.15-0.26

## After covariates adjustment - significant outcomes:
# 9.8% OBSTadj             OR 3.71    2.50-5.52
# 2.2% URCOMPINJadj        OR 0.36    0.27-0.49
# 28.5% SURGadj            OR 0.63    0.50-0.81
# 7.3% ILEUSadj            OR 0.20    0.17-0.24
# 23% GU90adj              OR 0.18    0.15-0.21
# 10.4% RECO11adj          OR 0.61    0.46-0.81
# 7% RECO11_bestadj        OR 0.55    0.45-0.69
# 39.3% RECO11_worstadj    OR 1.47    1.15-1.87
# 18.9% GU30adj            OR 0.03    0.01-0.08
# 4.7% PROC365adj          OR 0.71    0.56-0.89

# 72.6% PAIN_msadj         OR 0.32    0.15-0.69
# 37.9% PAIN_sadj          OR 0.14    0.10-0.19

evaluate_data <- data.frame(
  Outcome = c(

```

```

"OBST",
"URCOMPINJ",
"SURG",
"ILEUS",
"GU90",
"REC011",
"REC011_best",
"OBSTadj",
"URCOMPINJadj",
"SURGadj",
"ILEUSadj",
"GU90adj",
"REC011adj",
"REC011_bestadj",
"REC011_worstadj",
"GU30adj",
"PROC365adj",
"PAIN_ms",
"PAIN_s",
"PAIN_msadj",
"PAIN_sadj"
),
OR = c(
  3.67,
  0.43,
  0.72,
  0.10,
  0.20,
  0.57,
  0.50,
  3.71,
  0.36,
  0.63,
  0.20,
  0.18,
  0.61,
  0.55,
  1.47,
  0.03,
  0.71,
  0.32,
  0.20,
  0.32,
  0.14
),
CI_low = c(
  2.21,
  0.27,
  0.58,
  0.04,
  0.16,
  0.39,
  0.33,

```

```

    2.50,
    0.27,
    0.50,
    0.17,
    0.15,
    0.46,
    0.45,
    1.15,
    0.01,
    0.56,
    0.14,
    0.15,
    0.15,
    0.10
  ),
  CI_high = c(
    6.10,
    0.69,
    0.90,
    0.25,
    0.26,
    0.81,
    0.74,
    5.52,
    0.49,
    0.81,
    0.24,
    0.21,
    0.81,
    0.69,
    1.87,
    0.08,
    0.89,
    0.73,
    0.26,
    0.69,
    0.19
  )
)

# Specify rare outcome assumption
evaluate_data <- evaluate_data %>%
  mutate(
    rare = ifelse(
      Outcome %in% c("SURG", "GU90", "SURGadj", "GU90adj", "REC011_worstadj", "GU30adj", "Pain_ms", "Pa
      FALSE,
      TRUE
    )
  )

# Calculate E-values

```

```

evaluate_results <- evaluate_data %>%
  rowwise() %>%
  mutate(
    OR_input = ifelse(OR < 1, 1/OR, OR),
    CI_low_input = ifelse(OR < 1, 1/CI_high, CI_low),
    CI_high_input = ifelse(OR < 1, 1/CI_low, CI_high),

    E_value = list(
      values.OR(
        est = OR_input,
        lo = CI_low_input,
        hi = CI_high_input,
        rare = rare
      )
    )
  ) %>%
  ungroup()

evaluate_table <- evaluate_results %>%
  mutate(
    E_value_point = sapply(
      E_value,
      function(x) x["E-values", "point"]
    ),
    E_value_CI = sapply(
      E_value,
      function(x) x["E-values", "lower"]
    )
  ) %>%
  dplyr::select(
    Outcome,
    OR,
    CI_low,
    CI_high,
    E_value_point,
    E_value_CI
  )

evaluate_table

# A tibble: 21 x 6
  Outcome      OR CI_low CI_high E_value_point E_value_CI
  <chr>      <dbl> <dbl> <dbl>      <dbl>      <dbl>
1 OBST         3.67  2.21   6.1         6.80       3.85
2 URCOMPINJ    0.43  0.27   0.69        4.08       2.26
3 SURG         0.72  0.58   0.9         1.64       1.29
4 ILEUS        0.1   0.04   0.25       19.5       7.46
5 GU90         0.2   0.16   0.26        3.90       3.33
6 REC011       0.57  0.39   0.81        2.90       1.77
7 REC011_best  0.5   0.33   0.74        3.41       2.04
8 OBSTadj      3.71  2.5    5.52        6.88       4.44
9 URCOMPINJadj 0.36  0.27   0.49         5         3.50
10 SURGadj     0.63  0.5    0.81        1.83       1.46

```

```

# i 11 more rows

evaluate_table_formatted <- evaluate_table %>%
  mutate(
    OR = sprintf("%.2f", OR),
    CI = paste0(
      sprintf("%.2f", CI_low),
      "-",
      sprintf("%.2f", CI_high)
    ),
    E_value_point = sprintf("%.2f", E_value_point),
    E_value_CI = sprintf("%.2f", E_value_CI)
  ) %>%
  dplyr::select(
    Outcome,
    OR,
    CI,
    E_value_point,
    E_value_CI
  )

evaluate_table_formatted %>%
  flextable() %>%
  theme_booktabs() %>%
  autofit() %>%
  align(
    j = c("OR", "CI", "E_value_point", "E_value_CI"),
    align = "center",
    part = "all"
  ) %>%
  save_as_docx(
    path = "output/E_value_stent.docx"
  )

## The E-value answers: How strong would an unmeasured confounder need to be associated with both the exposure and the outcome?
# For example: E-value = 6.88
# Interpretation: An unmeasured confounder would need to be associated with both the treatment and the outcome with an effect size at least 6.88 times that of the treatment effect to explain away the observed association.

## Rosenbaum's sensitivity analysis addresses a slightly different question: How strong would an unmeasured confounder need to be associated with both the exposure and the outcome to explain away the observed association?
# The key parameter is  $\Gamma$  (Gamma). For example:  $\Gamma = 10 \rightarrow$  very robust
# Interpretation: An unmeasured confounder would need to increase the odds of treatment assignment by 10 times to explain away the observed association.

# There is no universal rule that says "calculate E-values only for significant unadjusted outcomes" or "calculate E-values only for adjusted outcomes"
# For our study, Report E-values for adjusted treatment effects of clinically important outcomes, regardless of whether the outcomes are statistically significant.

```

1.1. Plot E-value

```

# E-value sensitivity contour plot for Obstruction risk in overlap weighting before adjustment to procedure

# OBST values

```



```

RR_est <- 3.67
RR_CI  <- 2.21

E_value_point <- RR_est + sqrt(RR_est*(RR_est-1))
E_value_CI <- RR_CI + sqrt(RR_CI*(RR_CI-1))

# Function for contour
evaluate_curve <- function(RR){

  x <- seq(RR + 0.001, 20, length.out = 5000)

  y <- RR * (x - 1)/(x - RR)

  data.frame(
    RR_EU = x,
    RR_UY = y
  )
}

curve_est <- evaluate_curve(RR_est)
curve_CI <- evaluate_curve(RR_CI)

# Create plot
evaluate_plot <- ggplot() +

  # Point estimate curve
  geom_line(
    data = curve_est,
    aes(
      x = RR_EU,
      y = RR_UY
    ),
    color = "red",
    linewidth = 0.8
  ) +

  # CI curve
  geom_line(
    data = curve_CI,
    aes(
      x = RR_EU,
      y = RR_UY
    ),
    color = "blue",
    linewidth = 0.8
  ) +

  # E-value point
  geom_point(
    aes(
      x = E_value_point,
      y = E_value_point
    )
  )

```

```

    ),
    color = "red",
    size = 2.5
) +

# CI E-value point
geom_point(
  aes(
    x = E_value_CI,
    y = E_value_CI
  ),
  color = "blue",
  size = 2.5
) +

# Labels
annotate(
  "text",
  x = E_value_point,
  y = E_value_point + 0.35,
  label = paste0(
    "E value: (",
    round(E_value_point,2),
    ", ",
    round(E_value_point,2),
    ")"
  ),
  color = "black",
  size = 4,
  hjust = 0
) +

annotate(
  "text",
  x = E_value_CI,
  y = E_value_CI - 0.25,
  label = paste0(
    "E value (CI): (",
    round(E_value_CI,2),
    ", ",
    round(E_value_CI,2),
    ")"
  ),
  color = "black",
  size = 4,
  hjust = 0
) +

scale_x_continuous(
  limits = c(1,20),
  breaks = seq(5,20,5)
) +

```

```

scale_y_continuous(
  limits = c(1,20),
  breaks = seq(0,20,5)
) +

labs(
  x = "Risk ratio for exposure-confounder relationship",
  y = "Risk ratio for confounder-outcome relationship"
) +

theme_gray(base_size = 14)

# evaluate_plot

# ggsave(
#   filename = "figures/Evalue_OBST_contour.png",
#   plot = evaluate_plot,
#   width = 8,
#   height = 9,
#   dpi = 300
# )

# Figure. E-value demonstrating required strength of unmeasured confounder to explain observed associati.

# E-value sensitivity contour plot for Obstruction risk in overlap weighting after adjustment to procedu

# OBST values
RR_est_a <- 3.71
RR_CI_a  <- 2.50

E_value_point_a <- RR_est_a + sqrt(RR_est_a*(RR_est_a-1))
E_value_CI_a <- RR_CI_a + sqrt(RR_CI_a*(RR_CI_a-1))

# Function for contour
evaluate_curve <- function(RR){

  x <- seq(RR + 0.001, 20, length.out = 5000)

  y <- RR * (x - 1)/(x - RR)

  data.frame(
    RR_EU = x,
    RR_UY = y
  )
}

curve_est_a <- evaluate_curve(RR_est_a)
curve_CI_a  <- evaluate_curve(RR_CI_a)

```

```

# Create plot
evaluate_plot_a <- ggplot() +

# Point estimate curve
geom_line(
  data = curve_est_a,
  aes(
    x = RR_EU,
    y = RR_UY
  ),
  color = "red",
  linewidth = 0.8
) +

# CI curve
geom_line(
  data = curve_CI_a,
  aes(
    x = RR_EU,
    y = RR_UY
  ),
  color = "blue",
  linewidth = 0.8
) +

# E-value point
geom_point(
  aes(
    x = E_value_point_a,
    y = E_value_point_a
  ),
  color = "red",
  size = 2.5
) +

# CI E-value point
geom_point(
  aes(
    x = E_value_CI_a,
    y = E_value_CI_a
  ),
  color = "blue",
  size = 2.5
) +

# Labels
annotate(
  "text",
  x = E_value_point_a,
  y = E_value_point_a + 0.35,
  label = paste0(
    "E value: (",
    round(E_value_point_a,2),

```

```

      ", ",
      round(E_value_point_a,2),
      ")"
    ),
    color = "black",
    size = 4,
    hjust = 0
  ) +

  annotate(
    "text",
    x = E_value_CI_a,
    y = E_value_CI_a - 0.25,
    label = paste0(
      "E value (CI): (",
      round(E_value_CI_a,2),
      ", ",
      round(E_value_CI_a,2),
      ")"
    ),
    color = "black",
    size = 4,
    hjust = 0
  ) +

  scale_x_continuous(
    limits = c(1,20),
    breaks = seq(5,20,5)
  ) +

  scale_y_continuous(
    limits = c(1,20),
    breaks = seq(0,20,5)
  ) +

  labs(
    x = "Risk ratio for exposure-confounder relationship",
    y = "Risk ratio for confounder-outcome relationship"
  ) +

  theme_gray(base_size = 14)

# evaluate_plot_a

# ggsave(
#   filename = "figures/Evalue_OBST_adj_contour.png",
#   plot = evaluate_plot_a,
#   width = 8,
#   height = 9,
#   dpi = 300
# )

```

```

#-----
# Combine plot and risk table using patchwork
#-----
combined_plot <- (evaluate_plot | evaluate_plot_a) +
  plot_annotation(
    tag_levels = "A"
  )

#-----
# Save the figure
#-----
ggsave(
  filename = "figures/Evalue_OBST.png",
  plot = combined_plot,
  width = 20,
  height = 10,
  dpi = 300
)

```

Warning: Removed 184 rows containing missing values or values outside the scale range (``geom_line()``).

Warning: Removed 42 rows containing missing values or values outside the scale range (``geom_line()``).

Warning: Removed 190 rows containing missing values or values outside the scale range (``geom_line()``).

Warning: Removed 61 rows containing missing values or values outside the scale range (``geom_line()``).

SENSITIVITY ANALYSIS 2

1 Do matching 1:1 without replacement

```

# Fit matching model using propensity score (calculated inside MatchIt): 1:1 matching

set.seed(123) # Fixes the sequence of "random" operations = the same individuals are matched every time

m.out <- matchit(
  TRT2 ~ AGS + YEARS + GENDER + BMI + PREOPRNPLN + HISTOINJ +
    SURGIND2 + PREOPVURGRADE + VURSIDE + UDR7M + VURPHASE +
    PREOPEPISODE1 + PREOPTYPEU1 + PREOPRNSCR + SURGTECH2,
  data = Out,
  method = "nearest",
  ratio = 1, # 1:1 matching (1 treated : 1 control)
  distance = "logit",
  # caliper should be a numeric value in MatchIt
)

```

```

# This means: 0.2 SD of the logit of PS (MatchIt interprets correctly)
caliper = 0.2 # A caliper = maximum allowed distance between treated and control PS = 0.2 (Austin 2018)
) # When the variance of the linear propensity score in the treatment group is twice as large as that in the control group

mdata <- match.data(
  m.out,
  weights = "weights_psm" # give a new name for the matching weights because the original dataset Out a
)
# Extract the matched dataset with columns: distance(PS), weights, subclass
# weights = how much each matched subject contributes in analysis
# In 1:1 matching, weights usually = 1
# you can ignore weights in further analysis, but using them is safer and more standard
# glm(outcome ~ TRT2, data = mdata, weights = weights) (standard) or glm(outcome ~ TRT2, data = mdata)

readr::write_csv(mdata, "data/matched_data.csv")
# write.csv() = base R, simple, works everywhere: write.csv(mdata, file = "data/matched_data.csv", row.names = FALSE)
# write_csv() = modern, faster, cleaner, recommended for research workflows

# Table for balance
# This gives: SMD before and after matching, and flagged imbalance (>0.1)
balm <- cobalt::bal.tab(m.out, un = TRUE, m.threshold = 0.1)
# un=TRUE shows imbalance before AND after matching, un=FALSE shows only after matching
balm

```

Balance Measures

	Type	Diff.Un	Diff.Adj	M.Threshold
distance	Distance	1.5815	0.1163	
AGS	Contin.	0.3507	0.0642	Balanced, <0.1
YEARS	Contin.	-0.4776	0.0109	Balanced, <0.1
GENDER	Binary	-0.0471	0.0206	Balanced, <0.1
BMI	Contin.	-0.8345	0.0397	Balanced, <0.1
PREOPRNPLN	Binary	-0.2699	-0.0103	Balanced, <0.1
HISTOINJ	Binary	0.0198	-0.0206	Balanced, <0.1
SURGIND2_1	Binary	0.0856	0.0206	Balanced, <0.1
SURGIND2_2	Binary	0.0595	0.0103	Balanced, <0.1
SURGIND2_3	Binary	-0.0089	-0.0103	Balanced, <0.1
SURGIND2_4	Binary	-0.0465	0.0000	Balanced, <0.1
SURGIND2_5	Binary	-0.0897	-0.0206	Balanced, <0.1
PREOPVURGRADE_3	Binary	-0.0140	-0.0206	Balanced, <0.1
PREOPVURGRADE_4	Binary	0.0802	0.0619	Balanced, <0.1
PREOPVURGRADE_5	Binary	-0.0662	-0.0412	Balanced, <0.1
VURSIDE	Binary	0.0247	-0.0412	Balanced, <0.1
UDR7M	Binary	-0.1399	-0.0309	Balanced, <0.1
VURPHASE_1	Binary	-0.0383	0.0000	Balanced, <0.1
VURPHASE_2	Binary	0.3112	-0.0103	Balanced, <0.1
VURPHASE_3	Binary	-0.2729	0.0103	Balanced, <0.1
PREOPEPISODE1_1	Binary	-0.0440	-0.0206	Balanced, <0.1
PREOPEPISODE1_2	Binary	-0.1765	-0.0309	Balanced, <0.1
PREOPEPISODE1_3	Binary	0.2205	0.0515	Balanced, <0.1
PREOPTYPEU1	Binary	0.0309	-0.0103	Balanced, <0.1
PREOPRNSCR	Binary	-0.2674	-0.0309	Balanced, <0.1

SURGTECH2_1	Binary	0.2494	0.0412	Balanced, <0.1
SURGTECH2_2	Binary	-0.0629	0.0000	Balanced, <0.1
SURGTECH2_3	Binary	-0.1866	-0.0412	Balanced, <0.1

Balance tally for mean differences

	count
Balanced, <0.1	27
Not Balanced, >0.1	0

Variable with the greatest mean difference

Variable	Diff.Adj	M.Threshold
AGS	0.0642	Balanced, <0.1

Sample sizes

	Control	Treated
All	197	215
Matched	97	97
Unmatched	100	118

```
bal_dfm <- as.data.frame(balm$Balance)
```

```
smd_tablem <- bal_dfm %>%
```

```
  mutate(
    Variable = rownames(bal_dfm),
    SMD_Before = round(abs(Diff.Un), 3),
    SMD_After = round(abs(Diff.Adj), 3)
  ) %>%
```

```
  select(Variable, SMD_Before, SMD_After) # SMDs for each covariate individually, before and after match
```

```
smd_tablem %>%
```

```
  flextable() %>%
```

```
  save_as_docx(path = "output/SMD_psm.docx")
```

```
# how much effective sample size remains (overall ESS: calculated manually)?
```

```
summary(m.out$weights)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.4709	1.0000	1.0000

```
bal.tab(m.out)$Effective.Sample.Size
```

```
NULL
```

```
sum(m.out$weights)^2 / sum(m.out$weights^2)
```

```
[1] 194
```

```
# Figure for balance - Love plot
```

```
# Review guidelines to decide on plotting Standardized or Absolute Mean Differences
```

```
plm <- love.plot(
```

```
  m.out,
```



```

threshold = 0.1,
abs = TRUE, # As SMD sign is meaningless for balance, show only absolute SMDs using abs = TRUE (the c
stats = "mean.diffs", # forces SMD instead of raw differences
var.order = "unadjusted", # Orders variables based on imbalance before matching
drop.distance = TRUE, # To exclude the distance from the plot
sample.names = c("Before Matching", "After Matching"),
  var.names = c(
    AGS = "Age at surgery",
    YEARS = "Year of surgery",
    BMI = "Body mass index",
    GENDER = "Female gender",
    PREOPRNPLN = "Preop pyelonephritis",
    HISTOINJ = "History of injection",
    VURSIDE = "Right-sided VUR",
    UDR7M = "Ureteral dilation 7mm",
    PREOPTYPEU1 = "Preop febrile UTIs",
    PREOPRNSCR = "Preop renal scarring",
    VURPHASE_1 = "Reflux: voiding phase",
    VURPHASE_2 = "Reflux: filling phase",
    VURPHASE_3 = "Reflux: filling + voiding",
    SURGTECH2_1 = "Surgeon experience 100 operations",
    SURGTECH2_2 = "Surgeon experience 50-100 operations",
    SURGTECH2_3 = "Surgeon experience <50 operations",
    PREOPEPISODE1_1 = "Preop UTI episodes: 1",
    PREOPEPISODE1_2 = "Preop UTI episodes: 2",
    PREOPEPISODE1_3 = "Preop UTI episodes: 3",
    SURGIND2_1 = "Indication: breakthrough UTI",
    SURGIND2_2 = "Indication: high-grade reflux (III-V)",
    SURGIND2_3 = "Indication: poor compliance to medical treatment",
    SURGIND2_4 = "Indication: ineffective/poorly tolerated CAP",
    SURGIND2_5 = "Indication: other",
    PREOPVURGRADE_3 = "Preop VUR grade: III",
    PREOPVURGRADE_4 = "Preop VUR grade: IV",
    PREOPVURGRADE_5 = "Preop VUR grade: V"
  )
)

plm <- plm +
  labs(
    x = "Absolute Standardized Differences",
    subtitle = "Matched population"
  ) +
  theme(
    plot.margin = margin(10, 10, 10, 10),
    axis.text = element_text(size = 16),
    axis.title = element_text(size = 16),
    plot.title = element_text(size = 16),
    legend.text = element_text(size = 16),
    legend.title = element_text(size = 16),
    axis.ticks = element_line(size = 0.5, color = "black"),
    axis.line = element_line(size = 0.5, color = "black"),
    panel.background = element_blank(),
    panel.grid.major = element_blank(),

```

```

    panel.grid.minor = element_blank(),
    panel.border = element_blank(),
    axis.ticks.length = unit(0.3, "cm"),
    panel.spacing = unit(2, "lines")
  )

ggsave ("figures/love_plot_m.png", plot = plm, width = 12, height = 10, dpi = 300)

# To save the plot with white background, if requested for publication
# ggsave(
#   "figures/love_plot.png",
#   plot = pl,
#   width = 12,
#   height = 10,
#   dpi = 300,
#   bg = "white"
# )

# Note: A propensity score should include: only baseline variables measured before treatment assignment

# "Distance" in the Love plot is the propensity score itself, included to show how well matching balance
# Reviewers mainly check:
## All covariates < 0.1 after matching
## Distance reduced substantially
## Visual improvement in Love plot
# many papers exclude distance because: It is not a confounder, it is a derived variable, it often has v

# Compute mean SMD
smd_df_m <- balm$Balance
mean_smd_before_m <- mean(abs(smd_df_m$Diff.Un), na.rm = TRUE) # calculate mean SMD before matching
mean_smd_after_m <- mean(abs(smd_df_m$Diff.Adj), na.rm = TRUE) # calculate mean SMD before matching
mean_smd_m <- data.frame(
  Mean_SMD_Before = mean_smd_before_m,
  Mean_SMD_After = mean_smd_after_m
)

mean_smd_m

  Mean_SMD_Before Mean_SMD_After
1      0.2163242      0.02813468

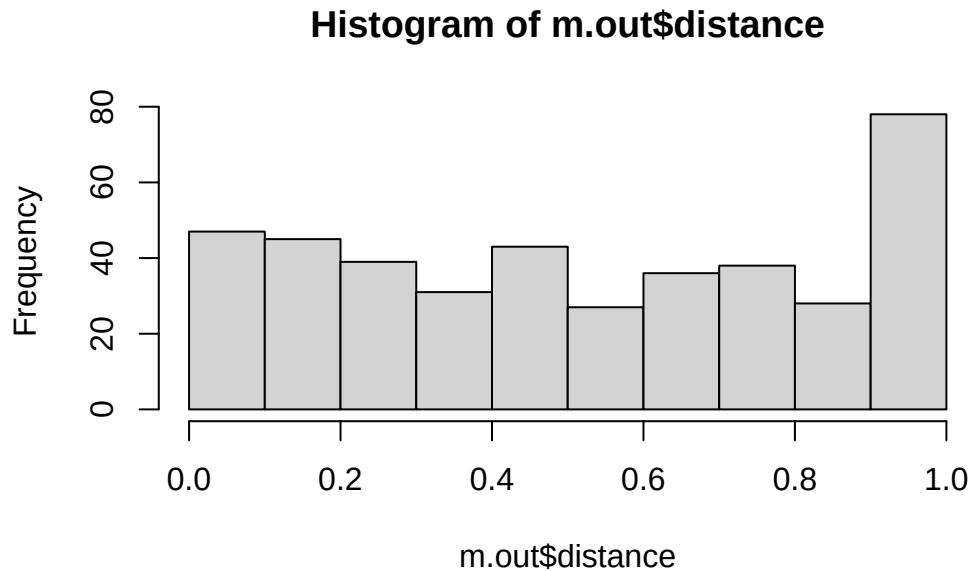
# Best reporting practices
# "All covariates had SMD < 0.1 (threshold) after matching": Covariate balance was assessed using stand
# (optionally) "Mean absolute SMD decreased from X to Y": It is acceptable as a summary metric but not a

# propensity score distribution: MatchIt does not store the propensity scores in m.out$ps. Instead, they
summary(m.out$distance)

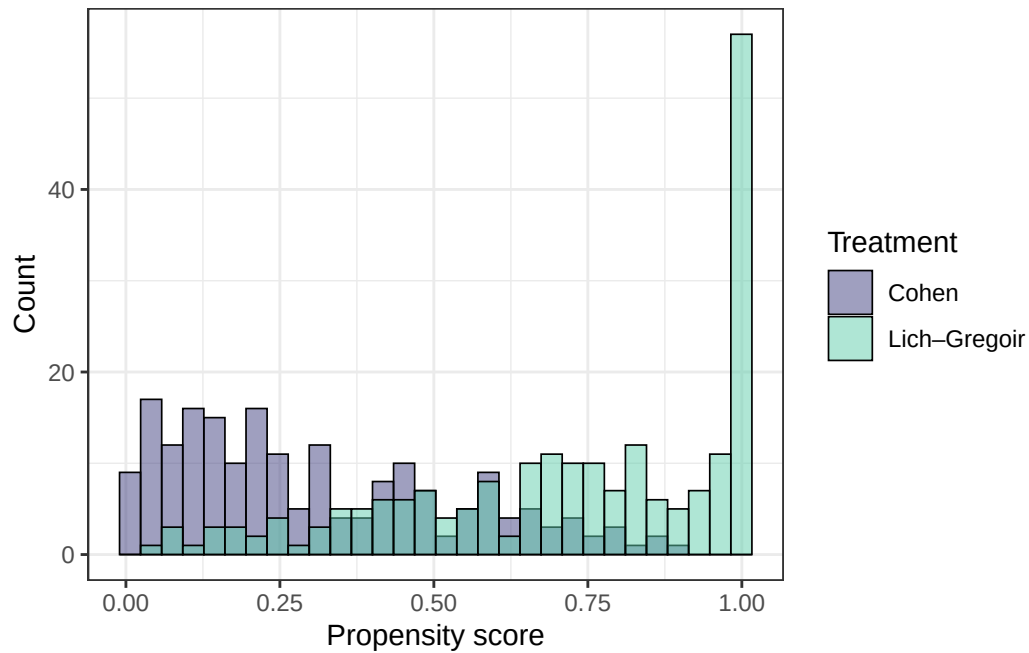
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.006799	0.218309	0.504244	0.521845	0.803450	0.999248

```
hist(m.out$distance)
```



```
ggplot(
  data.frame(
    ps = m.out$distance,
    Treatment = factor(Out$Treatment)
  ),
  aes(ps, fill = Treatment)
) +
  geom_histogram(alpha = 0.5, bins = 30, position = "identity", color = "black", linewidth = 0.3) +
  scale_fill_viridis_d(
    option = "mako",
    begin = 0.3,
    end = 0.8,
    labels = c("Cohen", "Lich-Gregoir")
  ) +
  # scale_fill_manual(
  #   values = c("#009E73", "#CC79A7") # Green vs purple fills (colorblind-friendly), c("#BDBDBD", "#2E86C1")
  # ) +
  labs(
    x = "Propensity score",
    y = "Count",
    fill = "Treatment"
  ) +
  theme_bw()
```



```
ggsave("figures/propensity_psm.png",
  width = 12,
  height = 10,
  dpi = 300)
```

```
# Distribution of weights: histogram and density
```

```
## Histogram
```

```
# summary(m.out$weights)
# png("figures/weights_histo_m.png",
#   width = 6,
#   height = 4,
#   units = "in",
#   res = 300)
```

```
# hist(
#   m.out$weights,
#   breaks = 30,
#   col = "lightblue",
#   border = "black",
#   main = "Distribution of Weights",
#   xlab = "Weight"
# )
# dev.off()
```

```
## Density
```

```
# owd<- ggplot(mdata,
#   aes(weights_psm,
#     colour = Treatment,
```

```

#           fill = Treatment)) +
#   geom_density(alpha = 0.3, linewidth = 1) +
#   labs(
#     x = "weight",
#     y = "Density"
#   ) +
#   theme_classic()
# ggsave ("figures/weights_dens_m.png", plot = owd, width = 12, height = 8, dpi = 300)

# Distributional balance for propensity score

# Before-After matching
p_psm <- bal.plot(
  m.out,
  var.name = "distance",
  type = "density",
  which = "both"      # "unadjusted" = unmatched (before matching); "adjusted" = matched (after matching)
)

p_psm <- p_psm +
  scale_color_viridis_d(
    option = "mako",
    begin = 0.3,
    end = 0.8,
    labels = c("Cohen group", "Lich-Gregoir group")
  ) +
  scale_fill_viridis_d(
    option = "mako",
    begin = 0.3,
    end = 0.8,
    labels = c("Cohen group", "Lich-Gregoir group")
  )

Scale for fill is already present.
Adding another scale for fill, which will replace the existing scale.

ggsave(
  "figures/PS_density_before_after_m.png",
  p_psm,
  width = 10,
  height = 6,
  dpi = 300
)

```

2 Descriptive analysis: AFTER matching

```

# mdata is the dataset with added weights_psm

# Convert variables to appropriate data types for analysis
mdata <- mdata %>%

```

```

mutate(
  across(c(PID, QCODE), as.character),

  across(c(AGS, YEARS, BLDLOSS, BMI, CATHPERIOD, FUPM, HOSPTIME, OPRTIME,
            POSTUTIFUP, POSTUTI_T), as.numeric),

  across(c(REC011, REC011_best, REC011_worst, ANASTOMO, BLDCOMPLEAK, BLDCOMPRETN,
            BLDCOMPVOID, BLDSPASM, CATH, CENTER, CLAVIENCLASS, EPIDURAL, GENDER,
            GU30, GU90N, GUER30, GUER90, HEMAT,
            HISTOINJ, ILEUS, INCCOMPBLEED, INCCOMPINF, INCCOMPSCAR, NARC,
            NONARC, PAIN, PREOPEPISODE1, PREOPRNPLN, PREOPRNSCR, PREOPTYPEU1,
            PREOPVURGRADE, PROC365, STENT, SURGIND, SURGTECH2, TRT2, UDR7M,
            URCOMPVCUR, URCOMPINJ, URCOMPSTEN, POSTUTI, VURPHASE,
            VURSIDE, RECGRADE_MAX, SURGIND2, LUTD, OBST, SURG, GU90,
            CATHPERIOD_G, ANY, MINOR, MAJOR), as.factor)
)

str(mdata) # 194 obs. of 71 variables

```

```

Classes 'matchdata' and 'data.frame': 194 obs. of 71 variables:
 $ PID      : chr  "P1" "P3" "P6" "P7" ...
 $ QCODE    : chr  "102" "3" "8" "9" ...
 $ REC011   : Factor w/ 2 levels "0","1": 1 1 2 2 1 2 1 1 1 1 ...
 $ AGS      : num  5 7 9 10 8 11 5 8 8 7 ...
 $ YEARS    : num  2011 2017 2018 2015 2017 ...
 $ ANASTOMO : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
 $ BLDCOMPLEAK : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
 $ BLDCOMPRETN : Factor w/ 2 levels "0","1": 1 2 2 1 2 2 1 2 1 1 ...
 $ BLDCOMPVOID : Factor w/ 2 levels "0","1": 2 1 1 1 1 1 1 1 1 1 ...
 $ BLDLOSS   : num  70 150 190 75 100 55 50 80 80 50 ...
 $ BLDSPASM  : Factor w/ 2 levels "0","1": 1 2 2 1 2 1 2 1 2 1 ...
 $ BMI       : num  13.5 12 15.8 15.3 15.3 15.3 13.7 15 15 14.4 ...
 $ CATH      : Factor w/ 2 levels "0","1": 2 2 2 2 2 2 2 2 2 2 ...
 $ CATHPERIOD : num  2 5 7 4 5 7 1 5 5 4 ...
 $ CENTER    : Factor w/ 4 levels "1","2","3","5": 1 1 1 1 1 1 1 1 1 1 ...
 $ CLAVIENCLASS : Factor w/ 4 levels "1","2","3","4": 2 2 2 2 1 3 1 2 1 2 ...
 $ EPIDURAL   : Factor w/ 2 levels "0","1": 1 1 1 2 1 1 1 1 1 1 ...
 $ FUPM       : num  24 44 36 36 37 36 24 36 24 24 ...
 $ GENDER     : Factor w/ 2 levels "0","1": 1 2 1 1 2 1 2 2 2 1 ...
 $ GU30       : Factor w/ 2 levels "0","1": 1 1 1 2 1 1 1 1 1 1 ...
 $ GU90N      : Factor w/ 3 levels "0","1","2": 1 2 2 2 1 1 1 1 1 1 ...
 $ GUER30     : Factor w/ 2 levels "0","1": 1 1 1 2 1 1 1 1 1 1 ...
 $ GUER90     : Factor w/ 2 levels "0","1": 2 1 1 2 1 1 1 1 1 1 ...
 $ HEMAT      : Factor w/ 2 levels "0","1": 1 2 2 2 1 1 1 1 1 1 ...
 $ HISTOINJ   : Factor w/ 2 levels "0","1": 2 1 1 1 2 1 1 1 1 1 ...
 $ HOSPTIME   : num  2 5 5 8 5 4 2 5 5 5 ...
 $ ILEUS      : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
 $ INCCOMPBLEED : Factor w/ 2 levels "0","1": 2 2 1 1 1 1 1 2 1 1 ...
 $ INCCOMPINF  : Factor w/ 2 levels "0","1": 1 1 2 1 1 1 1 1 1 1 ...
 $ INCCOMPSCAR : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
 $ NARC       : Factor w/ 2 levels "0","1": 1 2 2 1 1 1 1 1 1 1 ...
 $ NONARC     : Factor w/ 2 levels "0","1": 2 2 2 2 2 2 2 2 2 2 ...
 $ OPRTIME    : num  76 80 75 90 88 80 68 75 78 76 ...

```

```

$ PAIN          : Factor w/ 4 levels "0","1","2","3": 2 4 3 3 2 2 2 2 2 2 ...
$ PREOPEPISODE1: Factor w/ 3 levels "1","2","3": 2 3 3 3 2 3 3 3 3 3 ...
$ PREOPRNPLN    : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
$ PREOPRNSCR    : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
$ PREOPTYPEU1   : Factor w/ 2 levels "0","1": 2 2 2 2 2 2 2 2 2 2 ...
$ PREOPVURGRADE: Factor w/ 3 levels "3","4","5": 3 3 3 2 2 2 3 3 2 3 ...
$ PROC365       : Factor w/ 2 levels "0","1": 1 1 1 2 1 1 1 1 1 1 ...
$ STENT         : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 2 1 1 1 ...
$ SURGIND       : Factor w/ 11 levels "1","2","3","4",...: 3 3 4 4 3 3 4 3 3 4 ...
$ SURGTECH2     : Factor w/ 3 levels "1","2","3": 3 2 2 2 2 2 2 2 2 2 ...
$ TRT2          : Factor w/ 2 levels "0","1": 2 2 2 2 2 2 2 2 2 2 ...
$ UDR7M         : Factor w/ 2 levels "0","1": 2 2 2 2 2 2 2 2 2 2 ...
$ URCOMPCVUR    : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
$ URCOMPINJ     : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
$ URCOMPOBST    : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
$ URCOMPSTEN    : Factor w/ 2 levels "0","1": 1 1 1 1 1 2 1 1 1 2 ...
$ POSTUTI       : Factor w/ 2 levels "0","1": 2 1 1 2 1 2 1 1 1 1 ...
$ POSTUTIFUP    : num 1 44 36 1 37 1 24 36 24 24 ...
$ POSTUTI_T     : num 1 0 0 1 0 1 0 0 0 0 ...
$ VURPHASE      : Factor w/ 3 levels "1","2","3": 1 3 1 1 1 1 1 1 1 1 ...
$ VURSIDE       : Factor w/ 2 levels "0","1": 1 2 2 2 1 2 1 2 1 1 ...
$ RECGRADE_MAX  : Factor w/ 6 levels "0","1","2","3",...: 2 1 3 3 2 3 2 2 1 2 ...
$ SURGIND2      : Factor w/ 5 levels "1","2","3","4",...: 2 2 1 1 2 2 1 2 2 1 ...
$ LUTD          : Factor w/ 2 levels "0","1": 2 2 2 1 2 2 1 2 1 1 ...
$ OBST          : Factor w/ 2 levels "0","1": 1 1 1 1 1 2 1 1 1 2 ...
$ SURG          : Factor w/ 2 levels "0","1": 2 2 2 1 1 1 1 2 1 1 ...
$ GU90          : Factor w/ 2 levels "0","1": 1 2 2 2 1 1 1 1 1 1 ...
$ CATHPERIOD_G  : Factor w/ 3 levels "Early removal 3 days",...: 1 2 2 2 2 2 1 2 2 2 ...
$ REC011_best   : Factor w/ 2 levels "0","1": 1 1 2 2 1 2 1 1 1 1 ...
$ REC011_worst  : Factor w/ 2 levels "0","1": 1 1 2 2 1 2 1 1 1 1 ...
$ ANY           : Factor w/ 2 levels "0","1": 2 2 2 1 2 2 1 2 1 2 ...
$ MINOR         : Factor w/ 2 levels "0","1": 2 2 2 1 2 2 1 2 1 1 ...
$ MAJOR         : Factor w/ 2 levels "0","1": 1 1 1 1 1 2 1 1 1 2 ...
$ Treatment     : Factor w/ 2 levels "Cohen","Lich-Gregoir": 2 2 2 2 2 2 2 2 2 2 ...
$ weights       : num 0.578 0.393 0.577 0.26 0.282 ...
$ distance      : num 0.422 0.607 0.423 0.74 0.718 ...
$ weights_psm   : num 1 1 1 1 1 1 1 1 1 1 ...
$ subclass      : Factor w/ 97 levels "1","2","3","4",...: 1 2 3 4 5 6 7 8 9 10 ...

```

```
# Matched baseline characteristics
```

```
table1m<- mdata %>%
```

```
  select(YEARS, AGS, GENDER, BMI, HISTOINJ, SURGIND2, PREOPVURGRADE, VURSIDE, UDR7M,
         VURPHASE, PREOPTYPEU1, PREOPEPISODE1, PREOPRNPLN, PREOPRNSCR,
         SURGTECH2, CENTER, TRT2
```

```
) %>%
```

```
tbl_summary(
```

```
  by = TRT2, # for stratification
```

```
  statistic = list(
```

```
    all_continuous() ~ "{median} ({p25}-{p75})",
```

```
    all_categorical() ~ "{n} ({p}%)"
```

```
),
```

```
  digits = all_categorical() ~ c(0, 1), # n = 0 decimals, p = 1 decimal
```

```
  missing = "ifany", # shows missing only when present
```

```
  missing_text = "Missing"
```

```

) %>%
add_overall() %>%
# add_p(pvalue_fun = ~style_pvalue(.x, digits = 3)) %>% # add_p() is not informative after weighting
bold_labels()

table1m <- table1m %>%
  modify_caption(
    "Baseline characteristics after 1:1 propensity score matching"
  )

table1m %>%
  as_flex_table() %>%
  flextable::save_as_docx(path = "output/baseline_m.docx")

# Matched procedural factors
table2m <- mdata %>%
  select(TRT2, OPRTIME, STENT, CATH, CATHPERIOD, CATHPERIOD_G, HOSPTIME) %>%
  tbl_summary(
    by = TRT2, # for stratification
    statistic = list(
      all_continuous() ~ "{median} ({p25}-{p75})", # or ~ "{mean} ± {sd}",
      all_categorical() ~ "{n} ({p}%)"
    ),
    digits = all_categorical() ~ c(0, 1), # n = 0 decimals, p = 1 decimal
    missing = "ifany",
    missing_text = "Missing"
  ) %>%
  add_overall() %>%
  add_p(pvalue_fun = ~style_pvalue(.x, digits = 3)) %>% # 3 decimals
  bold_labels()

table2m

table2m %>%
  as_flex_table() %>%
  flextable::save_as_docx(path = "output/procedural_m.docx")

# Plot catheterization distribution: better visualization for weighted causal analysis: weighted density

p_m <- ggplot(
  mdata,
  aes(
    x = CATHPERIOD,
    weight = weights_psm, # all weights = 1 so they have no effect on the density estimate.
    fill = factor(
      TRT2,
      levels = c(0, 1),
      labels = c("Cohen Group", "Lich-Gregoir Group")
    )
  )
) +

```


Characteristic	Overall N = 194 ¹	0 N = 97 ¹	1 N = 97 ¹	p-value ²
OPRTIME	88 (70-135)	130 (100-159)	70 (68-80)	<0.001
STENT				0.127
0	130 (67.0%)	60 (61.9%)	70 (72.2%)	
1	64 (33.0%)	37 (38.1%)	27 (27.8%)	
CATH				>0.999
0	4 (2.1%)	2 (2.1%)	2 (2.1%)	
1	190 (97.9%)	95 (97.9%)	95 (97.9%)	
CATHPERIOD	4 (2-5)	4 (4-5)	3 (1-5)	<0.001
CATHPERIOD_G				<0.001
Early removal 3 days	63 (32.5%)	8 (8.2%)	55 (56.7%)	
Standard removal 4-7 days	94 (48.5%)	72 (74.2%)	22 (22.7%)	
Delayed removal >7 days	37 (19.1%)	17 (17.5%)	20 (20.6%)	
HOSPTIME	5.00 (3.00-6.00)	5.00 (5.00-6.00)	3.00 (2.00-5.00)	<0.001

¹Median (Q1-Q3); n (%)

²Wilcoxon rank sum test; Pearson's Chi-squared test; Fisher's exact test

```

scale_fill_viridis_d(option = "mako", begin = 0.3, end = 0.8) +
geom_density(alpha = 0.4) +
labs(
  x = "Catheterization Duration (days) After Matching ",
  y = "Density",
  fill = "Treatment Group"
) +
theme_bw() + #optional
theme(
  panel.grid = element_blank() # Removes all major and minor gridlines
)

ggsave(filename = "figures/catheterization_m.png",
  plot = p_m,
  width = 8, # width in inches
  height = 6, # height in inches
  dpi = 300) # resolution

# Outcomes
table3m<- mdata %>%
  select(TRT2, FUPM, LUTD, OBST, URCOMPINJ, SURG, ILEUS, CLAVIENCLASS, URCOMPVUR,
    GU30, GU90, GU90N, GUER30, GUER90, PROC365,
    REC011, REC011_best, REC011_worst, RECGRADE_MAX,
    BLDLOSS, BLDSPASM, HEMAT,
    POSTUTI) %>%
tbl_summary(
  by = TRT2, # for stratification
  statistic = list(
    all_continuous() ~ "{median} ({p25}-{p75})", # or ~ "{mean} ± {sd}",
    all_categorical() ~ "{n} ({p}%)")

```

```

    ),
    digits = all_categorical() ~ c(0, 1), # n = 0 decimals, p = 1 decimal
    missing = "ifany",
    missing_text = "Missing"
  ) %>%
  add_overall() %>%
  add_p(pvalue_fun = ~style_pvalue(.x, digits = 3)) %>% # 3 decimals
  bold_labels()

table3m

table3m %>%
  as_flex_table() %>%
  flextable::save_as_docx(path = "output/outcomes_m.docx")

##### Postoperative UTI outcome as time-to-event outcome: Weighted survival curve
fit_psm <- survfit(
  Surv(POSTUTIFUP, POSTUTI == 1) ~ Treatment,
  data = mdata
)

logrank_test <- survdiff(
  Surv(POSTUTIFUP, POSTUTI == 1) ~ TRT2,
  data = mdata
)

p_value <- 1 - pchisq(
  logrank_test$chisq,
  df = length(logrank_test$n) - 1
)

p_label <- paste0("Log-rank test p = ", format.pval(p_value, digits = 3, eps = 0.001))

p_u_psm <- ggsurvplot(
  fit_psm,
  data = mdata,
  fun = "event",
  pval = p_label,
  pval.size = 4,
  pval.coord = c(0, 0.9), # c(x,y) according to my x and y scale
  conf.int = TRUE,
  risk.table = TRUE,
  risk.table.y.text = FALSE, # to remove text from y-axis in the risk table
  risk.table.fontsize = 4, # controls numbers in table
  xlim = c(0, 135),
  break.time.by = 12,
  xlab = "Follow-up time",
  ylab = "Cumulative incidence of postoperative symptomatic UTI",
  palette = c("#ADD8E6", "#00008B"),
  legend.title = "Treatment",
  legend.labs = c("Cohen Group", "Lich-Gregoir Group"),
  ylim = c(0, 1)
)

```

Characteristic	Overall N = 194 ¹	0 N = 97 ¹	1 N = 97 ¹	p-value ²
FUPM	27 (24-37)	30 (24-38)	25 (24-36)	0.048
LUTD				0.460
0	158 (81.4%)	81 (83.5%)	77 (79.4%)	
1	36 (18.6%)	16 (16.5%)	20 (20.6%)	
OBST				0.059
0	174 (89.7%)	91 (93.8%)	83 (85.6%)	
1	20 (10.3%)	6 (6.2%)	14 (14.4%)	
URCOMPINJ				0.621
0	190 (97.9%)	94 (96.9%)	96 (99.0%)	
1	4 (2.1%)	3 (3.1%)	1 (1.0%)	
SURG				0.749
0	140 (72.2%)	69 (71.1%)	71 (73.2%)	
1	54 (27.8%)	28 (28.9%)	26 (26.8%)	
ILEUS				<0.001
0	180 (92.8%)	84 (86.6%)	96 (99.0%)	
1	14 (7.2%)	13 (13.4%)	1 (1.0%)	
CLAVIENCLASS				0.013
1	93 (60.0%)	56 (70.9%)	37 (48.7%)	
2	42 (27.1%)	14 (17.7%)	28 (36.8%)	
3	19 (12.3%)	9 (11.4%)	10 (13.2%)	
4	1 (0.6%)	0 (0.0%)	1 (1.3%)	
Missing	39	18	21	
URCOMPVCUR				0.026
0	178 (97.3%)	84 (94.4%)	94 (100.0%)	
1	5 (2.7%)	5 (5.6%)	0 (0.0%)	
Missing	11	8	3	
GU30				<0.001
0	156 (80.4%)	63 (64.9%)	93 (95.9%)	
1	38 (19.6%)	34 (35.1%)	4 (4.1%)	
GU90				<0.001
0	149 (76.8%)	64 (66.0%)	85 (87.6%)	
1	45 (23.2%)	33 (34.0%)	12 (12.4%)	
GU90N				0.001
0	114 (58.8%)	47 (48.5%)	67 (69.1%)	
1	45 (23.2%)	33 (34.0%)	12 (12.4%)	
2	35 (18.0%)	17 (17.5%)	18 (18.6%)	
GUER30				<0.001
0	137 (70.6%)	46 (47.4%)	91 (93.8%)	
1	57 (29.4%)	51 (52.6%)	6 (6.2%)	
GUER90				<0.001
0	130 (67.0%)	44 (45.4%)	86 (88.7%)	
1	64 (33.0%)	53 (54.6%)	11 (11.3%)	
PROC365				>0.999
0	184 (94.8%)	92 (94.8%)	92 (94.8%)	
1	10 (5.2%)	5 (5.2%)	5 (5.2%)	
REC011				0.563
0	125 (91.2%)	62 (89.9%)	63 (92.6%)	
1	12 (8.8%)	7 (10.1%)	5 (7.4%)	
Missing	57	28	29	
REC011_best				0.551

```

)

#-----
# Customize the plot
#-----
p_u_psm$plot <- p_u_psm$plot +
  scale_x_continuous(
    limits = c(0, 135),
    breaks = seq(0, 135, 12)
  ) +
  scale_y_continuous(
    labels = function(x) paste0(round(x * 100, 1), "%"),
    limits = c(0, 1)
  ) +
  theme_classic() + # Many clinical epidemiology journals prefer theme_classic() instead of theme_bw()
  theme(
    axis.title.x = element_text(size = 12),
    axis.title.y = element_text(size = 12),
    axis.text.x = element_text(size = 10),
    axis.text.y = element_text(size = 10),
    legend.title = element_text(size = 11),
    legend.text = element_text(size = 10)
  )

```

Scale for x is already present.

Adding another scale for x, which will replace the existing scale.

Scale for y is already present.

Adding another scale for y, which will replace the existing scale.

```

#-----
# Customize the risk table
#-----
p_u_psm$table <- p_u_psm$table +
  scale_x_continuous(
    limits = c(0, 135),
    breaks = seq(0, 135, 12)
  ) +
  theme(
    axis.title.x = element_blank(),
    axis.text.x = element_blank(),
    axis.ticks.x = element_blank(),
    axis.line.x = element_blank(),
    axis.line.y = element_blank(),
    plot.title = element_text(size = 12),
    text = element_text(size = 10),
    axis.title.y = element_text(size = 12)
  )

```

Scale for x is already present.

Adding another scale for x, which will replace the existing scale.

```

#-----
# Combine plot and risk table
#-----
final_plot_psm <- p_u_psm$plot / p_u_psm$table +
  plot_layout(heights = c(4, 1))

#-----
# Save the figure
#-----
ggsave(
  filename = "figures/UTI_m.png",
  plot = final_plot_psm,
  width = 8,
  height = 6,
  dpi = 300
)

Ignoring unknown labels:
* colour : "Treatment"

# "In the propensity score-matched cohort, weighted Cox proportional hazards regression with robust var:

# Figure: Weighted Kaplan-Meier cumulative incidence curves after matching.
# Table: Weighted Cox proportional hazards model using the matching weights estimating the association b

# Weighted Cox proportional hazards regression accounting for matched subclasses showed no significant c
# If you show Kaplan-Meier curves: Survival distributions were compared using the log-rank test (p=0.60)

```

3 Regression

Block 1

```

# GEE model after matching - Accounting for center-clustering

# Block 1: technique effect on postoperative outcomes = Univariable analyses

outcomes <- c(
  "LUTD",
  "OBST",
  "URCOMPINJ",
  "SURG",
  "ILEUS",
  "GU30",
  "GU90",
  "PROC365",
  "REC011",
  "REC011_best",
  "REC011_worst"
)

```

```

# convert outcomes to numeric
mdata[outcomes] <- lapply(
  mdata[outcomes],
  function(x) as.numeric(as.character(x))
)

# Apply GEE model after matching for all binary outcomes
gee_results_psm <- lapply(outcomes, function(y) {
  formula <- as.formula(
    paste(y, "~ TRT2")
  )
  model <- geeglm(
    formula,
    data = mdata,
    id = CENTER,
    weights = weights_psm,      # weights have no effect because weights are all equal
    family = binomial(link = "logit"),
    corstr = "exchangeable"
  )
  tidy(model, exponentiate = TRUE, conf.int = TRUE) # estimate = OR because exponentiate = TRUE
})
names(gee_results_psm) <- outcomes

# Combine results into one table
gee_table_psm <- do.call(
  rbind,
  lapply(names(gee_results_psm), function(x){
    tmp <- gee_results_psm[[x]]
    tmp <- tmp %>%
      filter(grepl("^TRT2", term)) %>%
      mutate(outcome = x)
    tmp
  })
)

# Create final manuscript table
gee_table_final_psm <- gee_table_psm %>%
  mutate(
    Effect_measure = "OR",
    Estimate = sprintf("%.2f", estimate),
    CI = paste0(
      sprintf("%.2f", conf.low),
      "-",
      sprintf("%.2f", conf.high)
    ),
    P_value = p.value
  ) %>%
  dplyr::select(
    Outcome = outcome,
    Effect_measure,
    Estimate,
    CI,

```

```

    P_value
  )

# Cox model for POSTUTI: In the propensity score-matched cohort, Cox proportional hazards regression with
# matching on propensity score

cox_uti_psm <- coxph(
  Surv(POSTUTIFUP, POSTUTI == 1) ~ TRT2 + strata(CENTER),
  data = mdata,
  weights = weights_psm, # weights have no effect
  robust = TRUE,
  cluster = subclass     # after matching, data structure is: Each treated patient is paired with a control
)

# The interpretation is: Among patients with similar propensity scores (matched pairs), what is the association
# between TRT2 and the outcome?

# `strata(CENTER)` allows each center to have its own baseline hazard function. It estimates the treatment effect
# within each center.

cox_summarypsm <- summary(cox_uti_psm)

cox_row <- grep(
  "~TRT2",
  rownames(cox_summarypsm$coefficients)
)

POSTUTI_table_psm <- data.frame(
  Outcome = "POSTUTI",
  Effect_measure = "HR",
  Estimate = sprintf(
    "%.2f",
    cox_summarypsm$coefficients[
      cox_row,
      "exp(coef)"
    ]
  ),
  CI = paste0(
    sprintf("%.2f",
      cox_summarypsm$conf.int[cox_row, "lower .95"]),
    "-",
    sprintf("%.2f",
      cox_summarypsm$conf.int[cox_row, "upper .95"])
  ),
  P_value = cox_summarypsm$coefficients[
    cox_row,
    "Pr(>|z|)"
  ]
)

# Combine final results
final_resultpsm <- rbind(
  gee_table_final_psm,
  POSTUTI_table_psm
)

```

```

# Multiple correction

## Bonferroni correction
final_resultspsm$p_bonferroni <- p.adjust(
  final_resultspsm$P_value,
  method = "bonferroni"
)

## Holm-Bonferroni
final_resultspsm$p_holm <- p.adjust(
  final_resultspsm$P_value,
  method = "holm"
)

## Benjamini-Hochberg FDR
final_resultspsm$p_FDR <- p.adjust(
  final_resultspsm$P_value,
  method = "BH"
)

final_resultspsm <- final_resultspsm %>%
  mutate(
    across(
      c(P_value, p_bonferroni, p_holm, p_FDR),
      ~case_when(
        .x < 0.001 ~ "<0.001",
        TRUE ~ sprintf("%.3f", .x)
      )
    )
  )

final_resultspsm

# A tibble: 12 x 8
  Outcome      Effect_measure Estimate CI      P_value p_bonferroni p_holm p_FDR
  <chr>        <chr>          <chr> <chr>    <chr>    <chr>      <chr> <chr>
1 LUTD        OR            1.38  1.01~ 0.044    0.523      0.305 0.087
2 OBST        OR            3.76  2.02~ <0.001  <0.001      <0.001 <0.0~
3 URCOMPINJ   OR            0.40  0.28~ <0.001  <0.001      <0.001 <0.0~
4 SURG        OR            0.86  0.64~ 0.333    1.000      1.000 0.400
5 ILEUS       OR            0.04  0.02~ <0.001  <0.001      <0.001 <0.0~
6 GU30        OR            0.03  0.00~ 0.015    0.184      0.123 0.037
7 GU90        OR            0.22  0.09~ <0.001  0.006      0.004 0.001
8 PROC365     OR            1.00  0.59~ 0.986    1.000      1.000 0.986
9 REC011      OR            0.70  0.45~ 0.114    1.000      0.683 0.195
10 REC011_best OR            0.71  0.42~ 0.187    1.000      0.935 0.281
11 REC011_worst OR           1.12  0.91~ 0.283    1.000      1.000 0.378
12 POSTUTI    HR            0.80  0.48~ 0.370    1.000      1.000 0.403

```



```

final_resultspsm %>%
  flextable() %>%
  theme_booktabs() %>%
  autofit() %>%
  align(
    j = c("Estimate", "CI", "P_value"),
    align = "center",
    part = "all"
  ) %>%
  save_as_docx(
    path = "output/treatment_unadj_m.docx"
  )

# Evaluate whether CENTER distribution is balanced after weighting, despite that CENTER was not included

# bal.tab(
#   m.out,
#   data = Out,
#   add1 = ~ CENTER
# )

# Results showed that all CENTER SMD < 0.05 → excellent balance
# Therefore, after ATO weighting, CENTER is well balanced between treatment groups.
# This is an important finding because your original data showed strong center-treatment imbalance.
# Why did CENTER become balanced although it was not in the propensity score model?
# This is possible because CENTER was correlated with the covariates you included in the propensity model

```

Block 2

Block 2: exposure-effect analysis (stent, operative time, catheterization) on postoperative outcomes :

```

outcomes <- c(
  "LUTD",
  "OBST",
  "URCOMPINJ",
  "SURG",
  "ILEUS",
  "GU30",
  "GU90",
  "PROC365",
  "REC011",
  "REC011_best",
  "REC011_worst"
)

# convert outcomes to numeric
mdata[outcomes] <- lapply(
  mdata[outcomes],
  function(x) as.numeric(as.character(x))
)

# Predictors to include in the outcome models

```

```

predictors <- c("TRT2", "STENT")

# Apply GEE model after overlap weighting for all binary outcomes
gee_results_psm2 <- lapply(outcomes, function(y) {
  formula <- as.formula(
    paste(y, "~",
          paste(predictors, collapse = " + ")
    )
  )
  model <- geeglm(
    formula,
    data = mdata,
    id = CENTER,
    weights = weights_psm, # weights have no effect because weights are all equal
    family = binomial(link = "logit"),
    corstr = "exchangeable"
  )
  tidy(model, exponentiate = TRUE, conf.int = TRUE) # estimate = OR because exponentiate = TRUE
})
names(gee_results_psm2) <- outcomes

# Combine results into one table
gee_table_psm2 <- do.call(
  rbind,
  lapply(names(gee_results_psm2), function(x){
    tmp <- gee_results_psm2[[x]]
    tmp %>%
      mutate(outcome = x)
  })
)

# Create final manuscript table
gee_table_final_psm2 <- gee_table_psm2 %>%
  mutate(
    Effect_measure = "OR",
    Estimate = sprintf("%.2f", estimate),
    CI = paste0(
      sprintf("%.2f", conf.low),
      "-",
      sprintf("%.2f", conf.high)
    ),
    P_value = p.value
  ) %>%
  dplyr::select(
    Outcome = outcome,
    Predictor = term,
    Effect_measure,
    Estimate,
    CI,
    P_value
  )

```

```

# Cox model for POSTUTI

cox_uti_psm2 <- coxph(
  Surv(POSTUTIFUP, POSTUTI == 1) ~ TRT2 + STENT + strata(CENTER),
  data = mdata,
  weights = weights_psm, # weights have no effect
  robust = TRUE,
  cluster = subclass
)

cox_summarypsm2 <- summary(cox_uti_psm2)

cox_row <- rownames(cox_summarypsm2$coefficients)

POSTUTI_table_psm2 <- data.frame(
  Outcome = "POSTUTI",
  Predictor = rownames(cox_summarypsm2$coefficients),
  Effect_measure = "HR",
  Estimate = sprintf(
    "%.2f",
    cox_summarypsm2$coefficients[
      cox_row,
      "exp(coef)"
    ]
  ),
  CI = paste0(
    sprintf("%.2f",
      cox_summarypsm2$conf.int[cox_row, "lower .95"]),
    "-",
    sprintf("%.2f",
      cox_summarypsm2$conf.int[cox_row, "upper .95"])
  ),
  P_value = cox_summarypsm2$coefficients[
    cox_row,
    "Pr(>|z|)"
  ]
)

# Combine final results
final_results_psm2 <- rbind(
  gee_table_final_psm2,
  POSTUTI_table_psm2
)

final_results_psm2

# A tibble: 35 x 6
  Outcome Predictor Effect_measure Estimate CI P_value
* <chr>    <chr>      <chr>      <chr>    <chr>    <dbl>
1 LUTD     (Intercept) OR        0.35    0.29-0.42 0
2 LUTD     TRT21       OR        1.15    0.91-1.46 0.245

```

3	LUTD	STENT1	OR	0.23	0.06-0.92	0.0371
4	OBST	(Intercept)	OR	0.02	0.00-0.08	0.00000127
5	OBST	TRT21	OR	3.90	2.49-6.09	0.00000000236
6	OBST	STENT1	OR	1.84	0.91-3.73	0.0904
7	URCOMPINJ	(Intercept)	OR	0.02	0.01-0.03	0
8	URCOMPINJ	TRT21	OR	0.41	0.28-0.60	0.00000394
9	URCOMPINJ	STENT1	OR	2.94	0.37-23.06	0.305
10	SURG	(Intercept)	OR	0.84	0.71-0.99	0.0323

i 25 more rows

```
final_results_psm2 <- final_results_psm2 %>%
  mutate(
    across(
      c(P_value),
      ~case_when(
        .x < 0.001 ~ "<0.001",
        TRUE ~ sprintf("%.3f", .x)
      )
    )
  )
```

```
final_results_psm2 %>%
  flextable() %>%
  theme_booktabs() %>%
  autofit() %>%
  align(
    j = c("Estimate", "CI", "P_value"),
    align = "center",
    part = "all"
  ) %>%
  save_as_docx(
    path = "treatment_adj_all_m.docx"
  )
```

```
# Multiple correction
```

```
# Extract TRT2 rows only
```

```
gee_trt2_psm2 <- gee_table_final_psm2 %>%
  filter(grepl("^TRT2", Predictor))
```

```
POSTUTI_trt2_psm2 <- POSTUTI_table_psm2 %>%
  filter(grepl("^TRT2", Predictor))
```

```
# Combine only treatment effects
```

```
treatment_results_psm2 <- rbind(
  gee_trt2_psm2,
  POSTUTI_trt2_psm2
)
```

```
# Apply Correction
```

```

treatment_results_psm2$p_bonferroni <- p.adjust(
  as.numeric(treatment_results_psm2$P_value),
  method = "bonferroni"
) # Bonferroni correction

treatment_results_psm2$p_holm <- p.adjust(
  as.numeric(treatment_results_psm2$P_value),
  method = "holm"
) # Holm-Bonferroni

treatment_results_psm2$p_FDR <- p.adjust(
  as.numeric(treatment_results_psm2$P_value),
  method = "BH"
) # Benjamini-Hochberg FDR

treatment_results_psm2 <- treatment_results_psm2 %>%
  mutate(
    across(
      c(P_value, p_bonferroni, p_holm, p_FDR),
      ~case_when(
        .x < 0.001 ~ "<0.001",
        TRUE ~ sprintf("%.3f", .x)
      )
    )
  )

treatment_results_psm2

# A tibble: 12 x 9
  Outcome Predictor Effect_measure Estimate CI P_value p_bonferroni p_holm
  <chr> <chr> <chr> <chr> <chr> <chr> <chr> <chr>
1 LUTD TRT21 OR 1.15 0.91~ 0.245 1.000 0.849
2 OBST TRT21 OR 3.90 2.49~ <0.001 <0.001 <0.001
3 URCOMPINJ TRT21 OR 0.41 0.28~ <0.001 <0.001 <0.001
4 SURG TRT21 OR 0.74 0.60~ 0.006 0.070 0.041
5 ILEUS TRT21 OR 0.08 0.06~ <0.001 <0.001 <0.001
6 GU30 TRT21 OR 0.06 0.02~ <0.001 <0.001 <0.001
7 GU90 TRT21 OR 0.19 0.09~ <0.001 <0.001 <0.001
8 PROC365 TRT21 OR 0.95 0.62~ 0.828 1.000 0.849
9 REC011 TRT21 OR 0.77 0.50~ 0.212 1.000 0.849
10 REC011_b~ TRT21 OR 0.67 0.48~ 0.021 0.256 0.128
11 REC011_w~ TRT21 OR 1.18 0.96~ 0.111 1.000 0.554
12 POSTUTI TRT21 HR 0.78 0.48~ 0.329 1.000 0.849
# i 1 more variable: p_FDR <chr>

treatment_results_psm2 %>%
  flextable() %>%
  theme_booktabs() %>%
  autofit() %>%
  align(
    j = c("Estimate", "CI", "P_value"),
    align = "center",

```

```

    part = "all"
  ) %>%
  save_as_docx(
    path = "output/treatment_adj_m.docx"
  )

# ADDITIONAL MULTIPLE COMPARISONS

# Extract STENT rows only

gee_st_psm2 <- gee_table_final_psm2 %>%
  filter(grepl("^STENT", Predictor))

POSTUTI_st_psm2 <- POSTUTI_table_psm2 %>%
  filter(grepl("^STENT", Predictor))

# Combine only stent effects
st_results_psm2 <- rbind(
  gee_st_psm2,
  POSTUTI_st_psm2
)

# Apply Correction

st_results_psm2$p_bonferroni <- p.adjust(
  as.numeric(st_results_psm2$P_value),
  method = "bonferroni"
) # Bonferroni correction

st_results_psm2$p_holm <- p.adjust(
  as.numeric(st_results_psm2$P_value),
  method = "holm"
) # Holm-Bonferroni

st_results_psm2$p_FDR <- p.adjust(
  as.numeric(st_results_psm2$P_value),
  method = "BH"
) # Benjamini-Hochberg FDR

st_results_psm2 <- st_results_psm2 %>%
  mutate(
    across(
      c(P_value, p_bonferroni, p_holm, p_FDR),
      ~case_when(
        .x < 0.001 ~ "<0.001",
        TRUE ~ sprintf("%.3f", .x)
      )
    )
  )

st_results_psm2

```

```
# A tibble: 12 x 9
  Outcome Predictor Effect_measure Estimate CI P_value p_bonferroni p_holm
  <chr> <chr> <chr> <chr> <chr> <chr> <chr> <chr>
1 LUTD STENT1 OR 0.23 0.06~ 0.037 0.445 0.260
2 OBST STENT1 OR 1.84 0.91~ 0.090 1.000 0.452
3 URCOMPINJ STENT1 OR 2.94 0.37~ 0.305 1.000 1.000
4 SURG STENT1 OR 0.11 0.03~ 0.002 0.020 0.017
5 ILEUS STENT1 OR 0.09 0.02~ 0.005 0.059 0.044
6 GU30 STENT1 OR 0.22 0.07~ 0.008 0.093 0.062
7 GU90 STENT1 OR 0.27 0.07~ 0.052 0.623 0.312
8 PROC365 STENT1 OR 0.65 0.09~ 0.677 1.000 1.000
9 REC011 STENT1 OR 2.59 1.95~ <0.001 <0.001 <0.001
10 REC011_b~ STENT1 OR 0.48 0.07~ 0.464 1.000 1.000
11 REC011_w~ STENT1 OR 1.80 1.54~ <0.001 <0.001 <0.001
12 POSTUTI STENT1 HR 0.79 0.33~ 0.601 1.000 1.000
# i 1 more variable: p_FDR <chr>
```

```
st_results_psm2 %>%
  flextable() %>%
  theme_booktabs() %>%
  autofit() %>%
  align(
    j = c("Estimate", "CI", "P_value"),
    align = "center",
    part = "all"
  ) %>%
  save_as_docx(
    path = "output/stent_adj_m.docx"
  )
```

4 Rosenbaum's sensitivity analyses

```
# Rosenbaum's primal sensitivity analysis
```

```
outcomes <- c(
  "LUTD",
  "OBST",
  "URCOMPINJ",
  "SURG",
  "ILEUS",
  "GU30",
  "GU90",
  "PROC365",
  "REC011",
  "REC011_best",
  "REC011_worst"
)
```

```
# Convert outcomes to numeric
mdata[outcomes] <- lapply(
  mdata[outcomes],
  function(x) as.numeric(as.character(x))
)
```

)

```
# Function for Rosenbaum sensitivity analysis using rbounds
rosenbaum_binary <- function(outcome, data, match_object,
                             gamma_seq = seq(1, 11, by = 0.1),
                             alpha = 0.05) {

  # matched pairs
  treated_ids <- rownames(match_object$match.matrix)
  control_ids <- match_object$match.matrix[, 1]

  pairs <- data.frame(
    treated = data[treated_ids, outcome],
    control = data[control_ids, outcome]
  )

  # remove missing outcome pairs
  pairs <- na.omit(pairs)

  # discordant pairs
  # x = control has outcome, treated does not
  # y = treated has outcome, control does not
  x <- sum(pairs$control == 1 & pairs$treated == 0)
  y <- sum(pairs$treated == 1 & pairs$control == 0)

  # run Rosenbaum sensitivity
  if ((x + y) == 0) {
    return(data.frame(
      Outcome = outcome,
      Discordant_pairs = 0,
      x = x,
      y = y,
      p_Gamma1 = NA,
      gamma_max = NA,
      p_gamma_max = NA
    ))
  }

  # Run sensitivity over Gamma values

  gamma_results <- sapply(gamma_seq, function(g) {

    res <- binarysens(
      x = x,
      y = y,
      Gamma = g,
      GammaInc = 0.1 # GammaInc defines the increment used to increase  $\Gamma$  values when the function eval
    )

    # extract upper bound p-value corresponding to Gamma g
    res$bounds$`Upper bound`[
      res$bounds$Gamma == g
    ]
  })
}
```



```

    ]
  })

  # p-value at Gamma = 1
  p_gamma1 <- gamma_results[gamma_seq == 1]

  # first Gamma where upper p-value >= alpha
  idx <- which(gamma_results < alpha)

  if(length(idx) == 0){

    if(p_gamma1 >= alpha){

      gamma_max <- "NS at Gamma=1"
      p_gamma_max <- NA

    } else {

      gamma_max <- paste0(">=", max(gamma_seq))
      p_gamma_max <- gamma_results[length(gamma_results)]

    }

  } else {

    gamma_max_value <- gamma_seq[max(idx)]

    # If still significant at the maximum tested Gamma
    if(gamma_max_value == max(gamma_seq)){

      gamma_max <- paste0(">=", max(gamma_seq))

    } else {

      gamma_max <- gamma_max_value

    }

    p_gamma_max <- gamma_results[max(idx)]
  }

  # Extract p-values from rbounds object
  data.frame(
    Outcome = outcome,
    Discordant_pairs = x + y,
    x = x,
    y = y,
    p_Gamma1 = round(p_gamma1, 3),
    gamma_max = gamma_max,
    p_gamma_max = round(p_gamma_max, 3)
  )
}

```

```

# Run all outcomes
rosenbaum_results <- do.call(
  rbind,
  lapply(
    outcomes,
    rosenbaum_binary,
    data = mdata,
    match_object = m.out
  )
)

rosenbaum_results

      Outcome Discordant_pairs  x  y p_Gamma1      gamma_max p_gamma_max
1      LUTD             28 12 16   0.172 NS at Gamma=1         NA
2      OBST             18  5 13   0.015          1.2       0.038
3  URCOMPINJ              4  3  1   0.062 NS at Gamma=1         NA
4      SURG            42 22 20   0.322 NS at Gamma=1         NA
5      ILEUS            12 12  0   0.000          >=11       0.000
6      GU30            32 31  1   0.000          10.1       0.049
7      GU90            33 27  6   0.000           2.4       0.048
8  PROC365            10  5  5   0.377 NS at Gamma=1         NA
9   REC011              8  5  3   0.145 NS at Gamma=1         NA
10 REC011_best           10  6  4   0.172 NS at Gamma=1         NA
11 REC011_worst          45 23 22   0.383 NS at Gamma=1         NA

# Rename columns for publication
rosenbaum_results_pub <- rosenbaum_results %>%
  rename(
    Outcome = Outcome,
    `Discordant pairs` = Discordant_pairs,
    `Control only events (x)` = x,
    `Treated only events (y)` = y,
    `P-value ( $\Gamma=1$ )` = p_Gamma1,
    `Gamma_max` = gamma_max,
    `P-value at Gamma_max` = p_gamma_max
  )

# Create publication table
rosenbaum_table <- rosenbaum_results_pub %>%
  flextable() %>%

  # Add title
  add_header_lines(
    "Rosenbaum's primal sensitivity analysis for postoperative outcomes in the matched cohort."
  ) %>%

  # Add footnote
  add_footer_lines(
    "The Gamma value  $\Gamma$  is a measure of sensitivity to hidden bias, representing the magnitude of differ
  ) %>%

```

```

theme_booktabs() %>%
autofit() %>%

align(
  j = c(
    "Discordant pairs",
    "Control only events (x)",
    "Treated only events (y)",
    "P-value ( $\Gamma=1$ )",
    " $\Gamma_{\max}$ ",
    "P-value at  $\Gamma_{\max}$ "
  ),
  align = "center",
  part = "all"
)

# Save as Word document
save_as_docx(
  rosenbaum_table,
  path = "output/Rosenbaum_effect.docx"
)

# A manual Rosenbaum primal sensitivity analysis for a single outcome

# Extract matched pairs for outcomes and filter those who did not get matched (NA values)

# m.pairs <- cbind(
#   mdata[row.names(m.out$match.matrix), 'ILEUS'], # Replace 'OUTCOME_VAR' with your binary outcome
#   mdata[m.out$match.matrix, 'ILEUS']
#)

# print(m.pairs)
# m.pairs <- na.omit(m.pairs)
# print(m.pairs)

# Count discordant pairs
# Discordant pairs are pairs where one unit (treated or control) has the outcome, but the other does not.
# Let x represent the number of pairs where the control has the outcome, but the treated does not.
# Let y represent the number of pairs where the treated has the outcome, but the control does not.
# x <- sum((m.pairs[, 1] == 0) & (m.pairs[, 2] == 1))
# y <- sum((m.pairs[, 1] == 1) & (m.pairs[, 2] == 0))
# print(x)
# print(y)

# Perform Rosenbaum primal sensitivity analysis
# primal <- binarysens(x=x, y=y, Gamma = 15, GammaInc = 2)
# print(primal)

# The matched pairs control for observed confounders, isolating the treatment effect.

```

```
# By focusing on discordant outcomes, Rosenbaum's method assesses the sensitivity of your results to un
```

TEMPORAL CHANGES

Number of procedures per year

Plots

```
# This plot shows: "What proportion of all Cohen procedures occurred in each year?" and separately "What
```

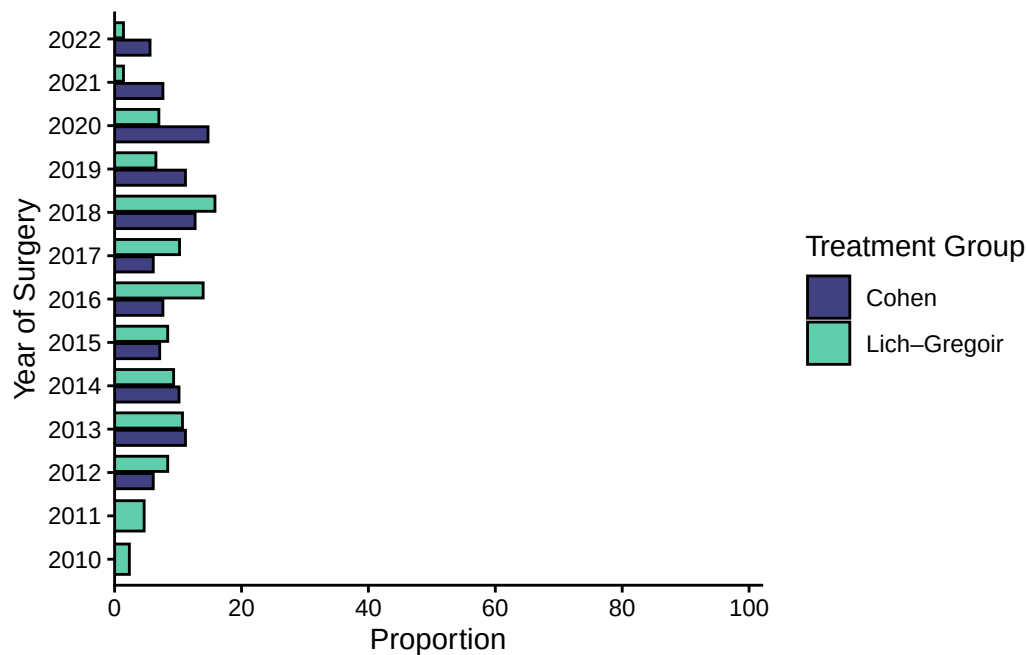
```
#-----  
# Horizontal bar for each year  
#-----
```

```
# -----  
# The whole cohort  
# -----
```

```
plot_data1 <- Out %>%  
  count(YEARS, Treatment) %>%  
  group_by(Treatment) %>%  
  mutate(  
    Percent = 100 * n / sum(n)  
  ) %>%  
  ungroup()
```

```
p_y_m1 <- ggplot(  
  plot_data1,  
  aes(  
    x = Percent,  
    y = factor(YEARS),  
    fill = Treatment  
  )  
) +  
  geom_col(  
    position = position_dodge(width = 0.8),  
    width = 0.7,  
    colour = "black"  
  ) +  
  scale_x_continuous(  
    limits = c(0, 100),  
    breaks = seq(0, 100, 20),  
    expand = expansion(mult = c(0, 0.02))  
  ) +  
  scale_fill_viridis_d(option = "mako", begin = 0.3, end = 0.8)  
  ) +  
  labs(x = "Proportion", y = "Year of Surgery", fill = "Treatment Group") +  
  theme_classic()
```

```
p_y_m1
```



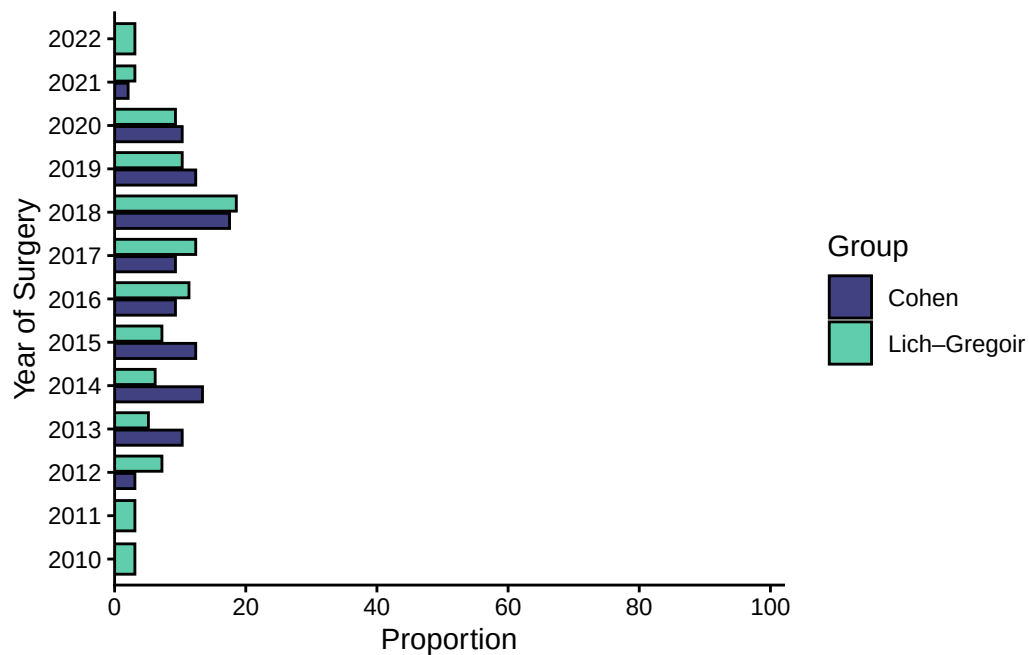
```
# -----
# After matching
# -----

plot_data <- mdata %>%
  count(YEARS, Treatment) %>%
  group_by(Treatment) %>%
  mutate(
    Percent = 100 * n / sum(n)
  ) %>%
  ungroup()

p_y_m <- ggplot(
  plot_data,
  aes(
    x = Percent,
    y = factor(YEARS),
    fill = Treatment
  )
) +
  geom_col(
    position = position_dodge(width = 0.8),
    width = 0.7,
    colour = "black"
  ) +
  scale_x_continuous(
    limits = c(0, 100),
    breaks = seq(0, 100, 20),
    expand = expansion(mult = c(0, 0.02))
  ) +
  scale_fill_viridis_d(option = "mako", begin = 0.3, end = 0.8)
  ) +
```

```
labs(x = "Proportion", y = "Year of Surgery", fill = "Group") +
theme_classic()
```

p_y_m



```
# -----
# After ATO weighting
# -----

plot_data_ato <- Out %>%
  group_by(YEARS, Treatment) %>%
  summarise(
    weighted_n = sum(weights),
    .groups = "drop"
  ) %>%
  group_by(Treatment) %>%
  mutate(
    Percent = 100 * weighted_n / sum(weighted_n)
  ) %>%
  ungroup()

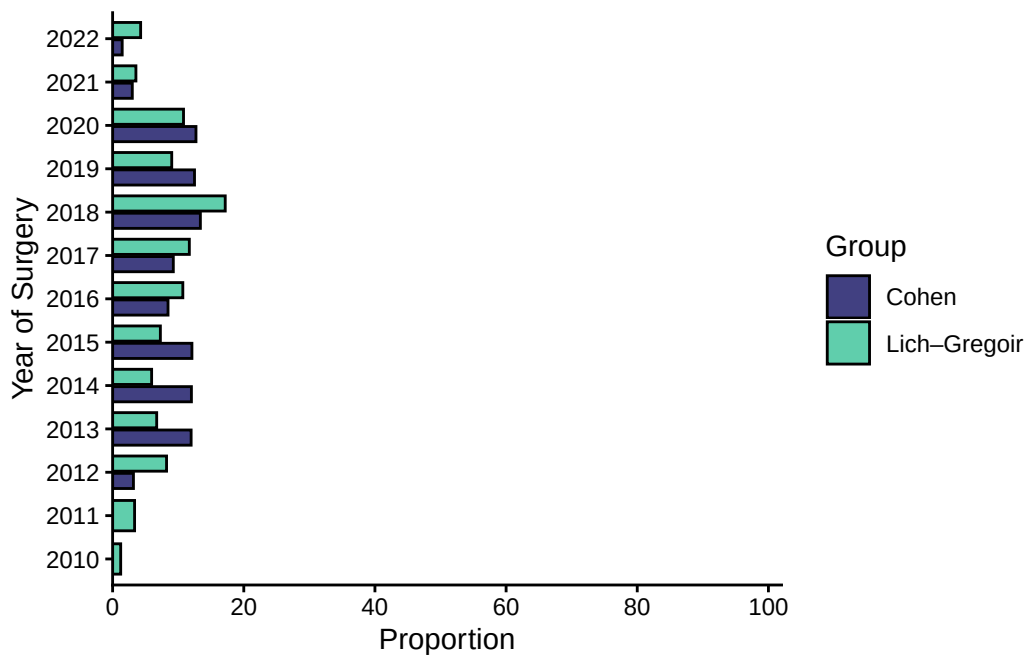
p_y_m_ato <- ggplot(
  plot_data_ato,
  aes(
    x = Percent,
    y = factor(YEARS),
    fill = Treatment
  )
) +
  geom_col(
    position = position_dodge(width = 0.8),
    width = 0.7,
```

```

    colour = "black"
  ) +
  scale_x_continuous(
    limits = c(0, 100),
    breaks = seq(0, 100, 20),
    expand = expansion(mult = c(0, 0.02))
  ) +
  scale_fill_viridis_d(
    option = "mako",
    begin = 0.3,
    end = 0.8
  ) +
  labs(
    x = "Proportion",
    y = "Year of Surgery",
    fill = "Group"
  ) +
  theme_classic()

```

p_y_m_ato



```

#-----
# Add subtitle (we can keep or remove them from the final figure)
#-----

```

```

p_y_m1 <- p_y_m1 +
  labs(subtitle = "Whole cohort")

```

```

p_y_m <- p_y_m +
  labs(subtitle = "Matched cohort")

```

```

p_y_m_ato <- p_y_m_ato +

```

```

    labs(subtitle = "Overlap weighting population")

#-----
# Combine plot
#-----

treatment_plot <-
  p_y_m1 /
  p_y_m /
  p_y_m_ato +
  plot_layout(heights = c(2,2,2)) +
  plot_annotation(
    tag_levels = "A"
  )

#-----
# Save the figure
#-----

ggsave(filename = "figures/treatment_across_years.png",
        plot = treatment_plot,
        width = 8,    # width in inches
        height = 10,  # height in inches
        dpi = 300)    # resolution

# Titles:
## Temporal distribution of surgical techniques across analysis populations
## Temporal distribution of Cohen and Lich-Gregoir ureteral reimplantation procedures before and after p
## Distribution of surgical cases over time (Year of Surgery) before and after propensity score matching

```

Table

```

# -----
# The whole cohort
# -----

year_table <- Out %>%
  count(YEARS, Treatment) %>%
  group_by(YEARS) %>%
  mutate(
    Percent = 100 * n / sum(n),
    n_percent = paste0(n, " (", round(Percent, 1), "%)")
  ) %>%
  select(YEARS, Treatment, n_percent) %>%
  pivot_wider(
    names_from = Treatment,
    values_from = n_percent
  ) %>%
  left_join(
    Out %>%
      count(YEARS, name = "Total_n") %>%
      mutate(
        Total_percent = 100 * Total_n / nrow(Out),

```



```

      Total = paste0(Total_n, " (", round(Total_percent, 1), "%)")
    ) %>%
      select(YEARS, Total),
      by = "YEARS"
    ) %>%
      arrange(YEARS)

year_table <- year_table %>%
  rename(
    `Year of Surgery` = YEARS,
    `Cohen n (%)` = Cohen,
    `Lich-Gregoir n (%)` = `Lich-Gregoir`,
    `Total surgeries` = Total
  )

year_table %>%
  gt()

year_table

# A tibble: 13 x 4
# Groups:   Year of Surgery [13]
  `Year of Surgery` `Lich-Gregoir n (%)` `Cohen n (%)` `Total surgeries`
      <dbl> <chr>                <chr>          <chr>
1         2010 5 (100%)          <NA>           5 (1.2%)
2         2011 10 (100%)         <NA>           10 (2.4%)
3         2012 18 (60%)           12 (40%)        30 (7.3%)
4         2013 23 (51.1%)         22 (48.9%)      45 (10.9%)
5         2014 20 (50%)           20 (50%)        40 (9.7%)
6         2015 18 (56.2%)         14 (43.8%)      32 (7.8%)
7         2016 30 (66.7%)         15 (33.3%)      45 (10.9%)
8         2017 22 (64.7%)         12 (35.3%)      34 (8.3%)
9         2018 34 (57.6%)         25 (42.4%)      59 (14.3%)
10        2019 14 (38.9%)         22 (61.1%)      36 (8.7%)
11        2020 15 (34.1%)         29 (65.9%)      44 (10.7%)
12        2021 3 (16.7%)          15 (83.3%)      18 (4.4%)
13        2022 3 (21.4%)          11 (78.6%)      14 (3.4%)

year_table <- year_table %>%
  mutate(
    `Year of Surgery` = as.character(`Year of Surgery`)
  )

year_table %>%
  flextable() %>%
  theme_booktabs() %>%
  autofit() %>%
  align(
    align = "center",
    part = "all"
  ) %>%
  save_as_docx(

```

Lich-Gregoir n (%)	Cohen n (%)	Total surgeries
2010		
5 (100%)	NA	5 (1.2%)
2011		
10 (100%)	NA	10 (2.4%)
2012		
18 (60%)	12 (40%)	30 (7.3%)
2013		
23 (51.1%)	22 (48.9%)	45 (10.9%)
2014		
20 (50%)	20 (50%)	40 (9.7%)
2015		
18 (56.2%)	14 (43.8%)	32 (7.8%)
2016		
30 (66.7%)	15 (33.3%)	45 (10.9%)
2017		
22 (64.7%)	12 (35.3%)	34 (8.3%)
2018		
34 (57.6%)	25 (42.4%)	59 (14.3%)
2019		
14 (38.9%)	22 (61.1%)	36 (8.7%)
2020		
15 (34.1%)	29 (65.9%)	44 (10.7%)
2021		
3 (16.7%)	15 (83.3%)	18 (4.4%)
2022		
3 (21.4%)	11 (78.6%)	14 (3.4%)

```

    path = "output/treatment_year_whole_cohort.docx"
  )

# -----
# After matching
# -----

year_tablem <- mdata %>%
  count(YEARS, Treatment) %>%
  group_by(YEARS) %>%
  mutate(
    Percent = 100 * n / sum(n),
    n_percent = paste0(n, " (", round(Percent, 1), "%)")
  ) %>%
  select(YEARS, Treatment, n_percent) %>%
  pivot_wider(
    names_from = Treatment,
    values_from = n_percent
  ) %>%
  left_join(
    Out %>%
      count(YEARS, name = "Total_n") %>%
      mutate(
        Total_percent = 100 * Total_n / nrow(Out),
        Total = paste0(Total_n, " (", round(Total_percent, 1), "%)")
      ) %>%
      select(YEARS, Total),
    by = "YEARS"
  ) %>%
  arrange(YEARS)

year_tablem <- year_tablem %>%
  rename(
    `Year of Surgery` = YEARS,
    `Cohen n (%)` = Cohen,
    `Lich-Gregoir n (%)` = `Lich-Gregoir`,
    `Total surgeries` = Total
  )

year_tablem %>%
  gt()

year_tablem

# A tibble: 13 x 4
# Groups:   Year of Surgery [13]
  `Year of Surgery` `Lich-Gregoir n (%)` `Cohen n (%)` `Total surgeries`
      <dbl> <chr>                <chr>          <chr>
1         2010 3 (100%)          <NA>           5 (1.2%)
2         2011 3 (100%)          <NA>          10 (2.4%)

```

Lich-Gregoir n (%)	Cohen n (%)	Total surgeries
2010		
3 (100%)	NA	5 (1.2%)
2011		
3 (100%)	NA	10 (2.4%)
2012		
7 (70%)	3 (30%)	30 (7.3%)
2013		
5 (33.3%)	10 (66.7%)	45 (10.9%)
2014		
6 (31.6%)	13 (68.4%)	40 (9.7%)
2015		
7 (36.8%)	12 (63.2%)	32 (7.8%)
2016		
11 (55%)	9 (45%)	45 (10.9%)
2017		
12 (57.1%)	9 (42.9%)	34 (8.3%)
2018		
18 (51.4%)	17 (48.6%)	59 (14.3%)
2019		
10 (45.5%)	12 (54.5%)	36 (8.7%)
2020		
9 (47.4%)	10 (52.6%)	44 (10.7%)
2021		
3 (60%)	2 (40%)	18 (4.4%)
2022		
3 (100%)	NA	14 (3.4%)

3	2012	7 (70%)	3 (30%)	30 (7.3%)
4	2013	5 (33.3%)	10 (66.7%)	45 (10.9%)
5	2014	6 (31.6%)	13 (68.4%)	40 (9.7%)
6	2015	7 (36.8%)	12 (63.2%)	32 (7.8%)
7	2016	11 (55%)	9 (45%)	45 (10.9%)
8	2017	12 (57.1%)	9 (42.9%)	34 (8.3%)
9	2018	18 (51.4%)	17 (48.6%)	59 (14.3%)
10	2019	10 (45.5%)	12 (54.5%)	36 (8.7%)
11	2020	9 (47.4%)	10 (52.6%)	44 (10.7%)
12	2021	3 (60%)	2 (40%)	18 (4.4%)
13	2022	3 (100%)	<NA>	14 (3.4%)

```
year_tablem <- year_tablem %>%
  mutate(
    `Year of Surgery` = as.character(`Year of Surgery`)
  )
```

```
year_tablem %>%
  flextable() %>%
  theme_booktabs() %>%
  autofit() %>%
  align(
    align = "center",
    part = "all"
  ) %>%
  save_as_docx(
    path = "output/treatment_year_matched_cohort.docx"
  )
```

```
# -----
# After ATO weighting
# -----
```

```
year_table_ato <- Out %>%
  group_by(YEARS, Treatment) %>%
  summarise(
    weighted_n = sum(weights),
    .groups = "drop"
  ) %>%
  group_by(YEARS) %>%
  mutate(
    Percent = 100 * weighted_n / sum(weighted_n),
    n_percent = paste0(
      round(weighted_n, 1),
      " (",
      round(Percent, 1),
      "%)"
    )
  ) %>%
  select(YEARS, Treatment, n_percent) %>%
  pivot_wider(
    names_from = Treatment,
    values_from = n_percent
```

```

) %>%
left_join(
  Out %>%
    group_by(YEARS) %>%
    summarise(
      weighted_total = sum(weights),
      .groups = "drop"
    ) %>%
    mutate(
      Total_percent = 100 * weighted_total / sum(weighted_total),
      Total = paste0(
        round(weighted_total, 1),
        " (",
        round(Total_percent, 1),
        "%)"
      )
    ),
  by = "YEARS"
) %>%
arrange(YEARS)

year_table_ato <- year_table_ato %>%
  rename(
    `Year of Surgery` = YEARS,
    `Cohen n (%)` = Cohen,
    `Lich-Gregoir n (%)` = `Lich-Gregoir`,
    `Total surgeries` = Total
  )

year_table_ato %>%
  gt()

year_table_ato

# A tibble: 13 x 6
# Groups:   Year of Surgery [13]
  `Year of Surgery` `Lich-Gregoir n (%)` `Cohen n (%)` weighted_total
    <dbl> <chr>                <chr>          <dbl>
1      2010 0.7 (100%)          <NA>           0.740
2      2011 2 (100%)            <NA>           2.01
3      2012 4.9 (72.1%)         1.9 (27.9%)     6.83
4      2013 4 (36%)             7.2 (64%)       11.2
5      2014 3.6 (33.1%)         7.2 (66.9%)     10.8
6      2015 4.4 (37.6%)         7.2 (62.4%)     11.6
7      2016 6.4 (55.9%)         5.1 (44.1%)     11.5
8      2017 7 (55.8%)           5.5 (44.2%)     12.6
9      2018 10.3 (56.2%)        8 (43.8%)       18.3
10     2019 5.4 (41.9%)         7.5 (58.1%)     12.9
11     2020 6.5 (46%)           7.6 (54%)       14.1
12     2021 2.1 (54.2%)         1.8 (45.8%)      3.93
13     2022 2.6 (74.5%)         0.9 (25.5%)      3.44
# i 2 more variables: Total_percent <dbl>, `Total surgeries` <chr>

```

Lich-Gregoir n (%)	Cohen n (%)	weighted_total	Total_percent	Total surgeries
2010				
0.7 (100%)	NA	0.7395953	0.6169953	0.7 (0.6%)
2011				
2 (100%)	NA	2.0084437	1.6755112	2 (1.7%)
2012				
4.9 (72.1%)	1.9 (27.9%)	6.8321610	5.6996181	6.8 (5.7%)
2013				
4 (36%)	7.2 (64%)	11.2030486	9.3459593	11.2 (9.3%)
2014				
3.6 (33.1%)	7.2 (66.9%)	10.7679299	8.9829687	10.8 (9%)
2015				
4.4 (37.6%)	7.2 (62.4%)	11.6140830	9.6888580	11.6 (9.7%)
2016				
6.4 (55.9%)	5.1 (44.1%)	11.4809209	9.5777697	11.5 (9.6%)
2017				
7 (55.8%)	5.5 (44.2%)	12.5586325	10.4768329	12.6 (10.5%)
2018				
10.3 (56.2%)	8 (43.8%)	18.3131712	15.2774623	18.3 (15.3%)
2019				
5.4 (41.9%)	7.5 (58.1%)	12.8868221	10.7506197	12.9 (10.8%)
2020				
6.5 (46%)	7.6 (54%)	14.0985203	11.7614591	14.1 (11.8%)
2021				
2.1 (54.2%)	1.8 (45.8%)	3.9291750	3.2778497	3.9 (3.3%)
2022				
2.6 (74.5%)	0.9 (25.5%)	3.4380010	2.8680959	3.4 (2.9%)

```

year_table_ato <- year_table_ato %>%
  mutate(
    `Year of Surgery` = as.character(`Year of Surgery`)
  )

year_table_ato %>%
  flextable() %>%
  theme_booktabs() %>%
  autofit() %>%
  align(
    align = "center",
    part = "all"
  ) %>%
  save_as_docx(
    path = "output/treatment_year_overlap_cohort.docx"
  )

# Footnote: Percentages for Cohen and Lich-Gregoir represent the distribution of surgical techniques wi

#-----
# Horizontal bar for categorized years
#-----

# -----
# After matching
# -----

# Surgery year was categorized into two eras (arbitrary) (<2015 vs 2015) to explore temporal changes in

# Create year categories
plot_data <- mdata %>%
  mutate(
    Year_g = ifelse(YEARS < 2015, "Before 2015", "2015 and beyond")
  ) %>%
  count(Year_g, Treatment) %>%
  group_by(Treatment) %>%
  mutate(
    Percent = 100 * n / sum(n) # the denominator is the total number of surgeries within each treatm
  ) %>%
  ungroup()

# Plot
p_y_m <- ggplot(
  plot_data,
  aes(
    x = Percent,
    y = factor(
      Year_g,
      levels = c("Before 2015", "2015 and beyond")
    ),
    fill = Treatment
  )

```

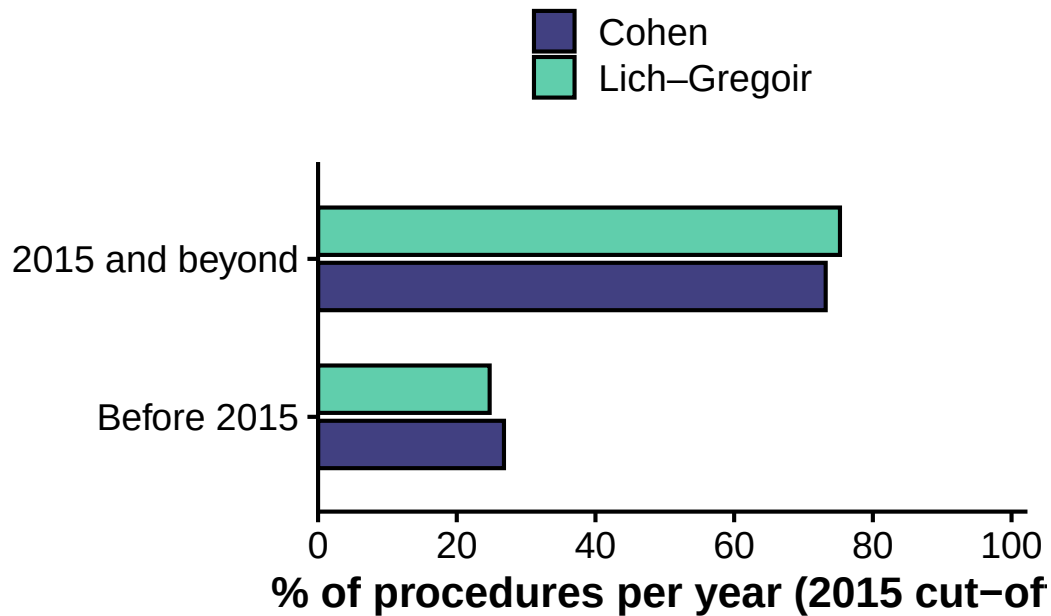


```

)
) +
  geom_col(
    position = position_dodge(width = 0.7),
    width = 0.6,
    colour = "black"
  ) +
  scale_fill_viridis_d(option = "mako", begin = 0.3, end = 0.8)
) +
  scale_x_continuous(
    limits = c(0, 100),
    breaks = seq(0, 100, 20),
    expand = expansion(mult = c(0, 0.02))
  ) +
  labs(
    x = "% of procedures per year (2015 cut-off)",
    y = NULL,
    fill = NULL
  ) +
  theme_classic(base_size = 16) +
  theme(
    legend.position = "top",
    legend.direction = "vertical",
    legend.text = element_text(size = 14),
    axis.text = element_text(size = 14),
    axis.title = element_text(size = 16, face = "bold")
  )
)

```

p_y_m



```

ggsave(
  "figures/procedures_2015_cutoff_after_matching.png",
  p_y_m,

```

```

width = 8,
height = 5,
dpi = 300
)

```

A cumulative enrollment curve

```

# -----
# The whole cohort
# -----
cumulative_data <- Out %>%
  count(YEARS, name = "n_year") %>%
  arrange(YEARS) %>%
  mutate(
    cumulative_n = cumsum(n_year),
    cumulative_percent = 100 * cumulative_n / nrow(Out)
  )

```

```

cumulative_data

```

```

p_cumulative <- ggplot(
  cumulative_data,
  aes(
    x = YEARS,
    y = cumulative_percent
  )
) +
  geom_line(
    linewidth = 1.2
  ) +
  geom_point(
    size = 3
  ) +
  scale_y_continuous(
    limits = c(0,100),
    breaks = seq(0,100,20),
    labels = function(x) paste0(x,"%")
  ) +
  scale_x_continuous(
    breaks = seq(min(Out$YEARS), max(Out$YEARS), 2)
  ) +
  labs(
    x = "Year of Surgery",
    y = "Cumulative percentage of cohort"
  ) +
  theme_classic()

```

```

p_cumulative

```

```

# show cumulative number instead

```

```

p_cumulative_n <- ggplot(

```

```

    cumulative_data,
    aes(
      x = YEARS,
      y = cumulative_n
    )
  ) +
  geom_line(linewidth = 1.2) +
  geom_point(size = 3) +
  labs(
    x = "Year of Surgery",
    y = "Cumulative number of procedures"
  ) +
  theme_classic()

p_cumulative_n

```

```

# Figure title: Cumulative accrual of ureteral reimplantation procedures over the study period

```

Transition in practice: Observed transition in practice

```

# This plot shows "What proportion of surgeries in each year were Cohen vs Lich-Gregoir."
# Among all unilateral VUR repairs performed in a given year, what proportion used Cohen vs Lich-Gregoir.
# This directly measures the change in practice preference.

```

```

# -----
# The whole cohort
# -----

```

```

plot_data1 <- Out %>%
  count(YEARS, Treatment) %>%
  group_by(YEARS) %>%
  mutate(
    Percent = 100 * n / sum(n)
  ) %>%
  ungroup()

p_transition1 <- ggplot(
  plot_data1,
  aes(
    x = YEARS,
    y = Percent,
    color = Treatment,
    group = Treatment
  )
) +
  geom_line(linewidth = 1.2) +
  geom_point(size = 2) +
  scale_y_continuous(
    limits = c(0, 100),
    breaks = seq(0, 100, 20),
    labels = function(x) paste0(x, "%")
  ) +

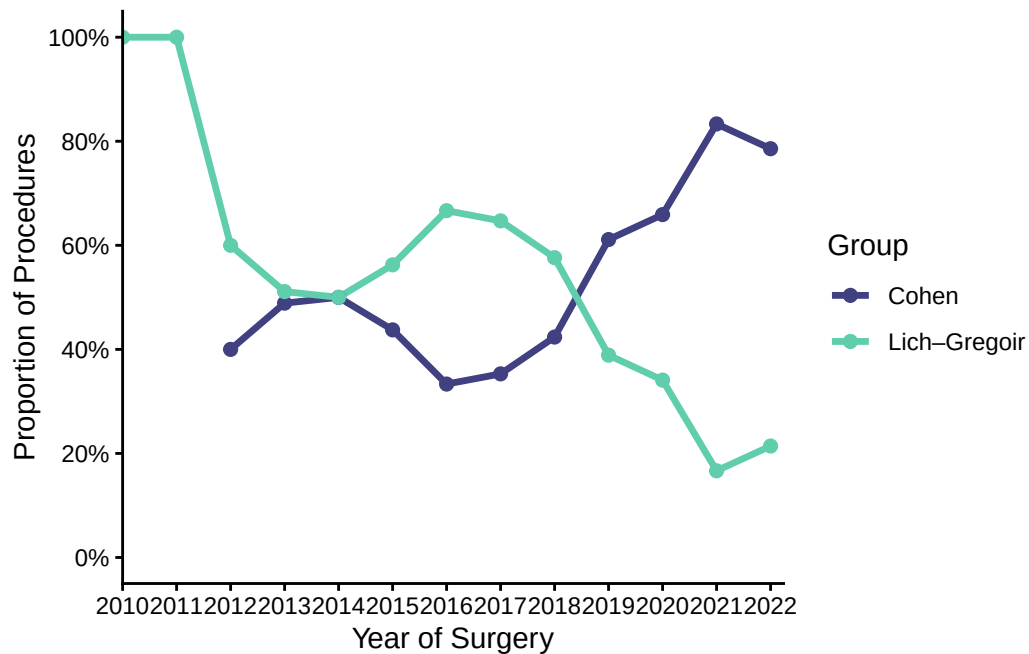
```

```

scale_x_continuous(
  limits = c(2010, 2022),
  breaks = seq(2010, 2022, 1),
  expand = expansion(mult = c(0, 0.02))
) +
scale_color_viridis_d(
  option = "mako",
  begin = 0.3,
  end = 0.8
) +
labs(
  x = "Year of Surgery",
  y = "Proportion of Procedures",
  color = "Group"
) +
theme_classic()

```

p_transition1



```

# -----
# After matching
# -----

```

```

plot_data <- mdata %>%
  count(YEARS, Treatment) %>%
  group_by(YEARS) %>%
  mutate(
    Percent = 100 * n / sum(n)
  ) %>%
  ungroup()

```

```

p_transition <- ggplot(

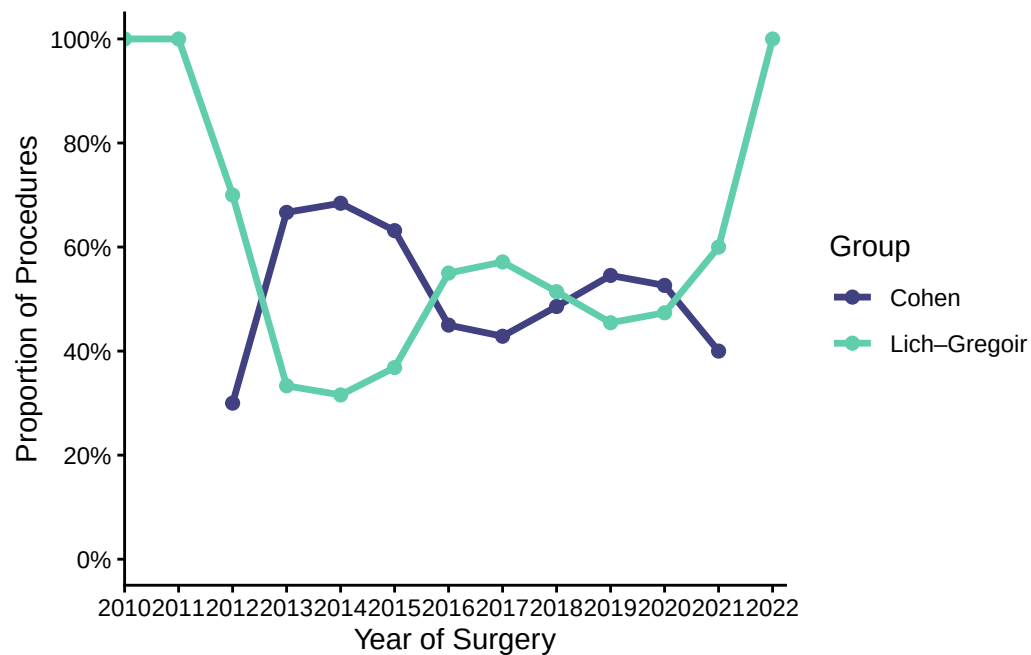
```

```

plot_data,
aes(
  x = YEARS,
  y = Percent,
  color = Treatment,
  group = Treatment
)
) +
geom_line(linewidth = 1.2) +
geom_point(size = 2) +
scale_y_continuous(
  limits = c(0, 100),
  breaks = seq(0, 100, 20),
  labels = function(x) paste0(x, "%")
) +
scale_x_continuous(
  limits = c(2010, 2022),
  breaks = seq(2010, 2022, 1),
  expand = expansion(mult = c(0, 0.02))
) +
scale_color_viridis_d(
  option = "mako",
  begin = 0.3,
  end = 0.8
) +
labs(
  x = "Year of Surgery",
  y = "Proportion of Procedures",
  color = "Group"
) +
theme_classic()

p_transition

```



```
# -----
# After ATO weighting
# -----

plot_data_ato <- Out %>%
  group_by(YEARS, Treatment) %>%
  summarise(
    weighted_n = sum(weights),
    .groups = "drop"
  ) %>%
  group_by(YEARS) %>%
  mutate(
    Percent = 100 * weighted_n / sum(weighted_n)
  ) %>%
  ungroup()

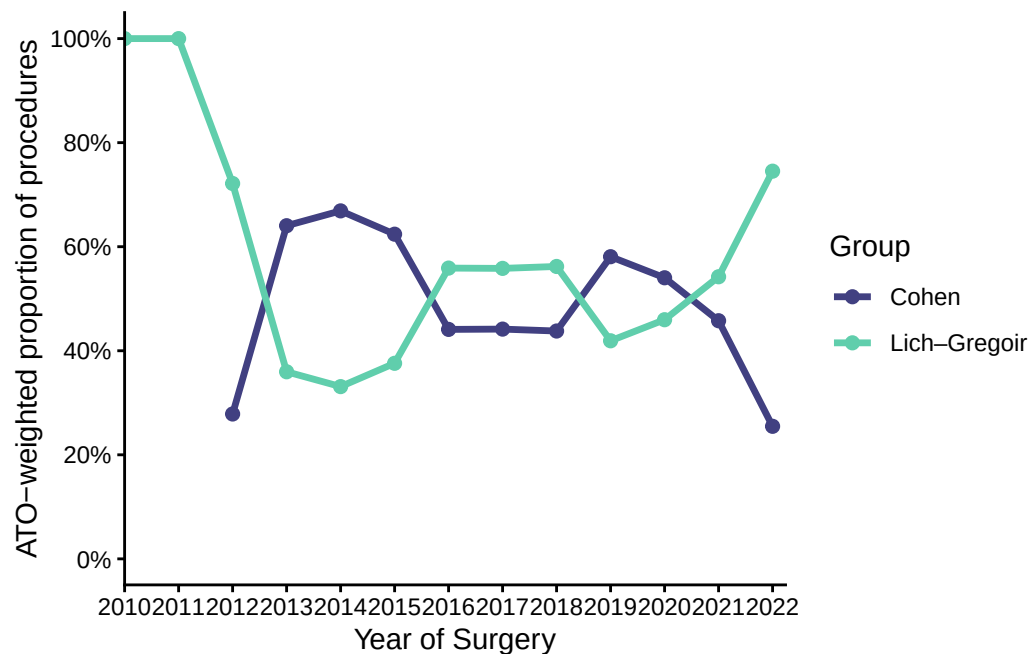
p_transition_ato <- ggplot(
  plot_data_ato,
  aes(
    x = YEARS,
    y = Percent,
    color = Treatment,
    group = Treatment
  )
) +
  geom_line(linewidth = 1.2) +
  geom_point(size = 2) +
  scale_y_continuous(
    limits = c(0, 100),
    breaks = seq(0, 100, 20),
    labels = function(x) paste0(x, "%")
  ) +
```

```

scale_x_continuous(
  limits = c(2010, 2022),
  breaks = seq(2010, 2022, 1),
  expand = expansion(mult = c(0, 0.02))
) +
scale_color_viridis_d(
  option = "mako",
  begin = 0.3,
  end = 0.8
) +
labs(
  x = "Year of Surgery",
  y = "ATO-weighted proportion of procedures",
  color = "Group"
) +
theme_classic()

```

p_transition_ato



```

#-----
# Add subtitle (we can keep or remove them from the final figure)
#-----

p_transition1 <- p_transition1 +
  labs(subtitle = "Whole cohort")

p_transition <- p_transition +
  labs(subtitle = "Matched cohort")

p_transition_ato <- p_transition_ato +
  labs(subtitle = "Overlap weighting population")

```

```

#-----
# Combine plot
#-----

transition_plot <-
  p_transition1 /
  p_transition /
  p_transition_ato +
  plot_layout(heights = c(2,2,2)) +
  plot_annotation(
    tag_levels = "A"
  )

#-----
# Save the figure
#-----
ggsave(filename = "figures/transition_in practice.png",
  plot = transition_plot,
  width = 8,    # width in inches
  height = 10,  # height in inches
  dpi = 300)   # resolution

# Title: Figure X. Temporal evolution of surgical technique use. (A) Whole cohort. (B) Propensity score

```